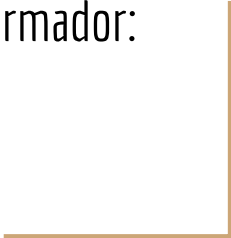




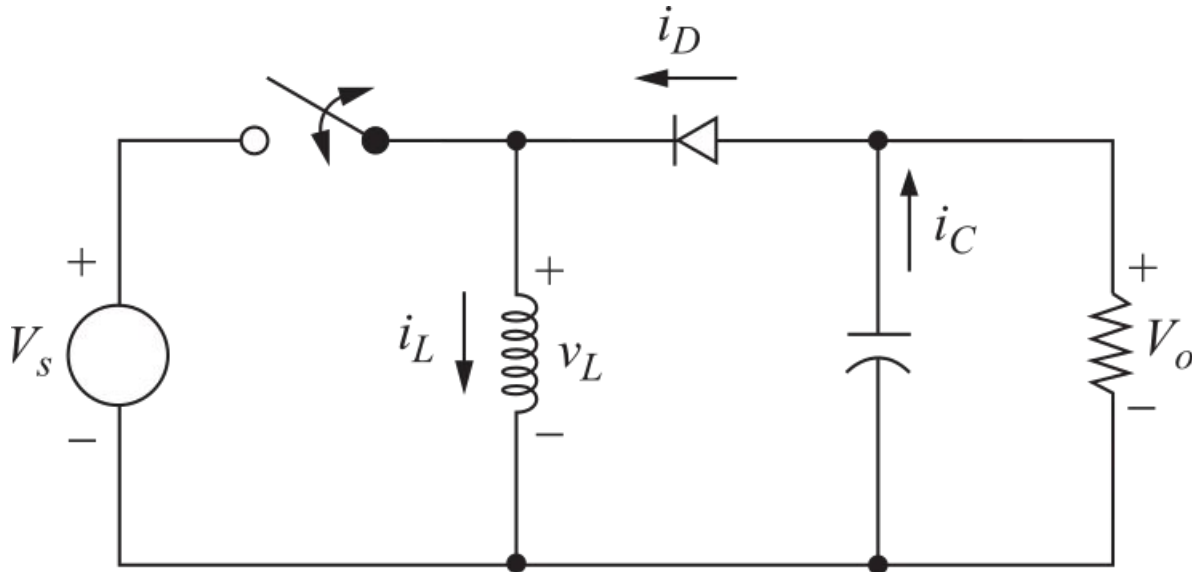
Conversión CC/CC

Convertidores con Transformador:
Flyback

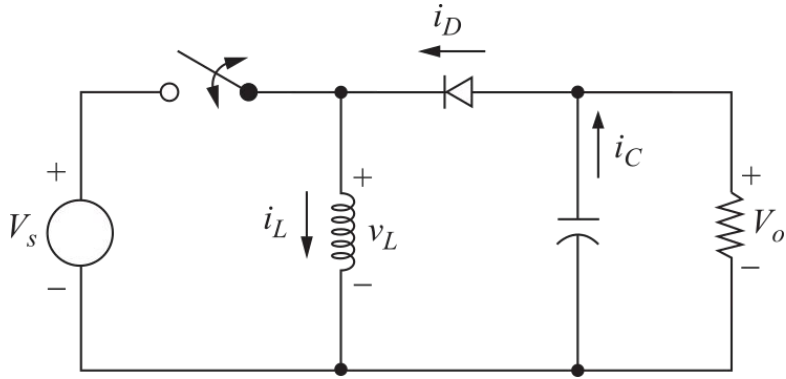


Convertidor Flyback: origen

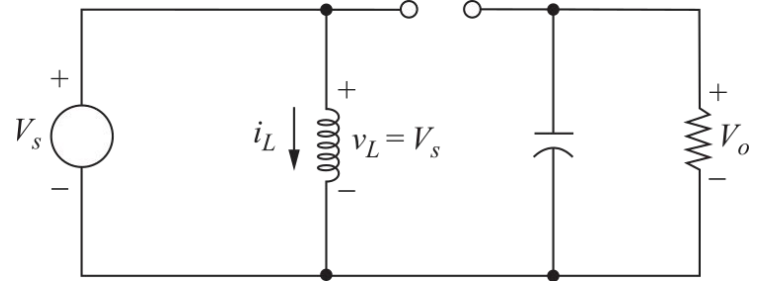
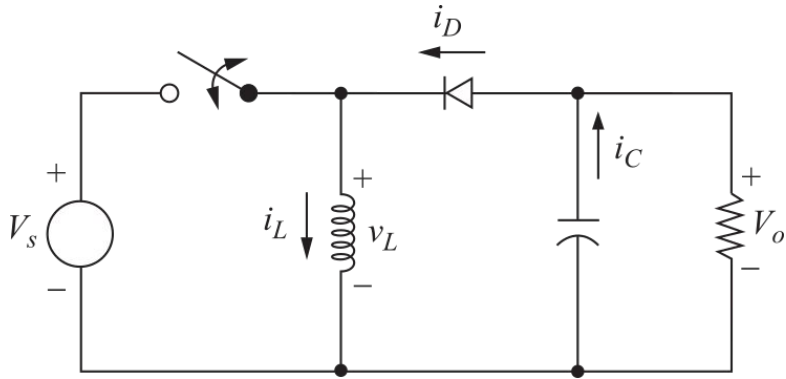
El convertidor *flyback* tiene su origen en el convertidor sin transformador llamado “Reductor-Elevador”, esencialmente cambiando el inductor por un transformador.



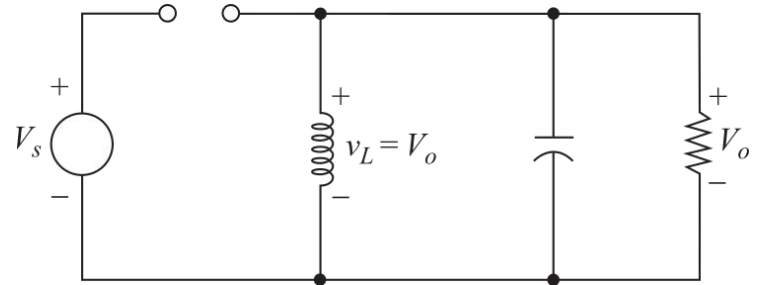
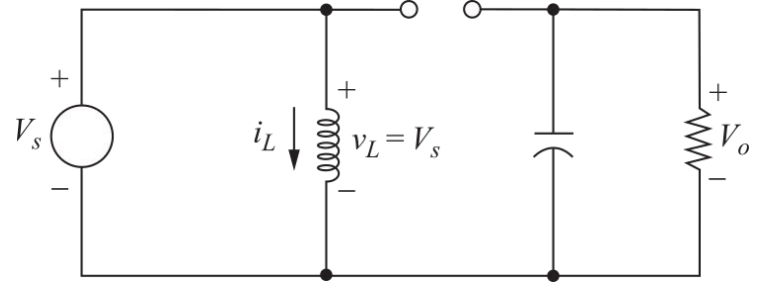
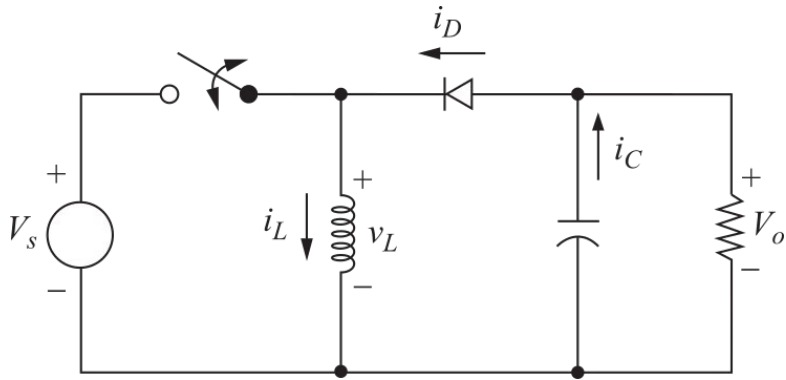
Convertidor Flyback: origen



Convertidor Flyback: origen

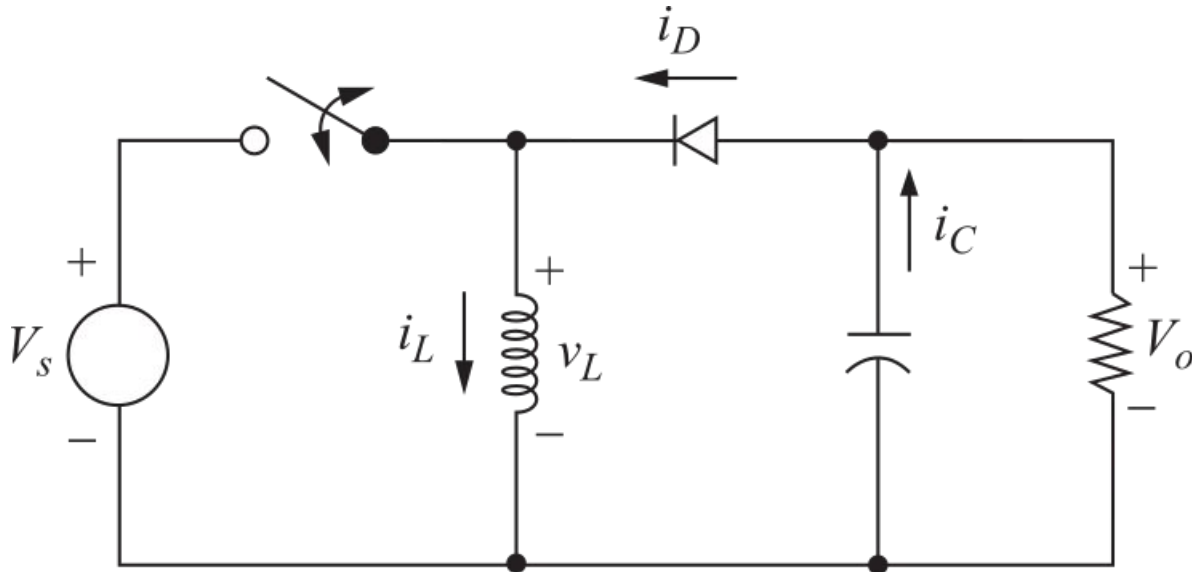


Convertidor Flyback: origen



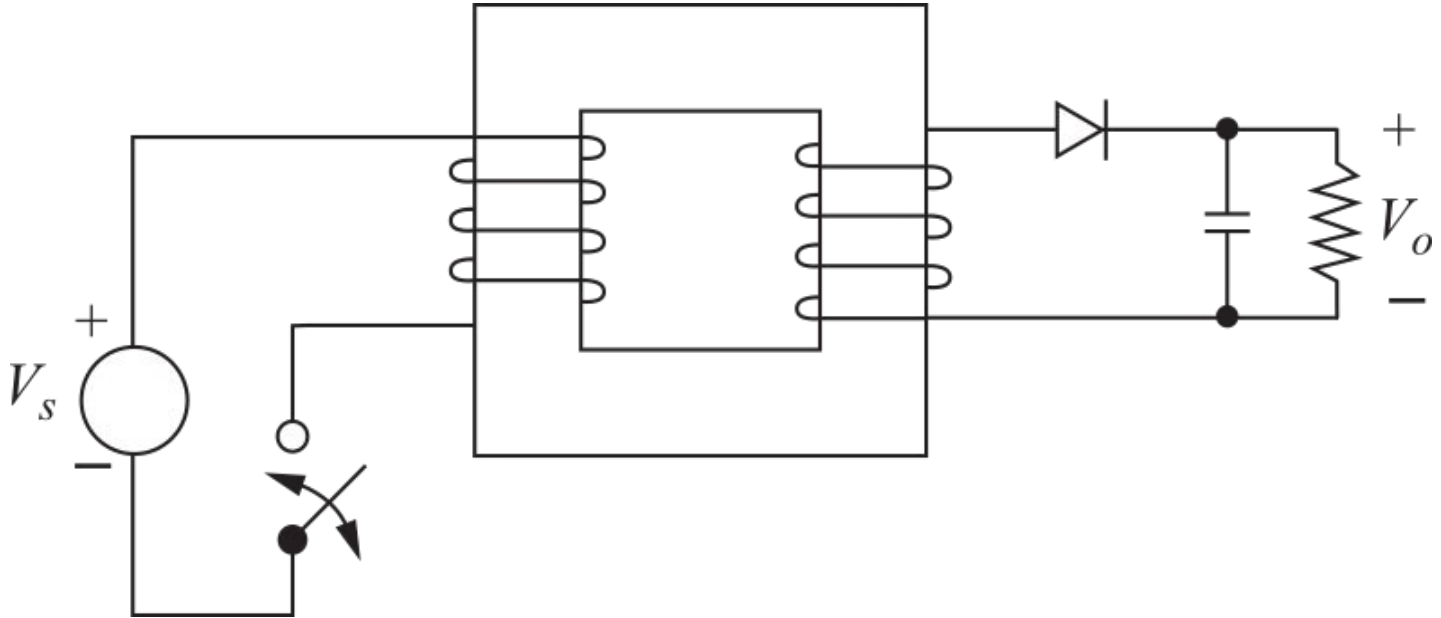
Convertidor Flyback: origen

El convertidor *flyback* tiene su origen en el convertidor sin transformador llamado “Reductor-Elevador”, esencialmente cambiando el inductor por un transformador.



Convertidor Flyback: esquema circuital

Un esquema básico resulta entonces:



Modelo del transformador

Un transformador ideal es un cuadripolo dónde se cumple que:

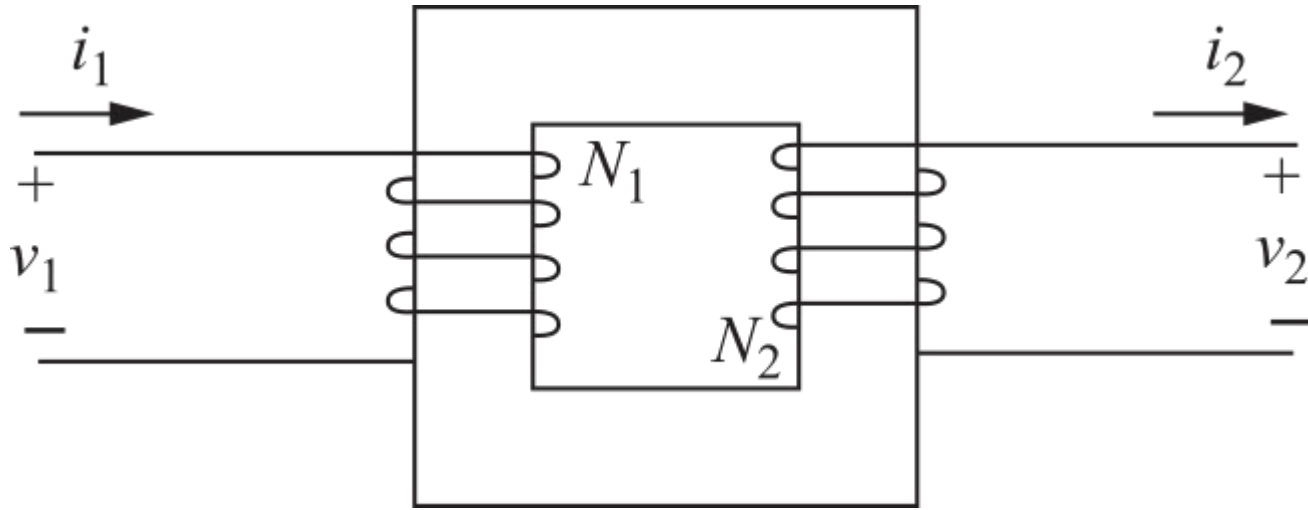
$$v_1 \cdot i_1 = v_2 \cdot i_2$$

$$v_2/v_1 = n$$



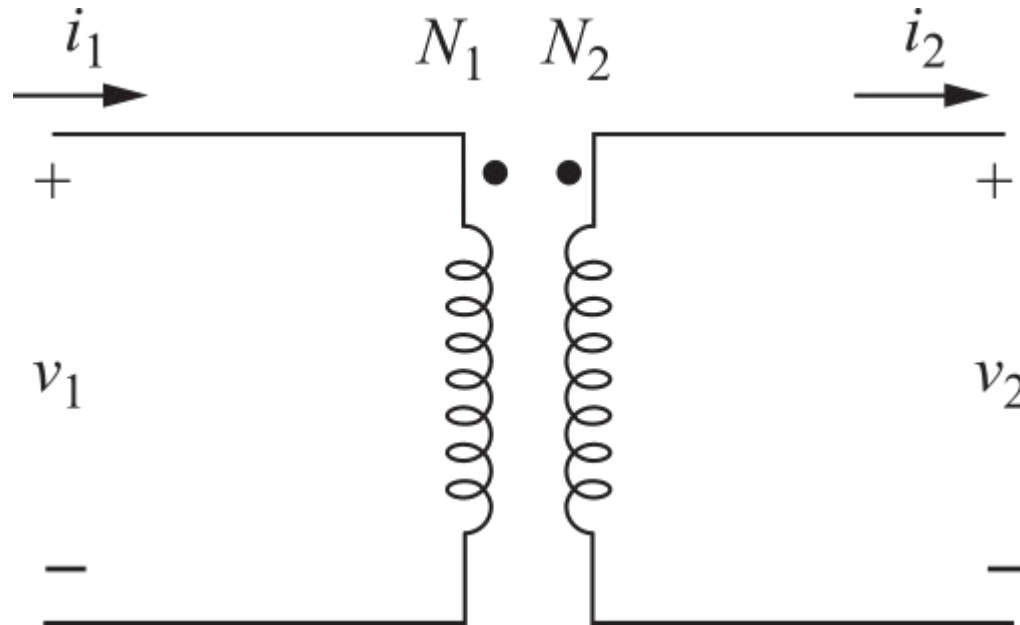
Modelo del transformador

La implementación de este cuadripolo se hace mediante arrollamientos de cobre sobre algún núcleo magnético:



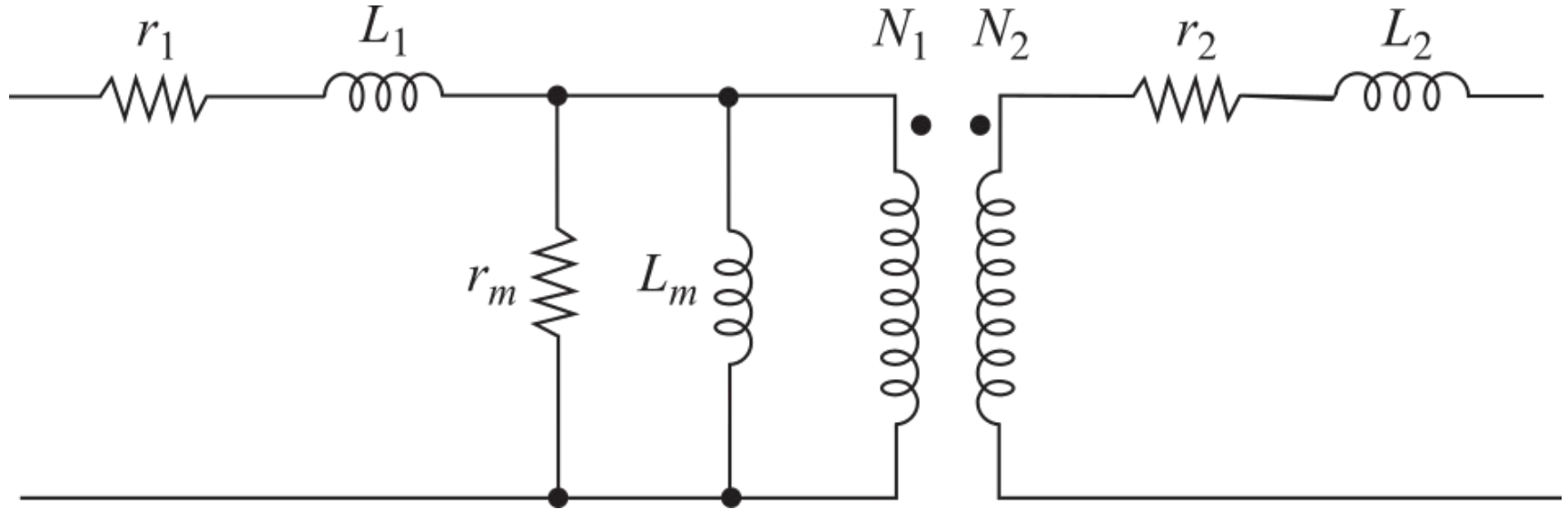
Modelo del transformador

Esquemáticamente:



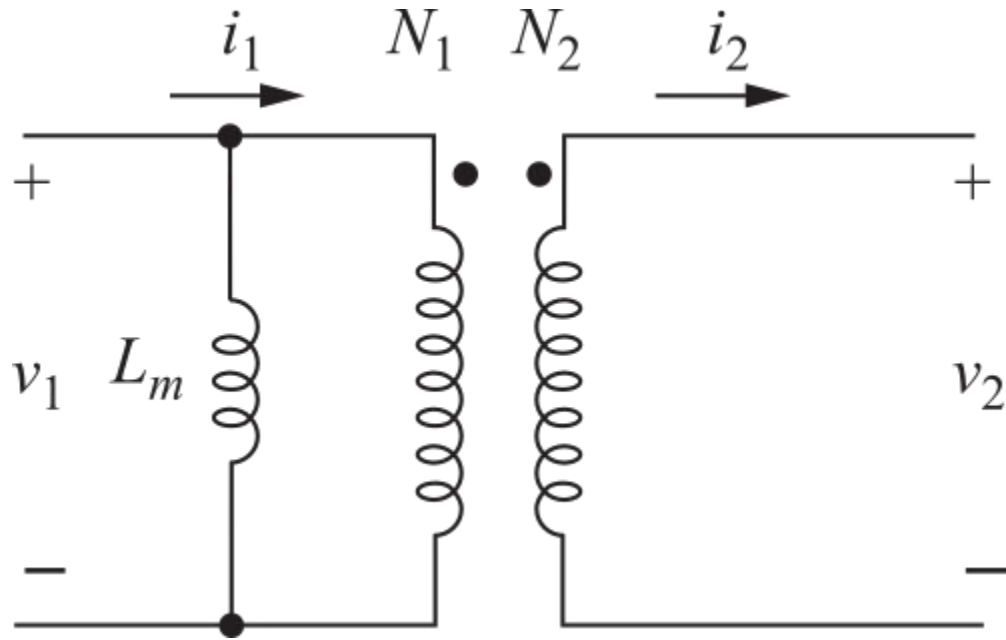
Modelo del transformador

Los bobinados tienen resistencia, existen inductancias de dispersión, pérdidas y acumulación de energía en el núcleo magnético.



Modelo del transformador

Vamos a despreciar todas las pérdidas y las inductancias de dispersión:



¿Por qué usar un transformador?

¿Por qué usar un transformador?

Hay dos razones principales por las que usar un transformador:

¿Por qué usar un transformador?

Hay dos razones principales por las que usar un transformador:

- Adaptar niveles de tensión (bajar o subir el nivel de tensión);

¿Por qué usar un transformador?

Hay dos razones principales por las que usar un transformador:

- Adaptar niveles de tensión (bajar o subir el nivel de tensión);
- Aislación galvánica;

¿Por qué usar un transformador?

Hay dos razones principales por las que usar un transformador:

- Adaptar niveles de tensión (bajar o subir el nivel de tensión);
- Aislación galvánica;

Recordemos: ¿por qué usar transformadores en alta y no en baja frecuencia?

¿Por qué usar un transformador?

Hay dos razones principales por las que usar un transformador:

- Adaptar niveles de tensión (bajar o subir el nivel de tensión);
- Aislación galvánica;

Recordemos: ¿por qué usar transformadores en alta y no en baja frecuencia?

Baja frecuencia = frecuencia de red
Alta frecuencia = orden del KHz

¿Por qué usar un transformador?

Hay dos razones principales por las que usar un transformador:

- Adaptar niveles de tensión (bajar o subir el nivel de tensión);
- Aislación galvánica;

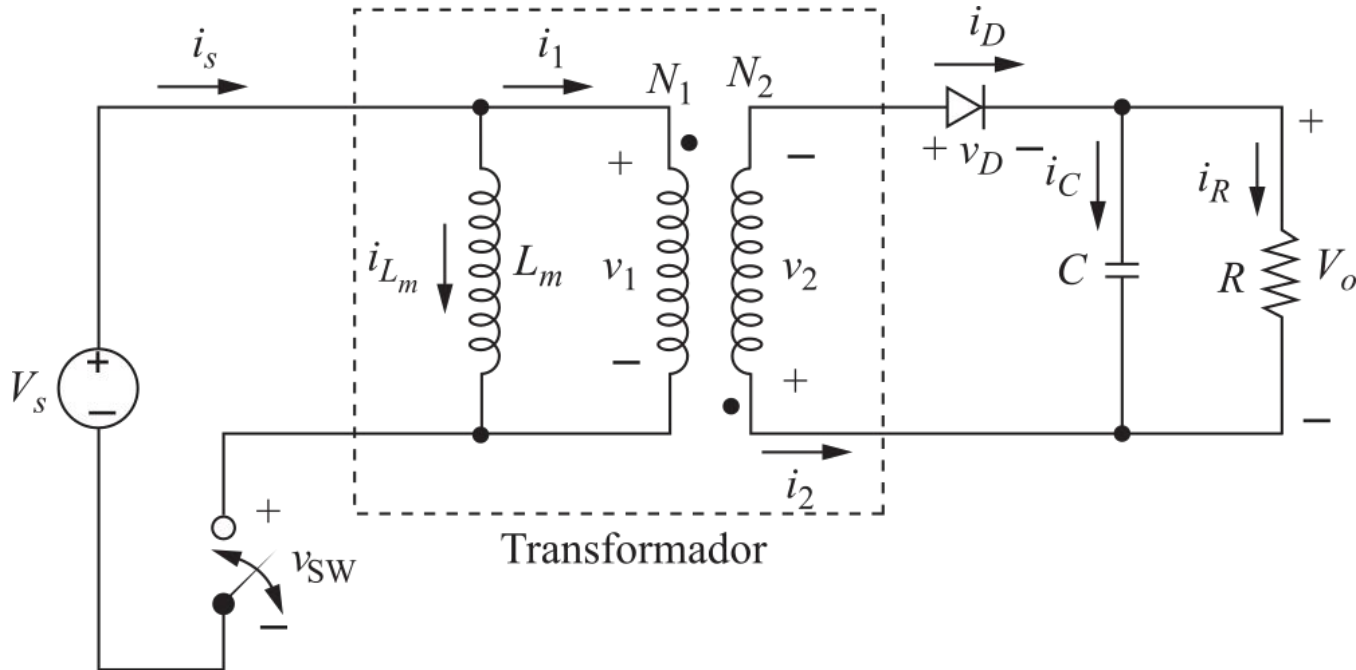
Recordemos: ¿por qué usar transformadores en alta y no en baja frecuencia?

→ Reducción de volumen, peso y costo.

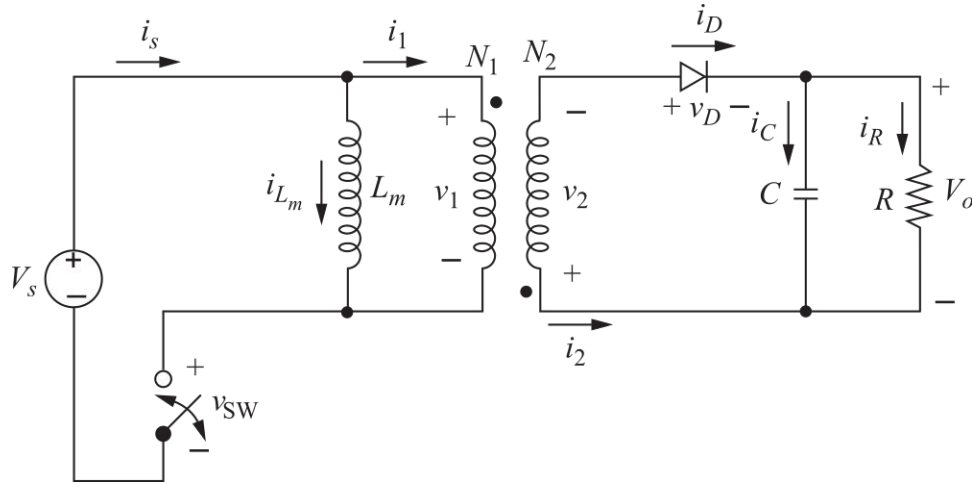
Baja frecuencia = frecuencia de red
Alta frecuencia = orden del KHz

Convertidor Flyback: esquema circuital

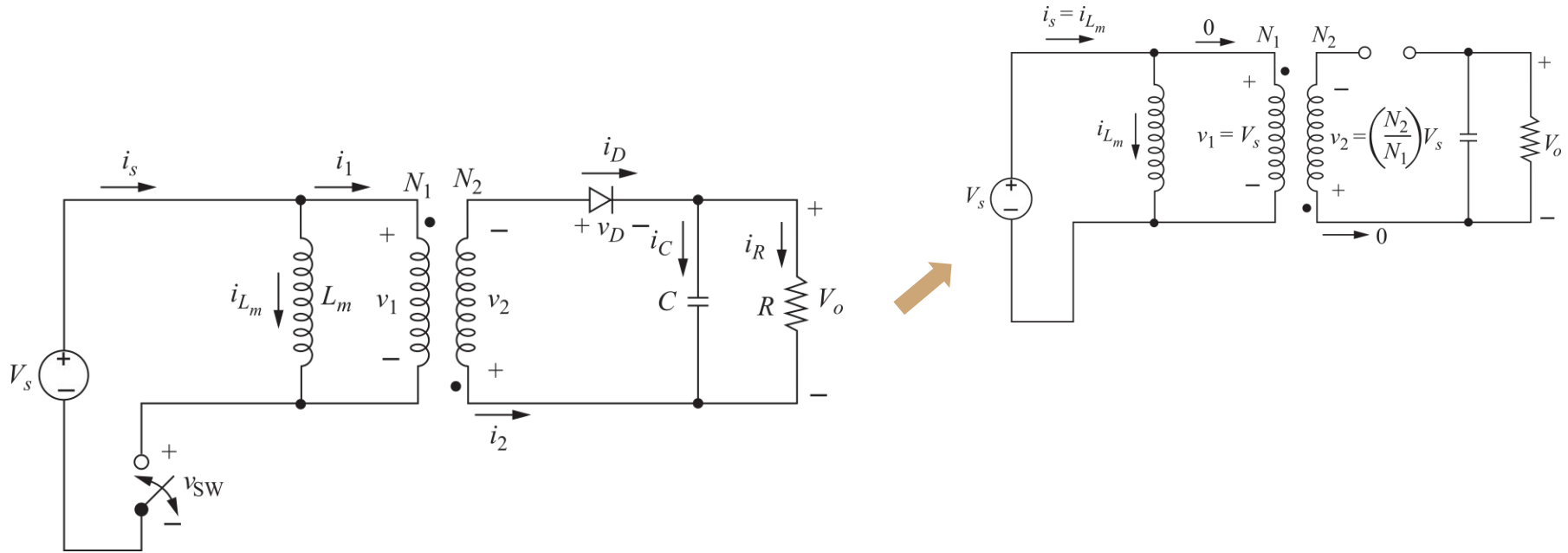
Para el estudio del circuito, utilizaremos el modelo simplificado del transformador.



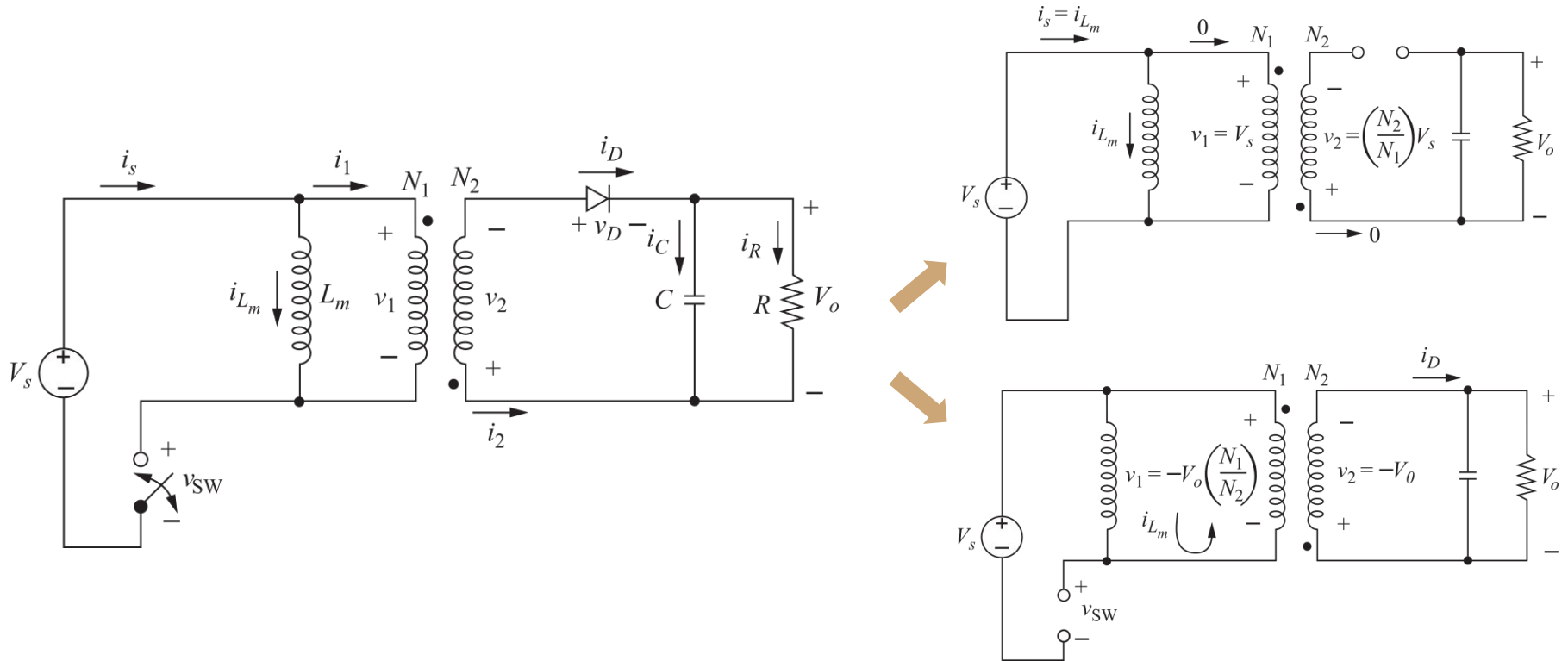
Convertidor Flyback: funcionamiento



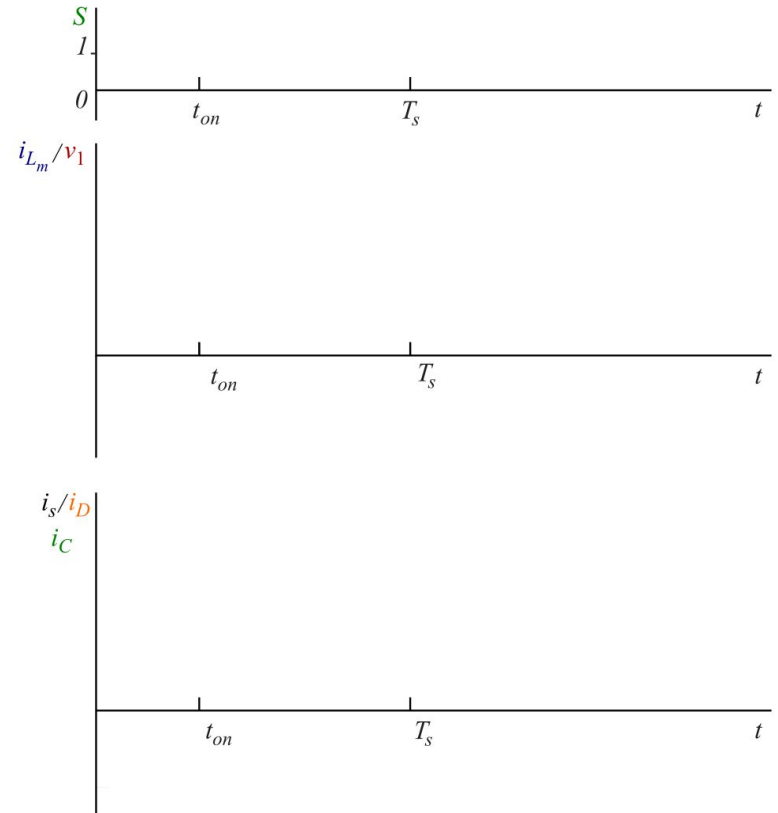
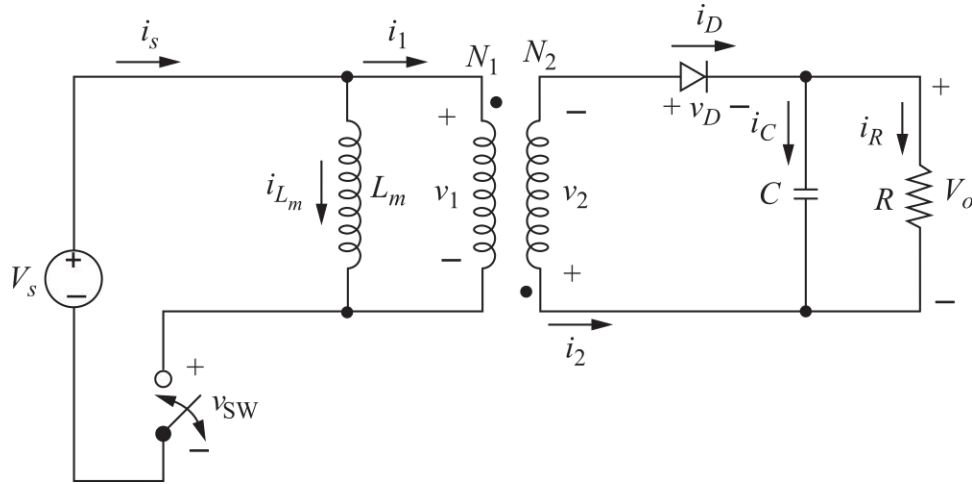
Convertidor Flyback: funcionamiento



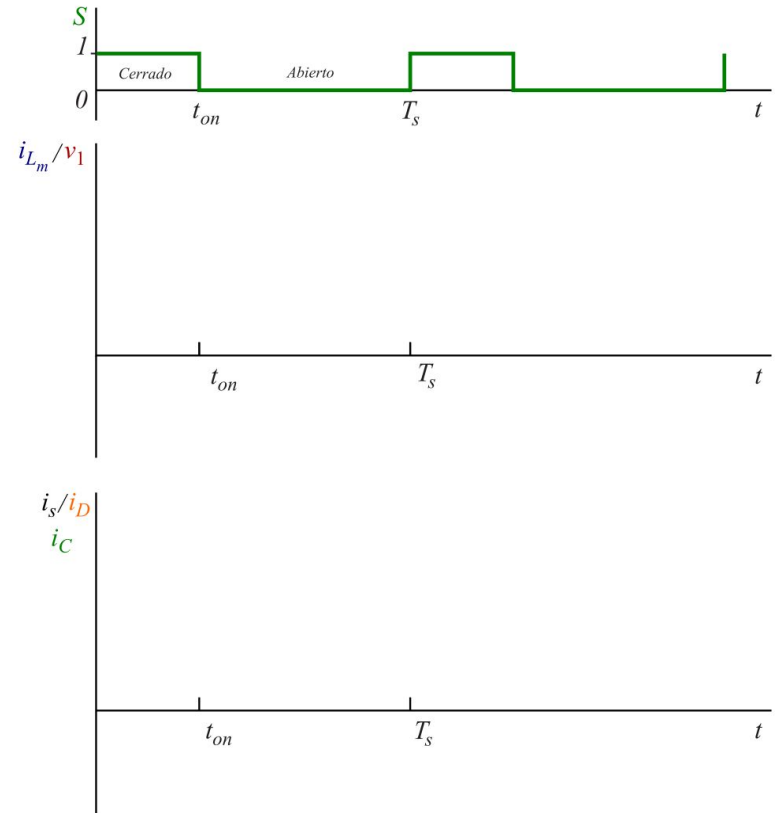
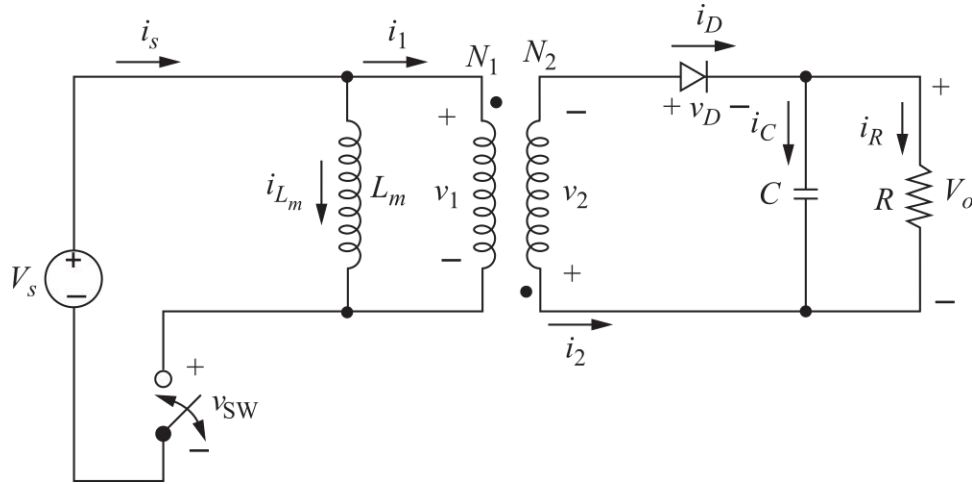
Convertidor Flyback: funcionamiento



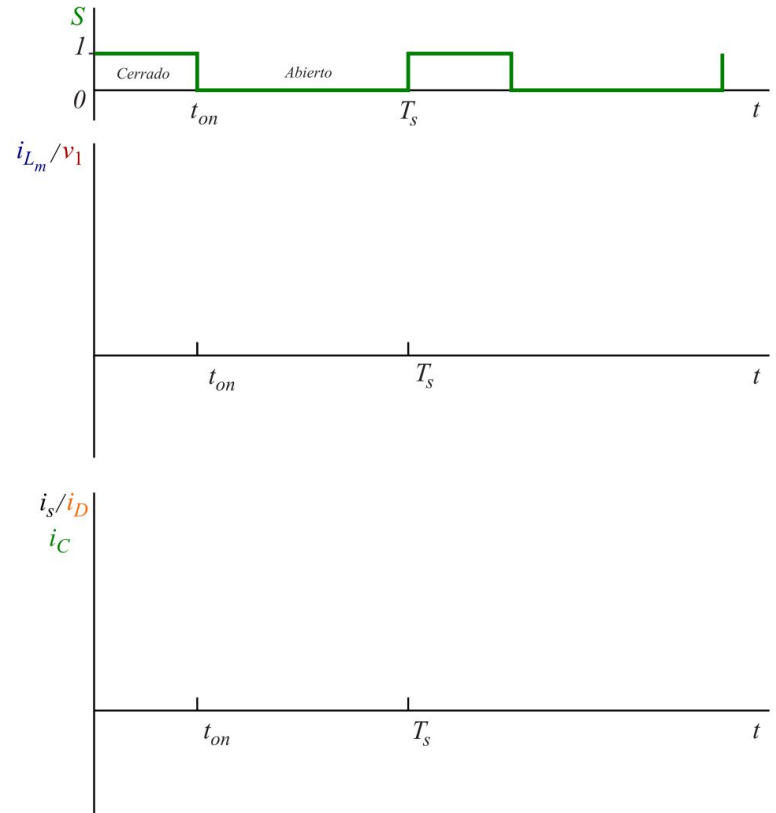
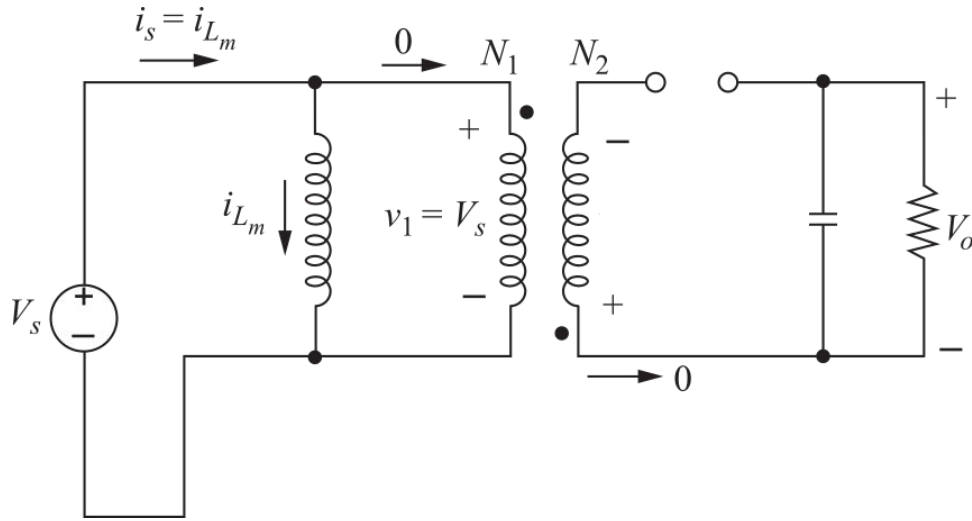
Convertidor Flyback: formas de onda



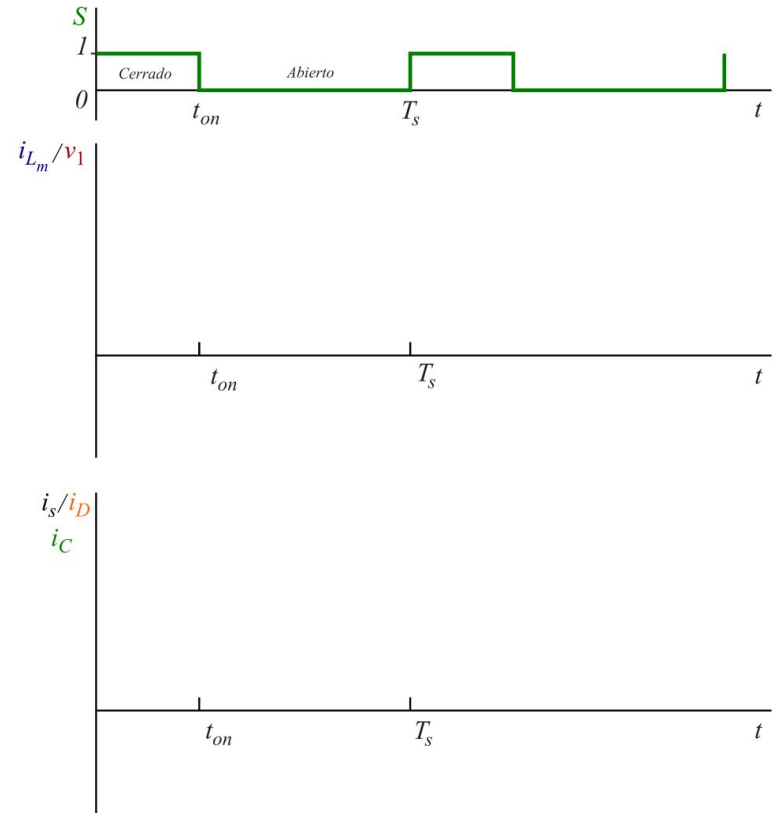
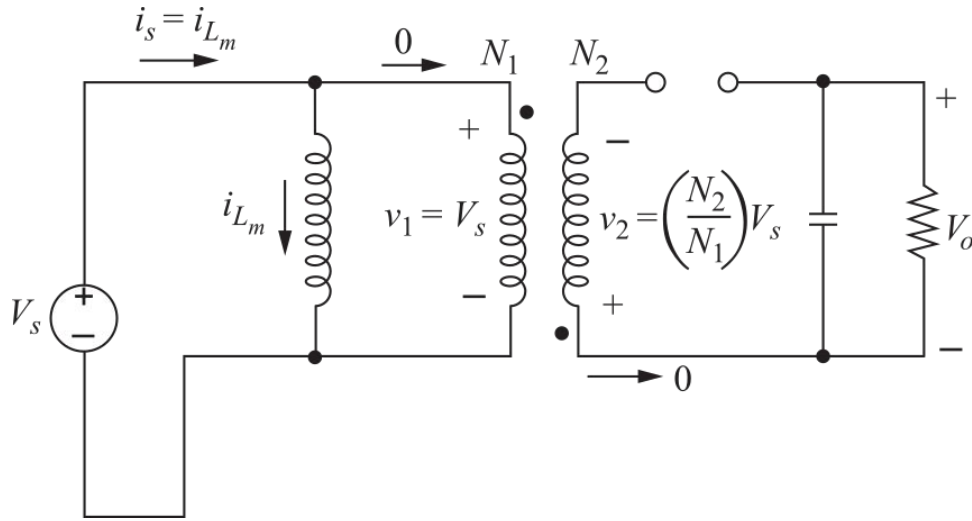
Convertidor Flyback: formas de onda



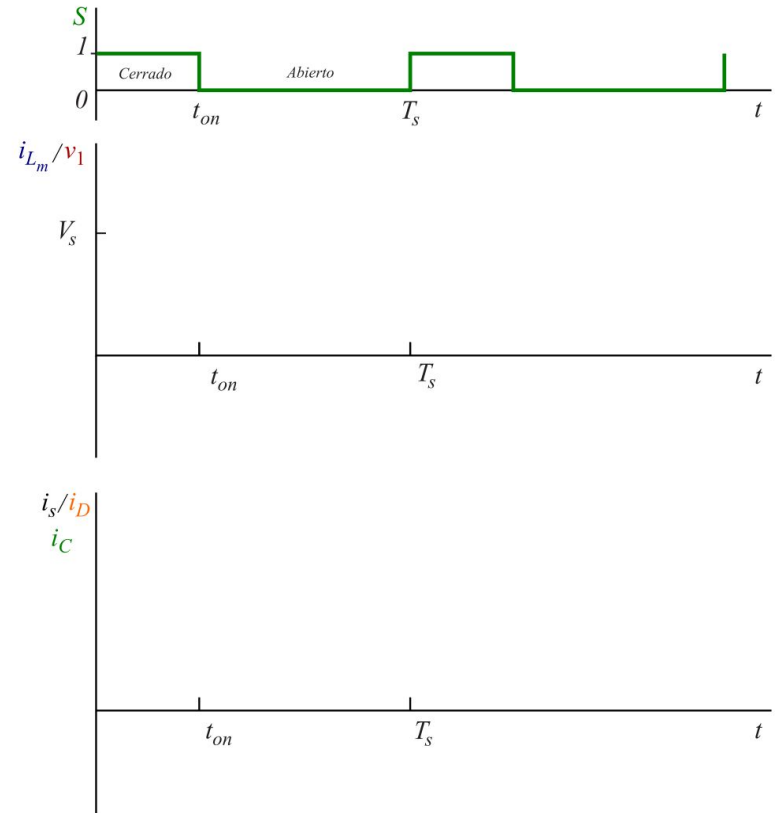
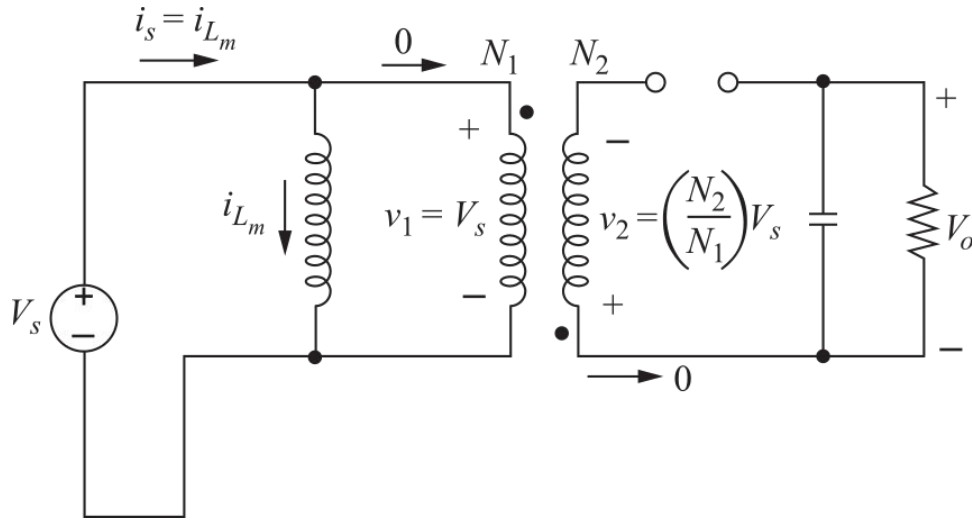
Convertidor Flyback: formas de onda



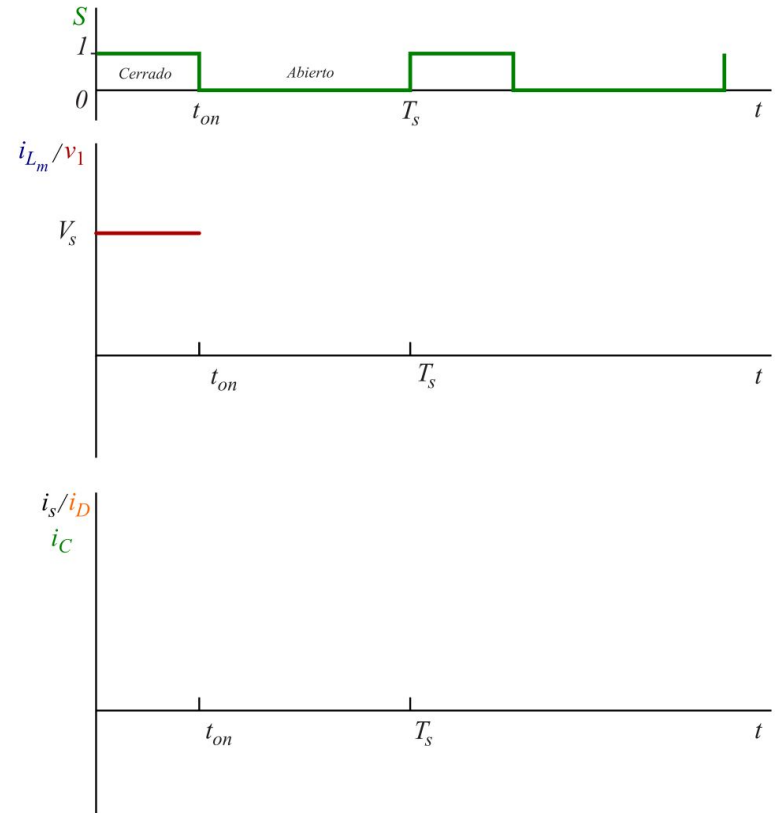
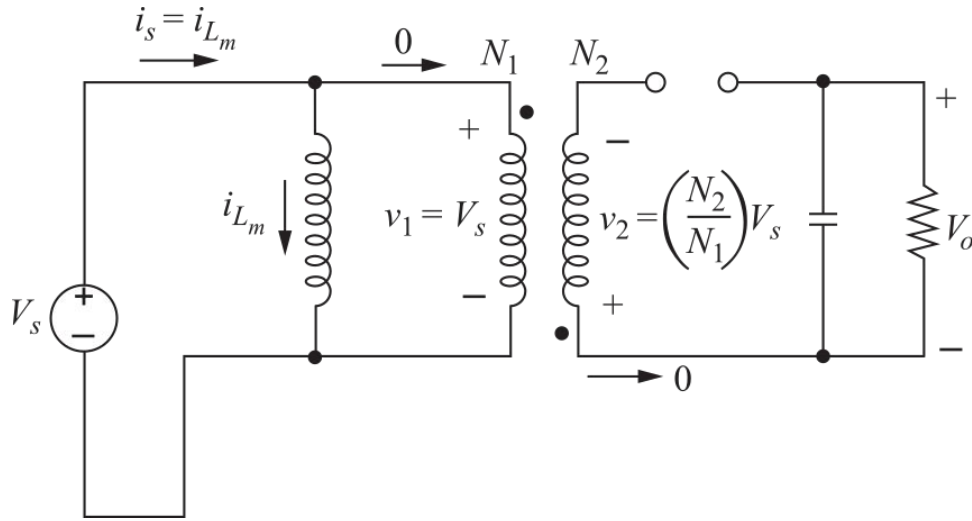
Convertidor Flyback: formas de onda



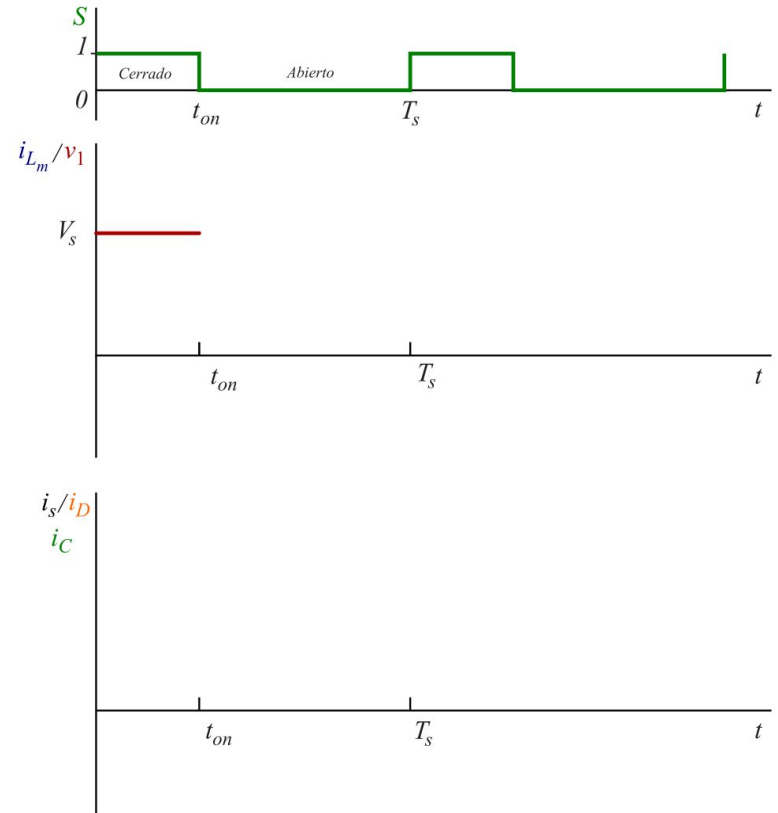
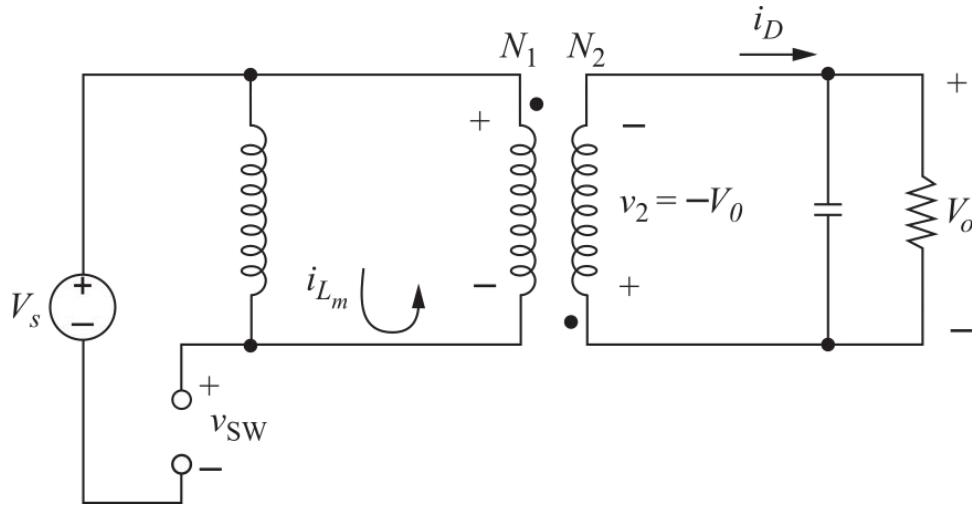
Convertidor Flyback: formas de onda



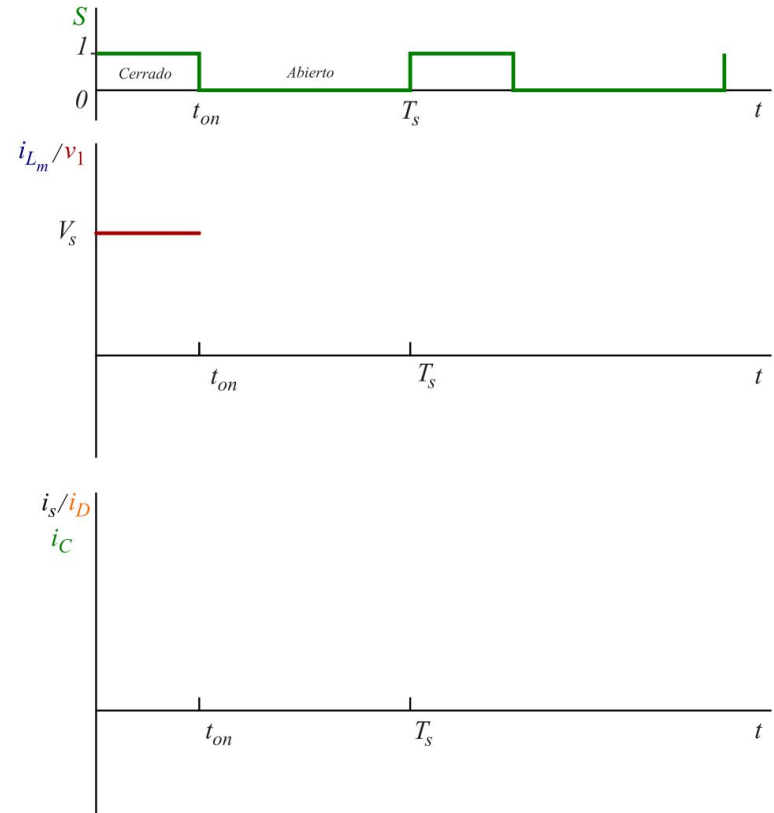
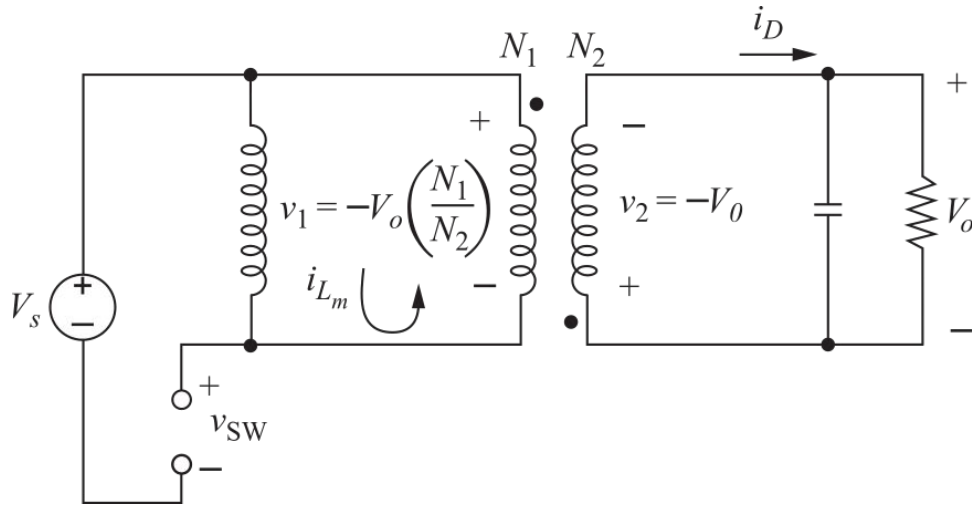
Convertidor Flyback: formas de onda



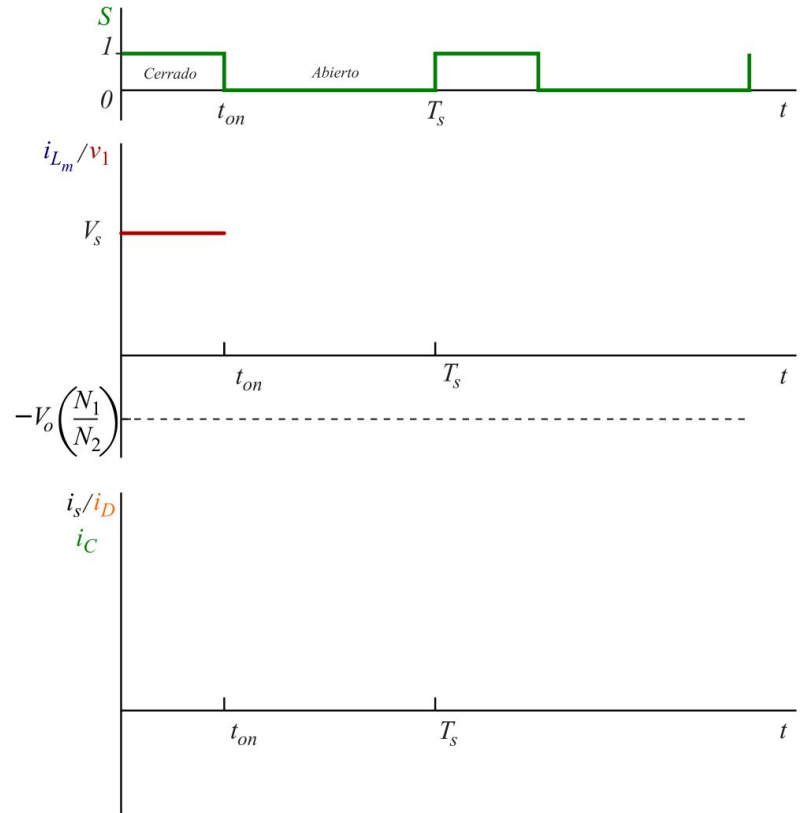
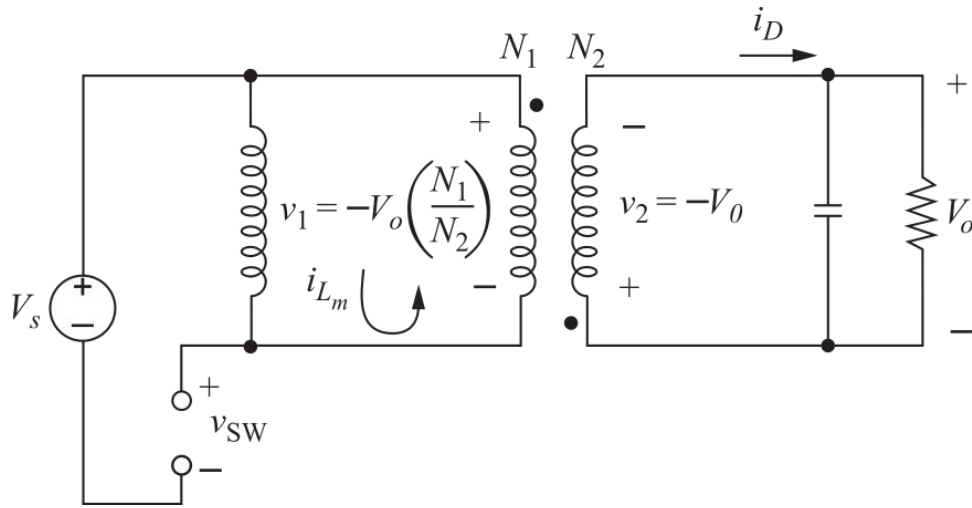
Convertidor Flyback: formas de onda



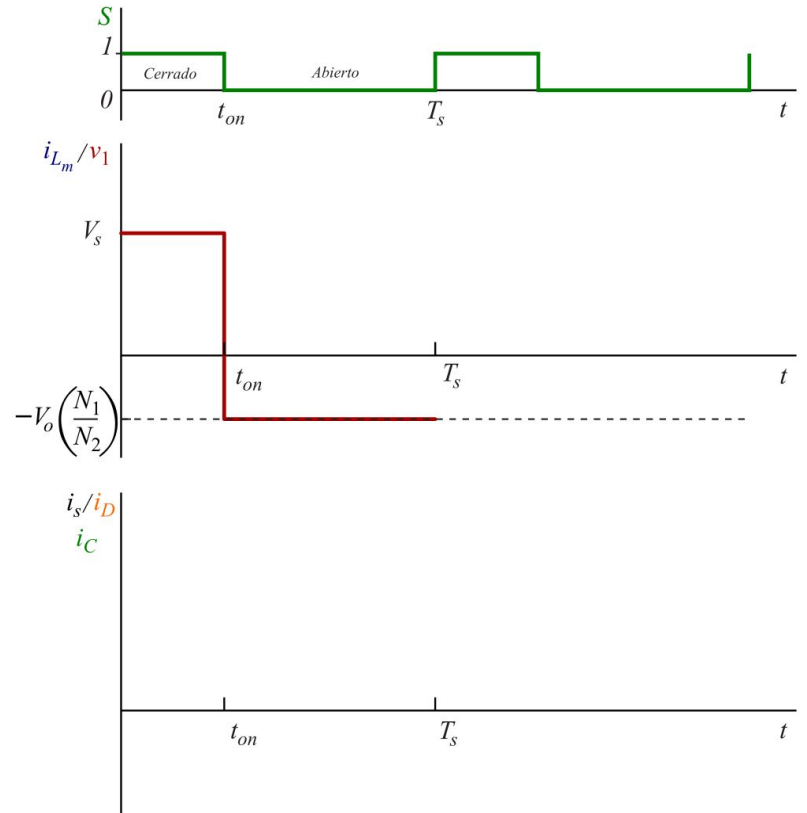
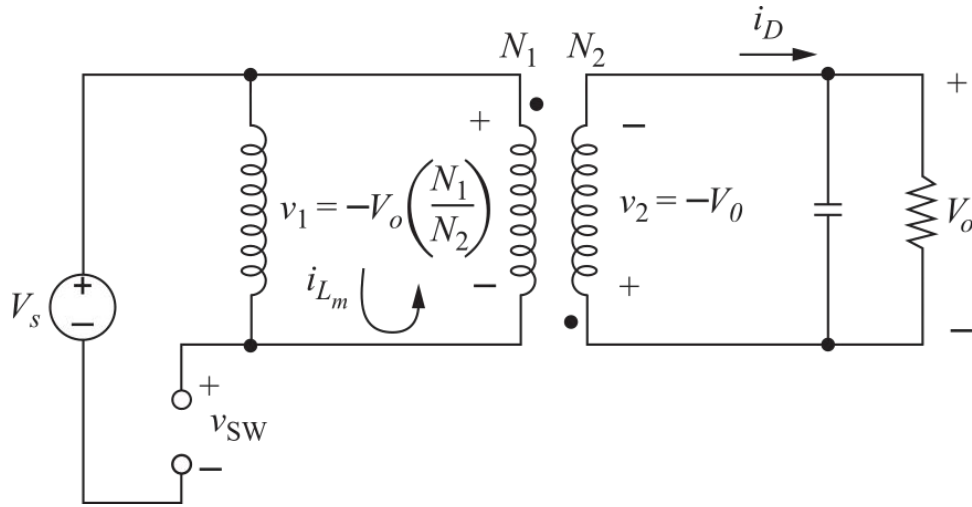
Convertidor Flyback: formas de onda



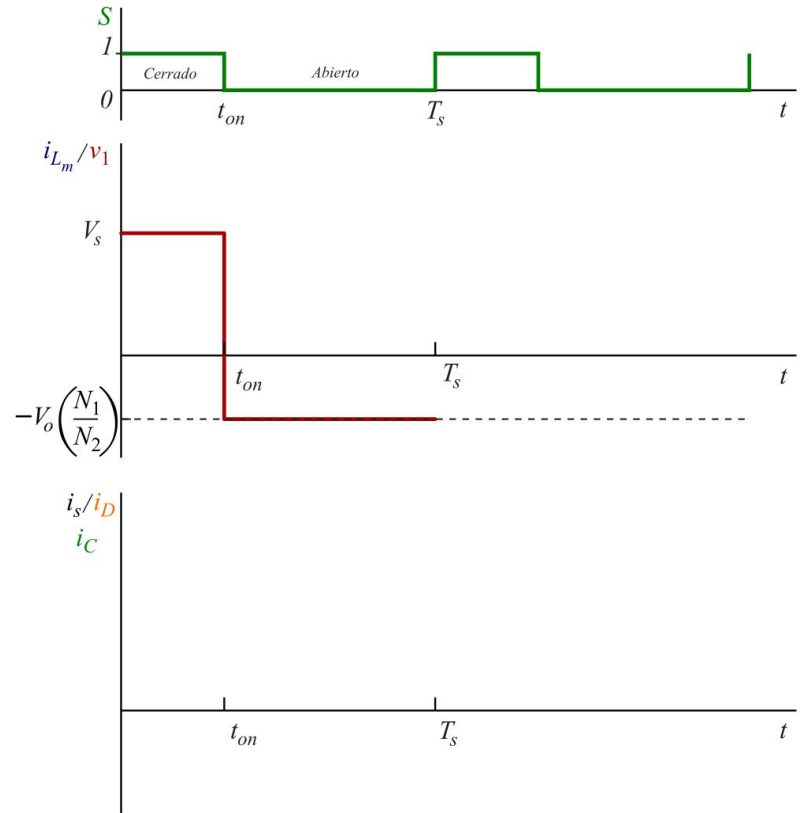
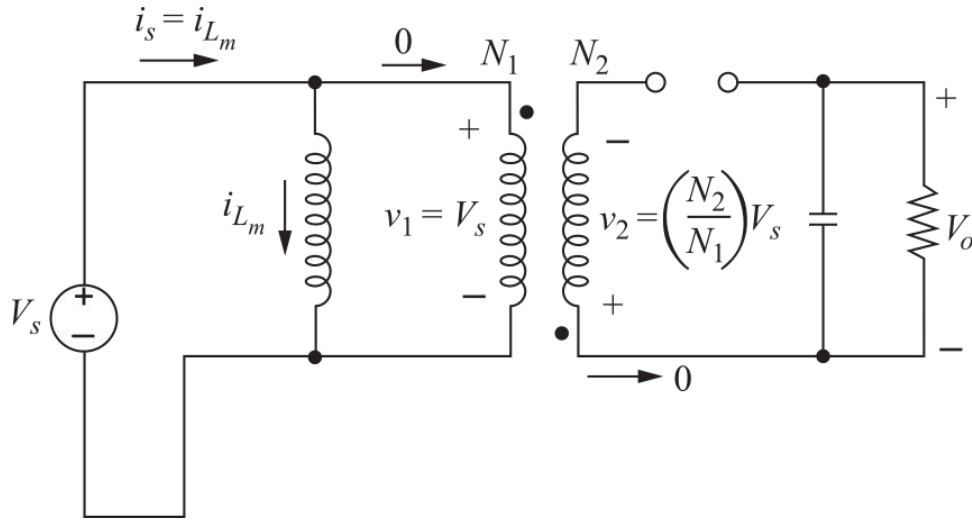
Convertidor Flyback: formas de onda



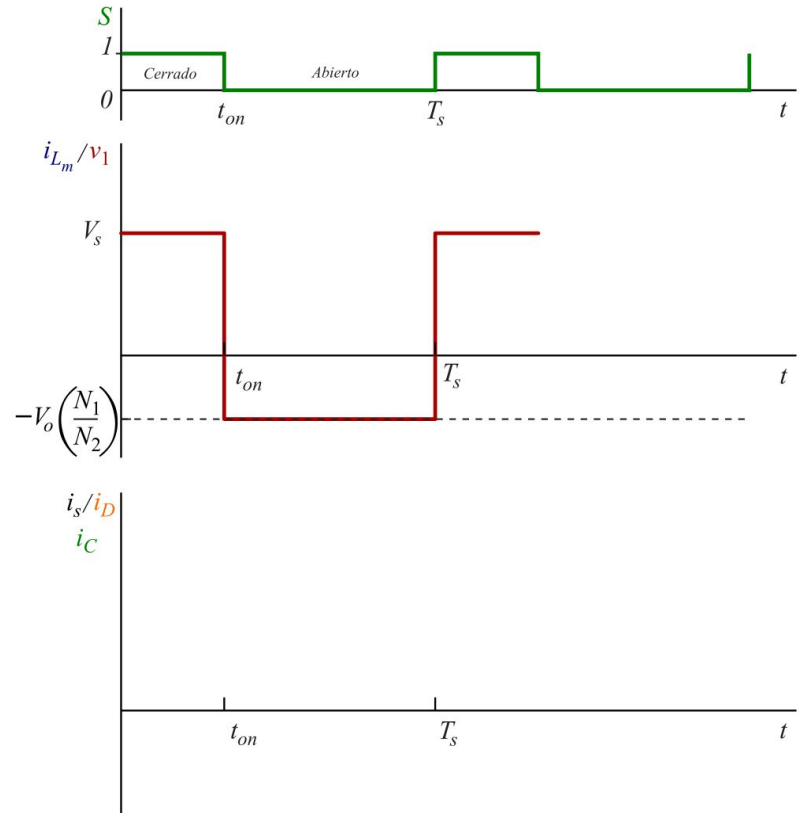
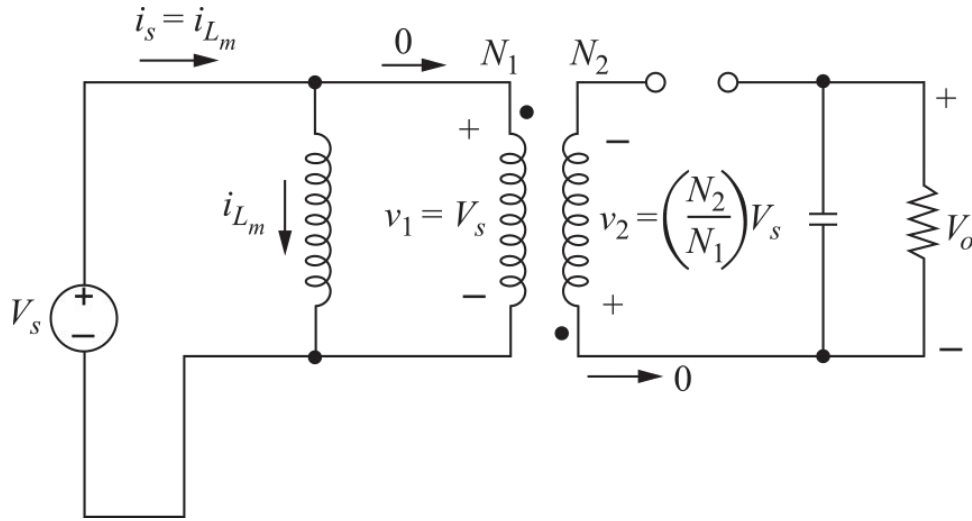
Convertidor Flyback: formas de onda



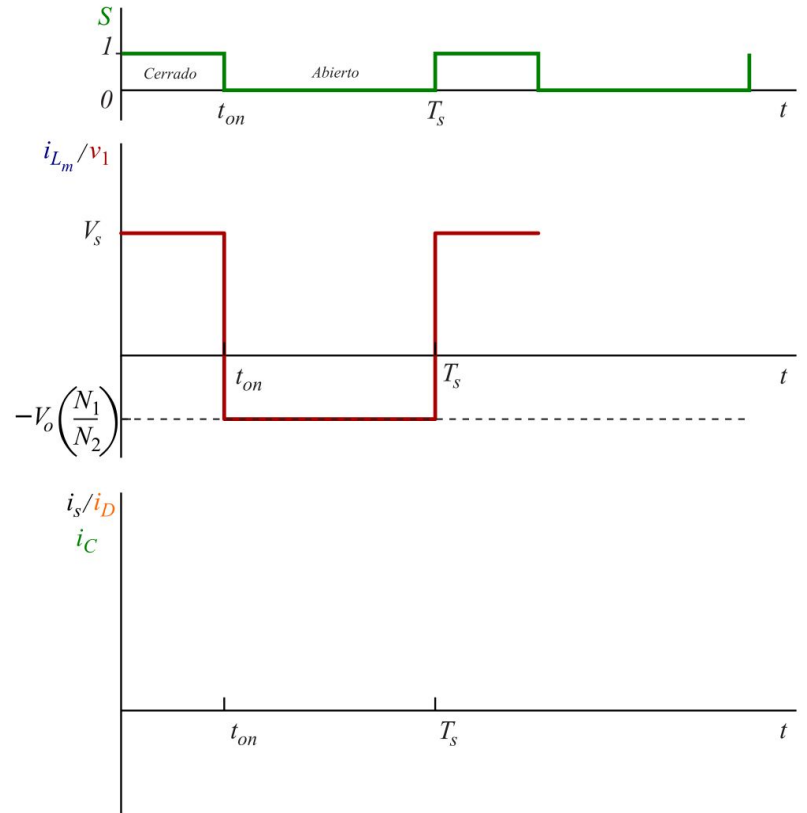
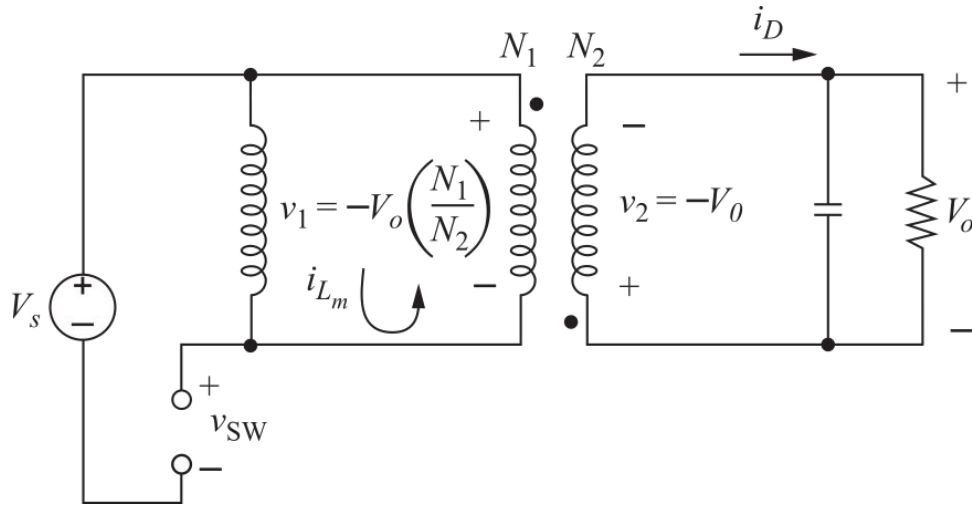
Convertidor Flyback: formas de onda



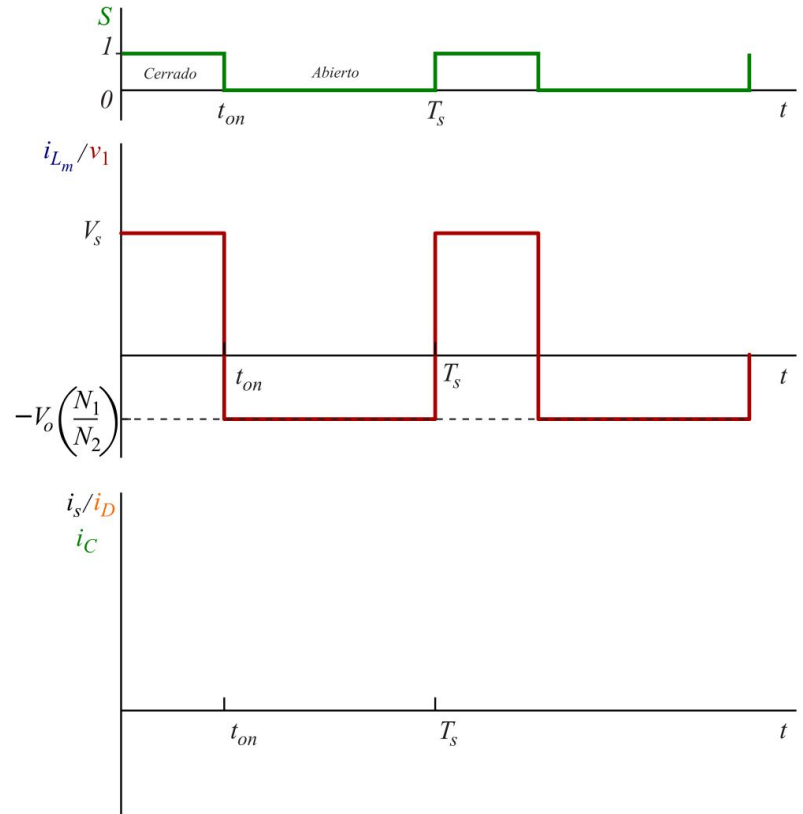
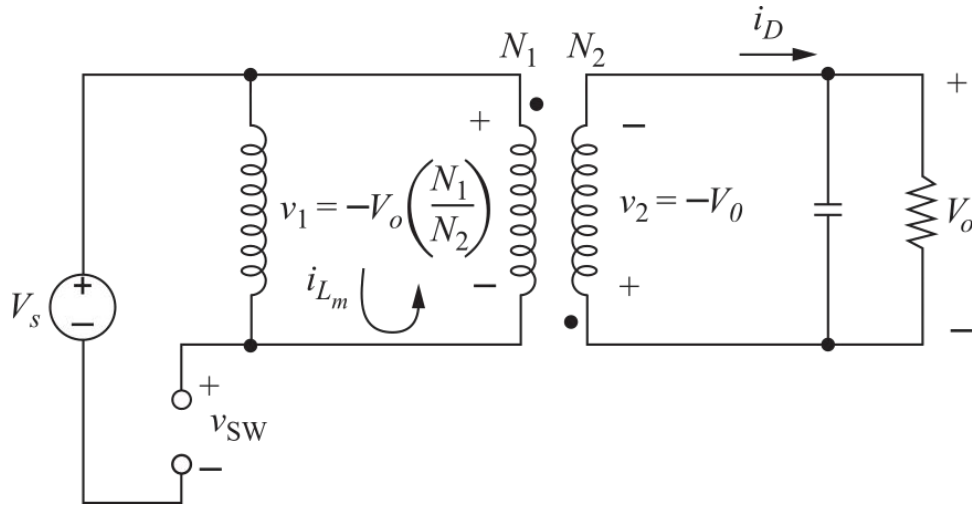
Convertidor Flyback: formas de onda



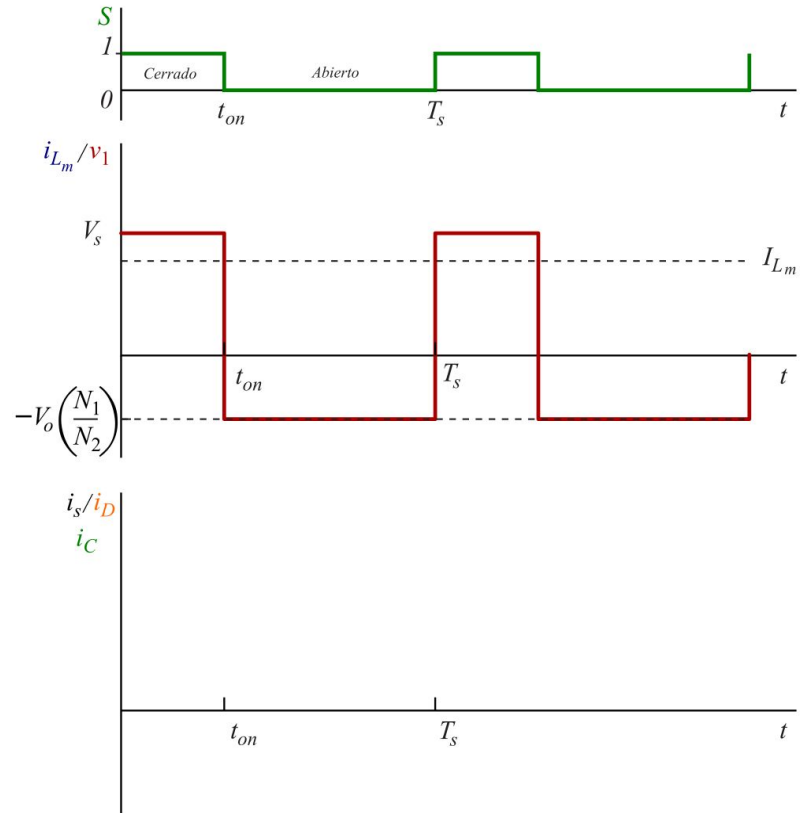
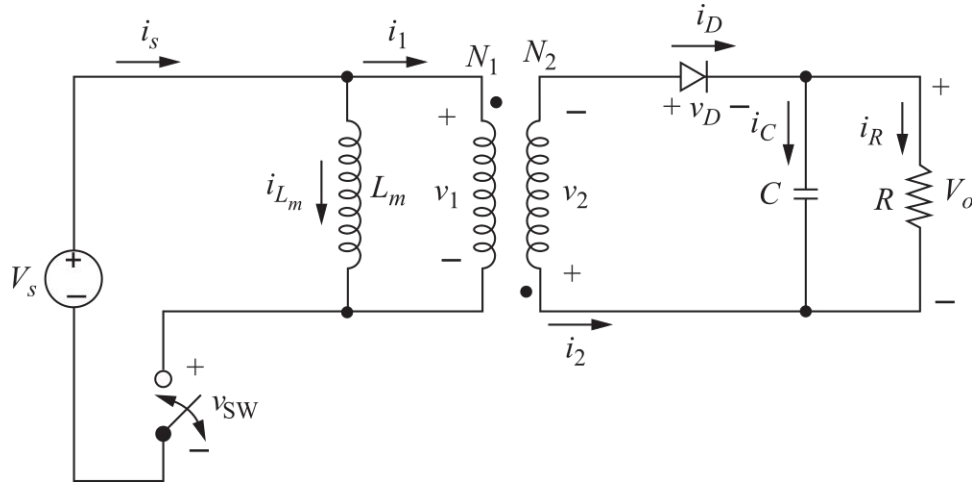
Convertidor Flyback: formas de onda



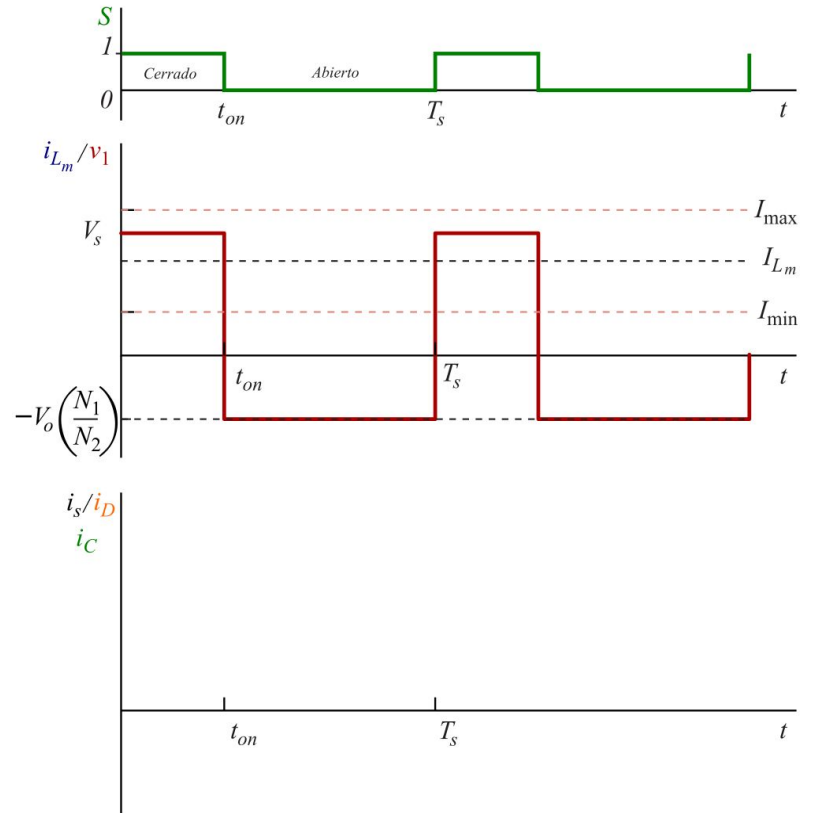
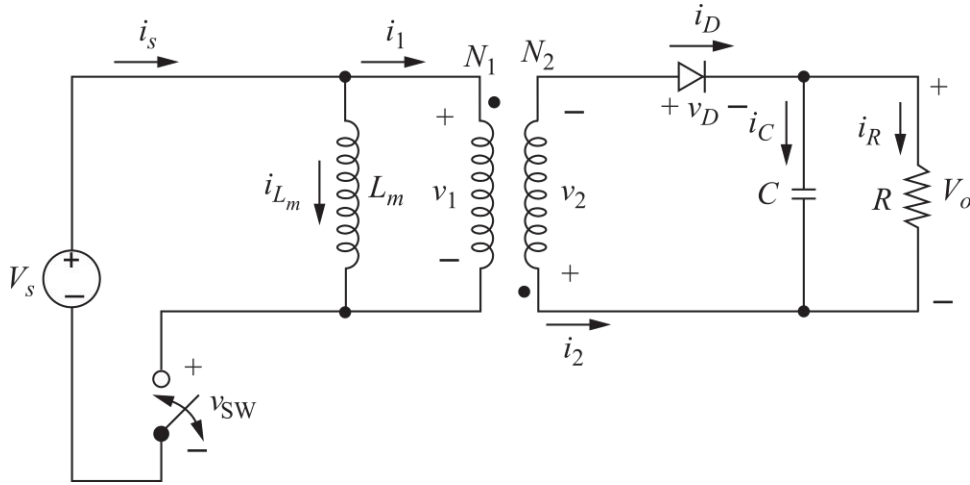
Convertidor Flyback: formas de onda



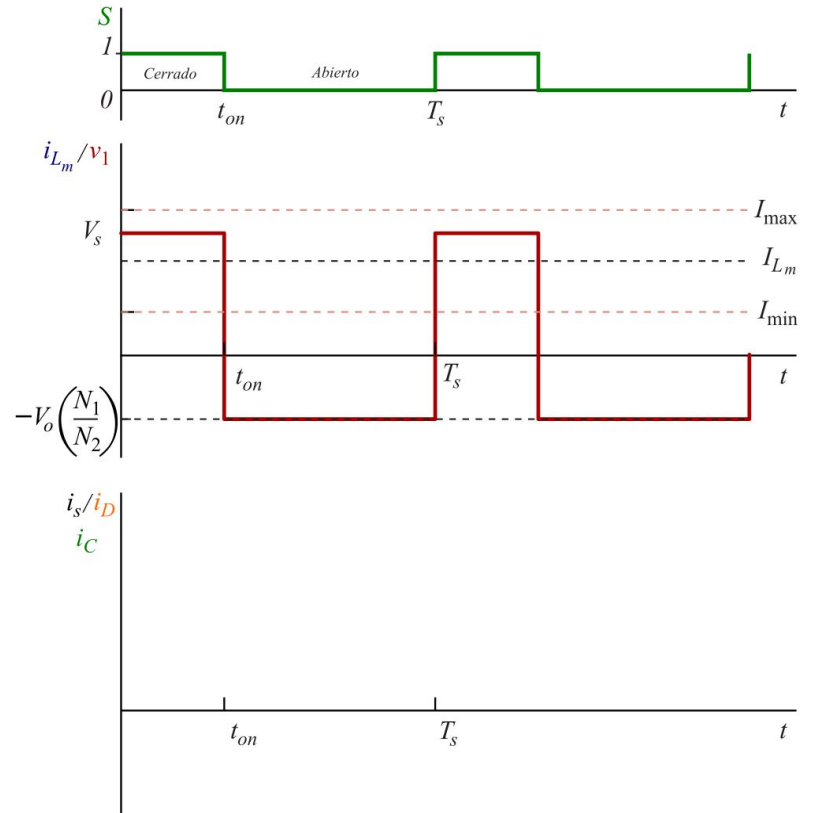
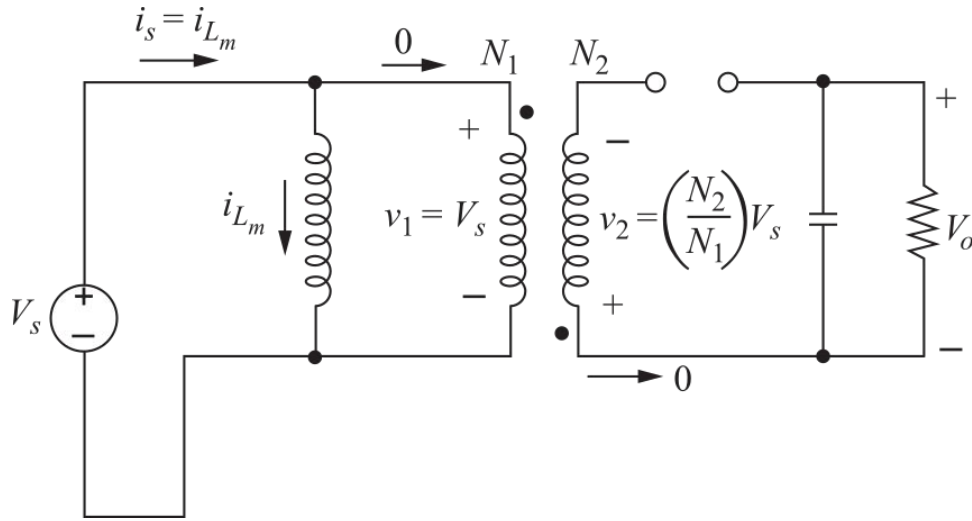
Convertidor Flyback: formas de onda



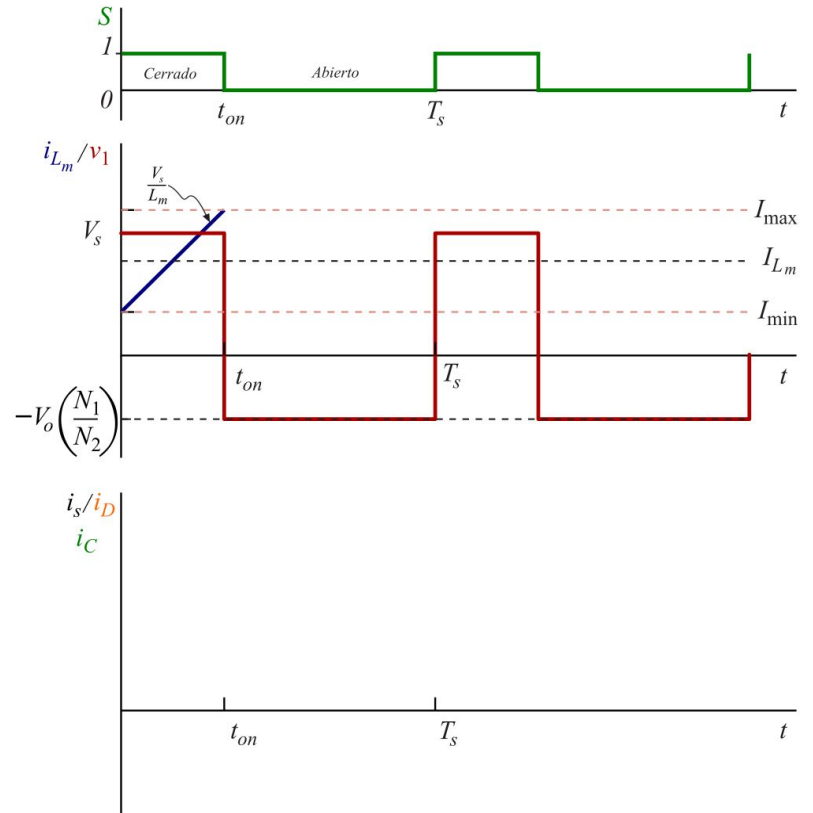
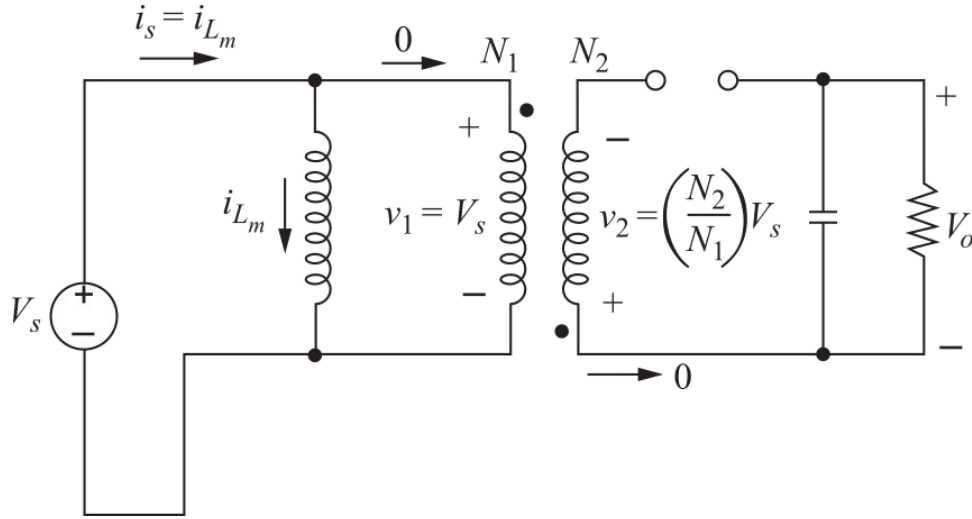
Convertidor Flyback: formas de onda



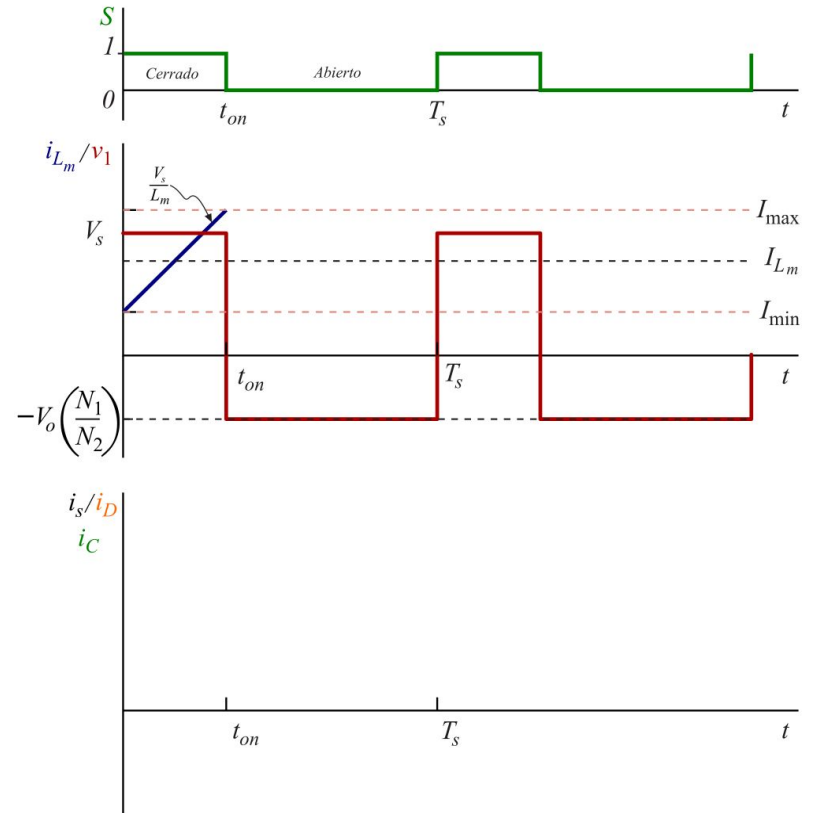
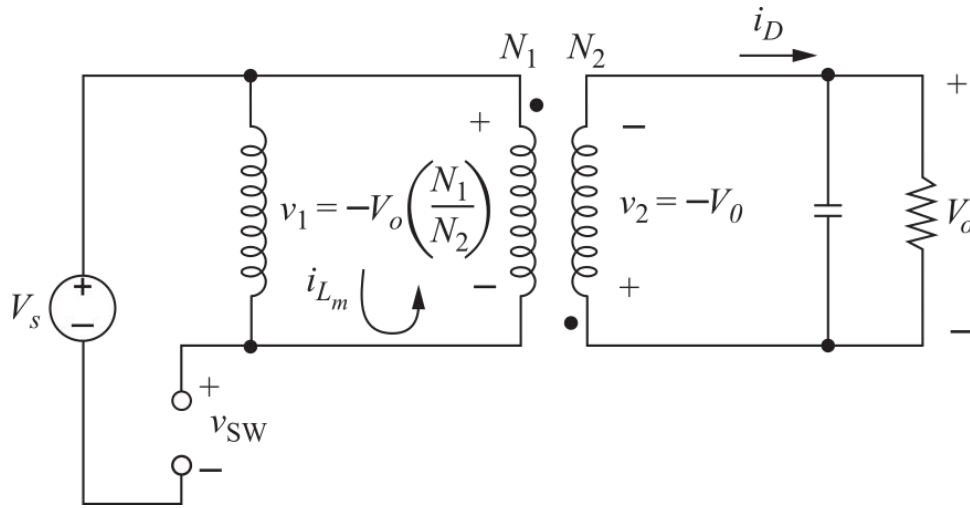
Convertidor Flyback: formas de onda



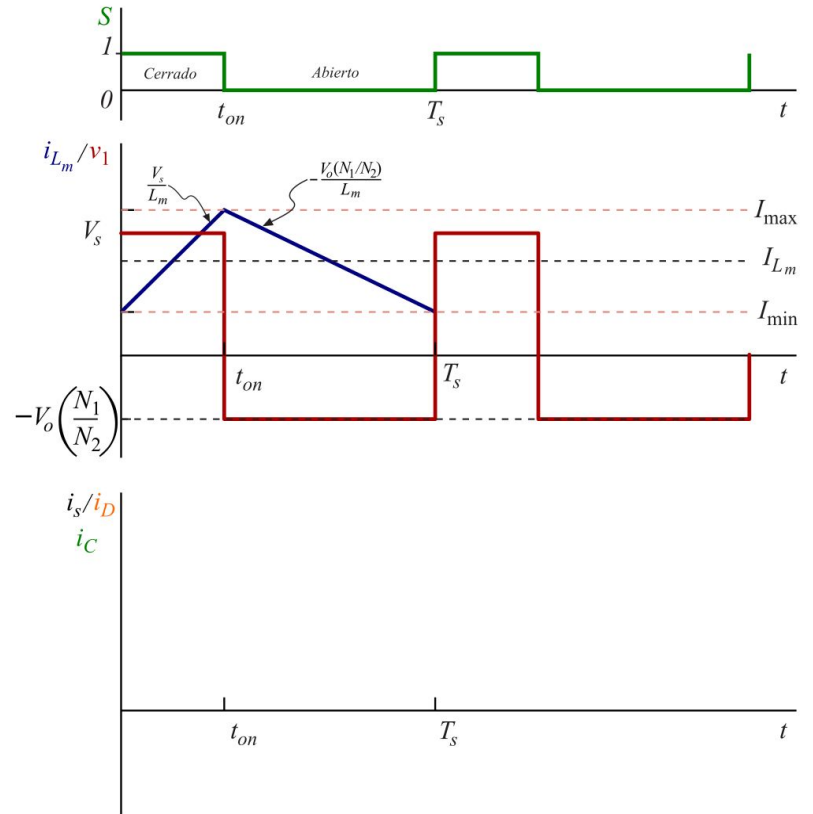
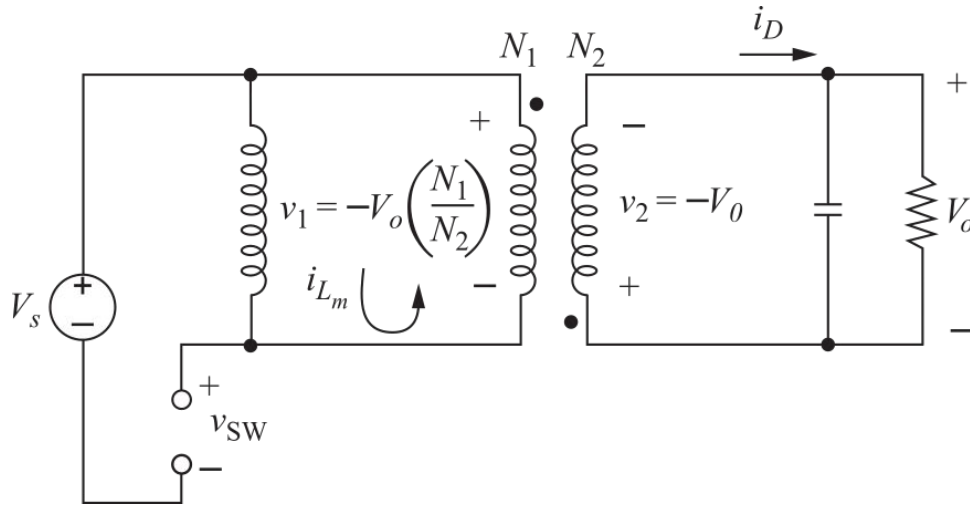
Convertidor Flyback: formas de onda



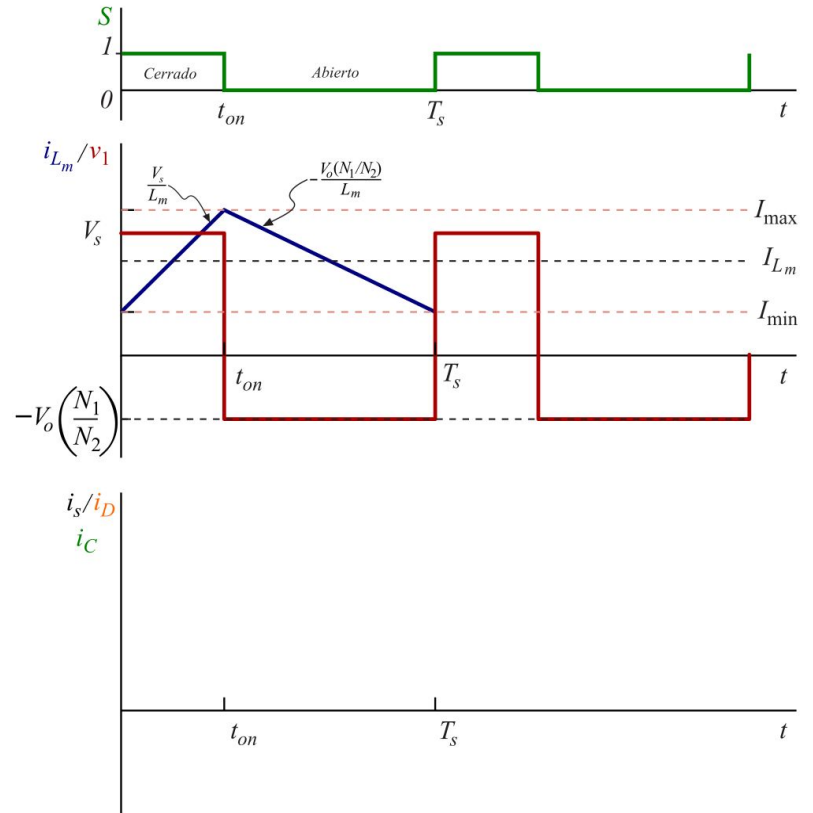
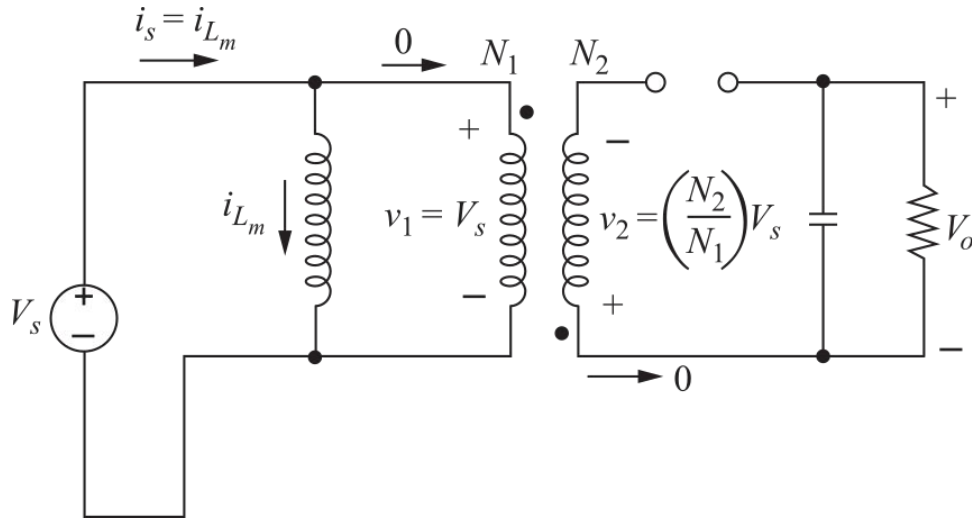
Convertidor Flyback: formas de onda



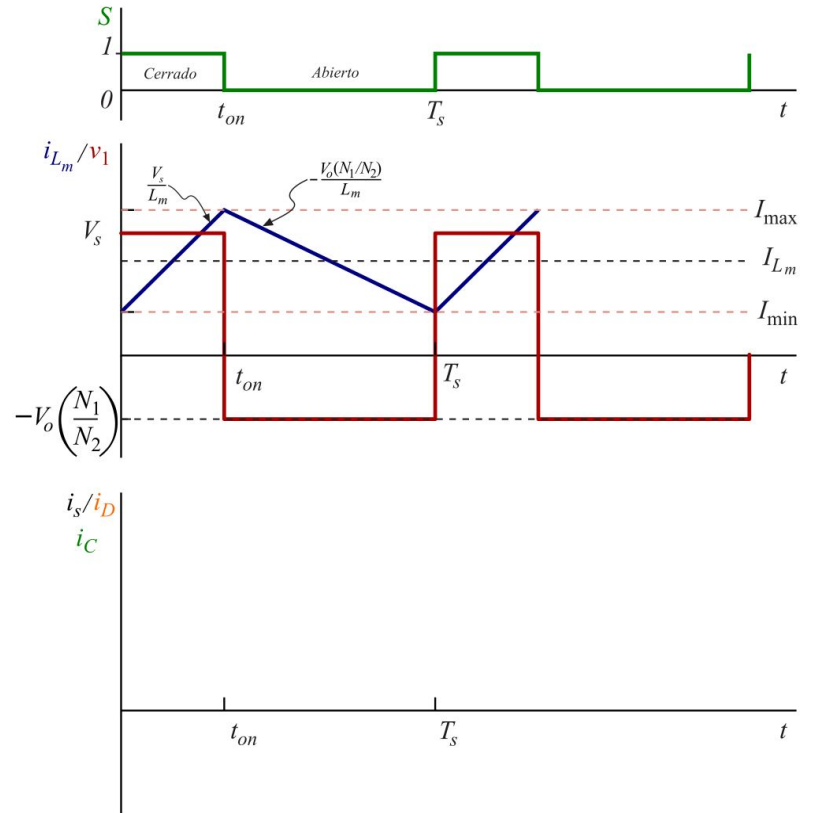
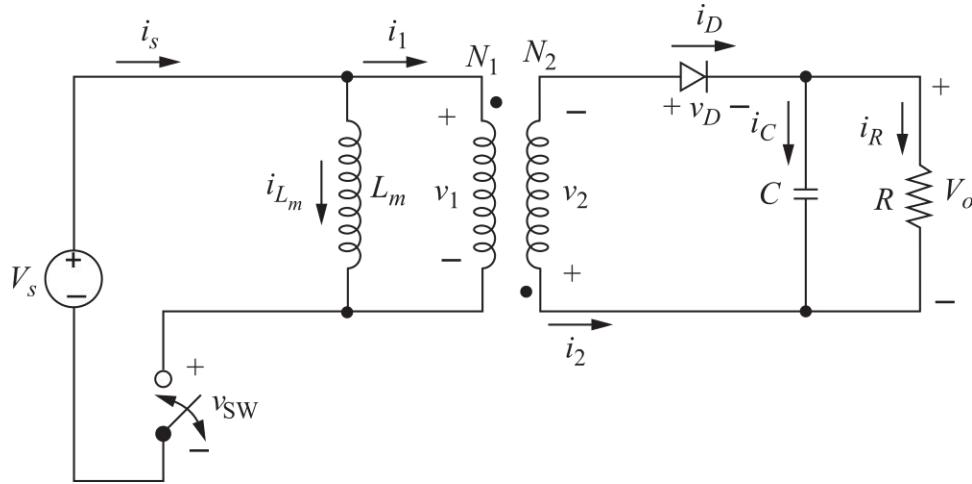
Convertidor Flyback: formas de onda



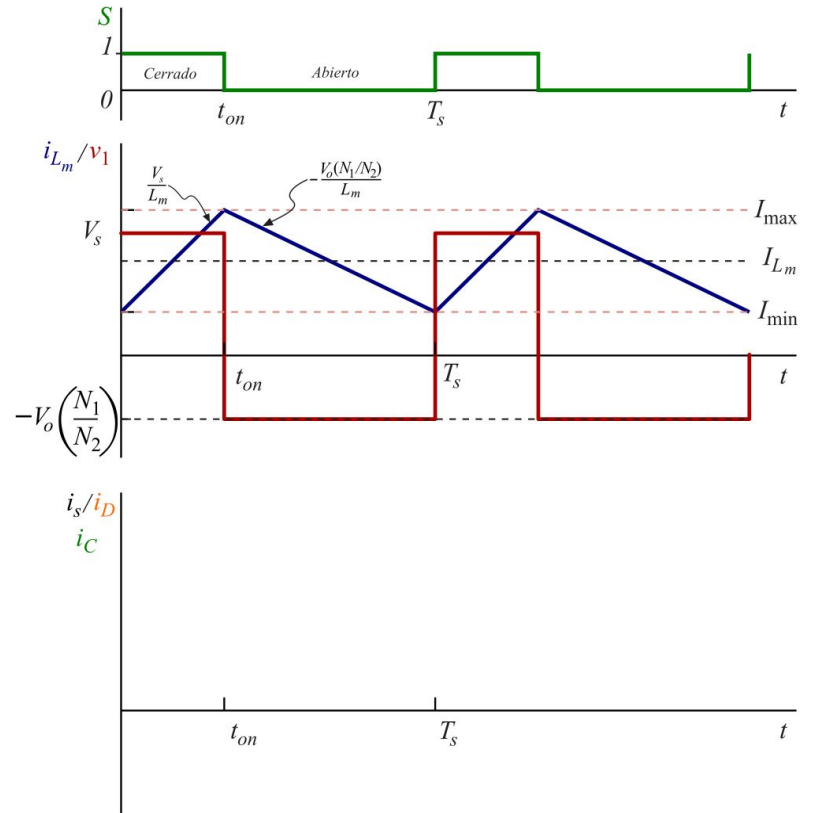
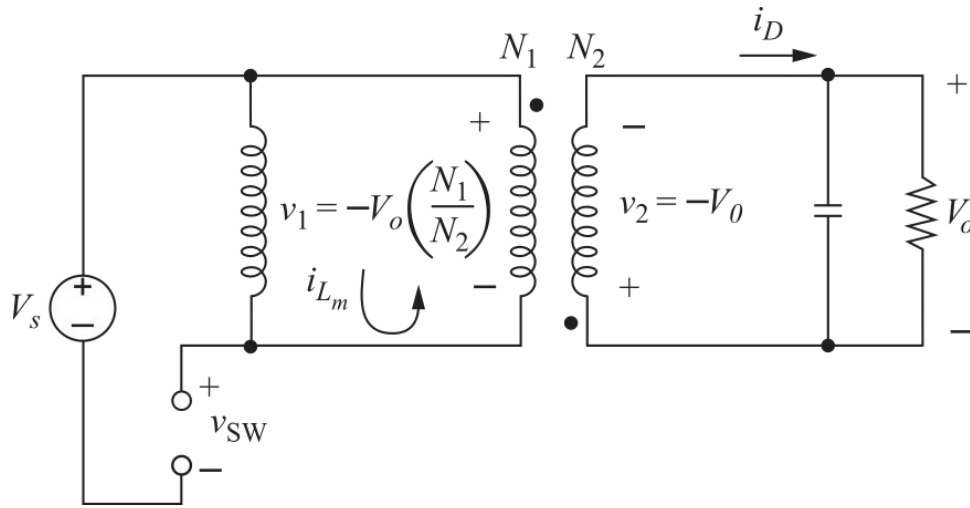
Convertidor Flyback: formas de onda



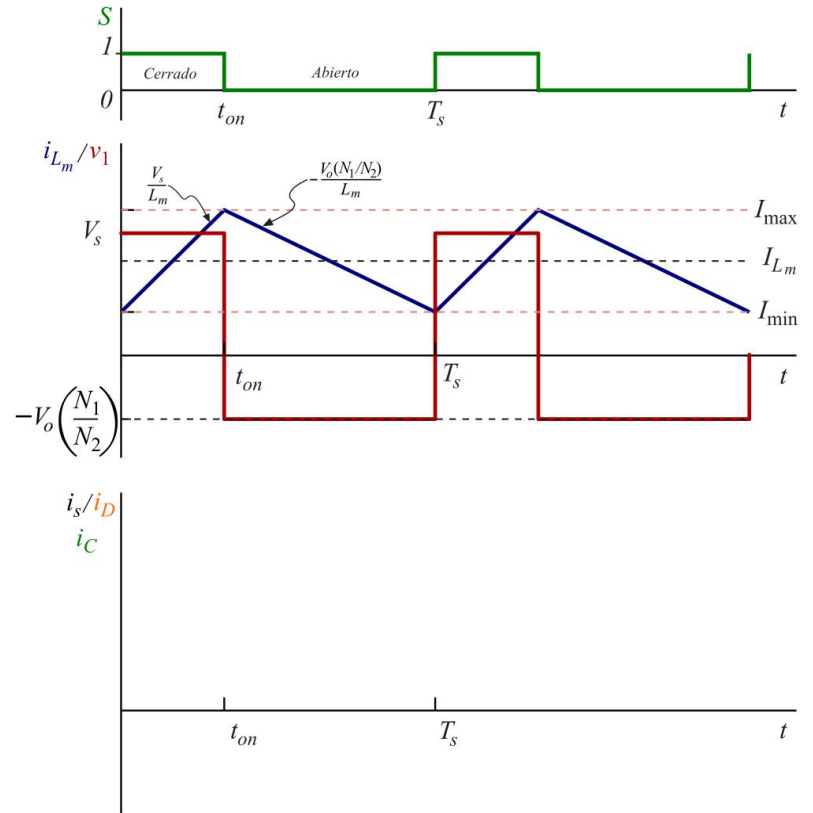
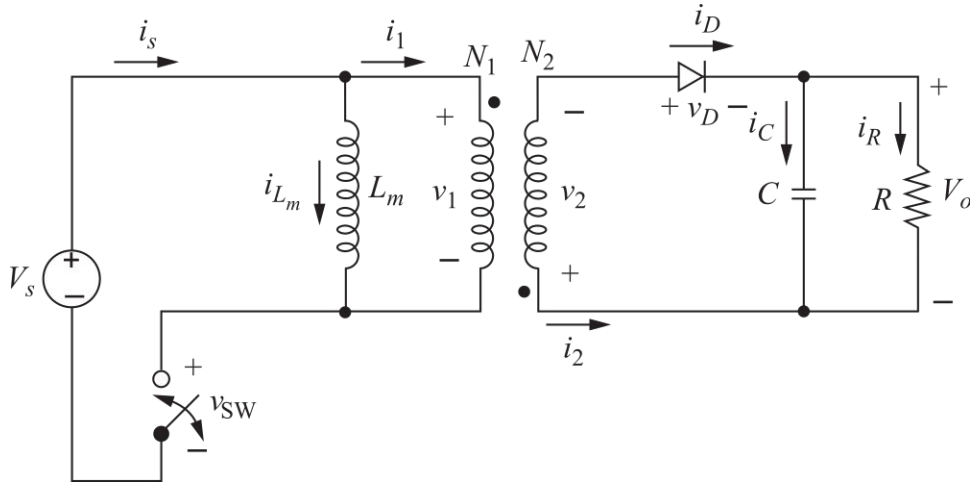
Convertidor Flyback: formas de onda



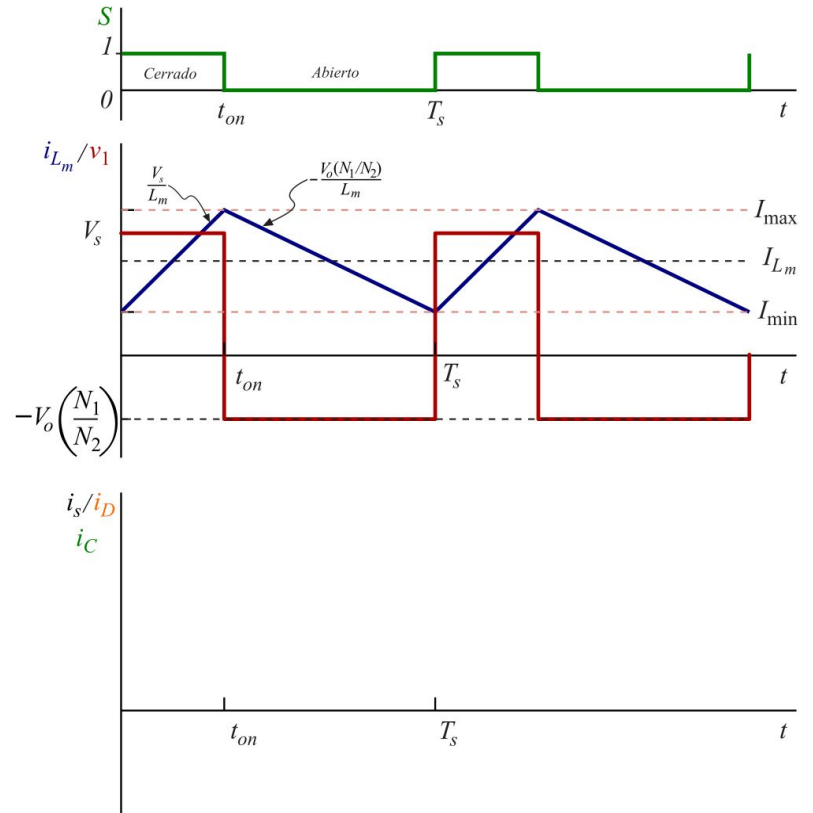
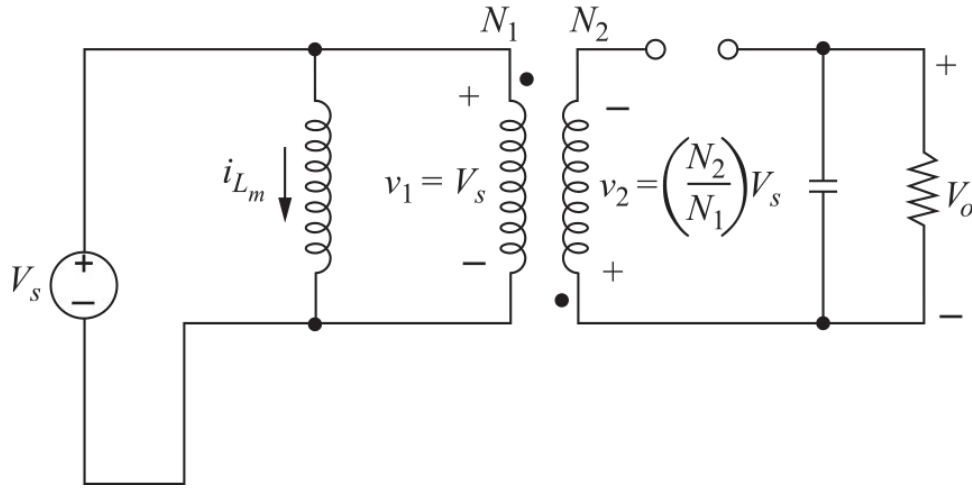
Convertidor Flyback: formas de onda



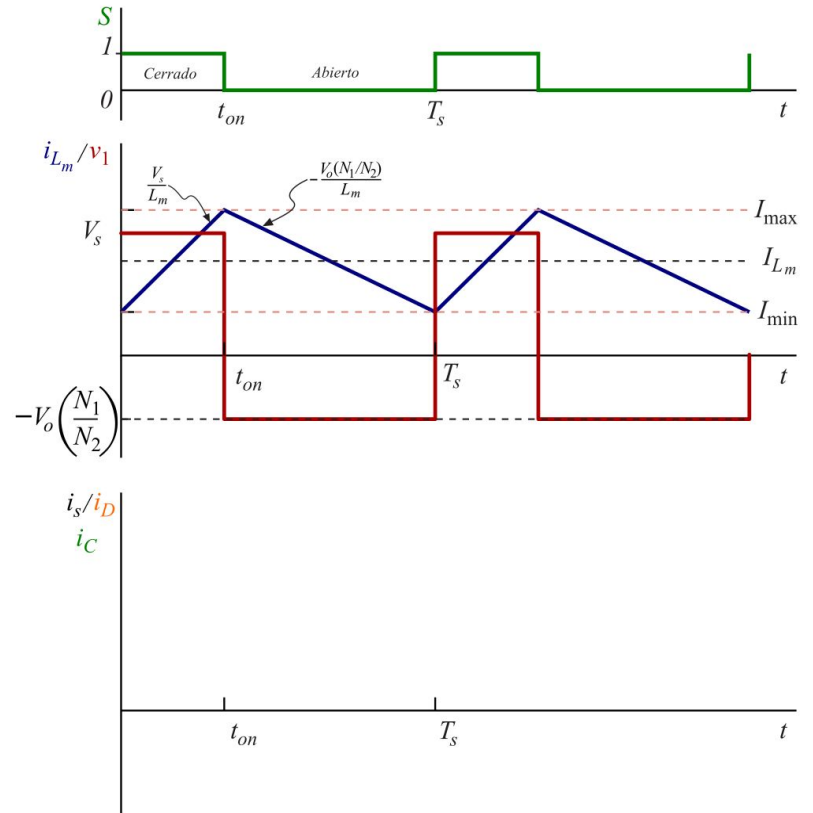
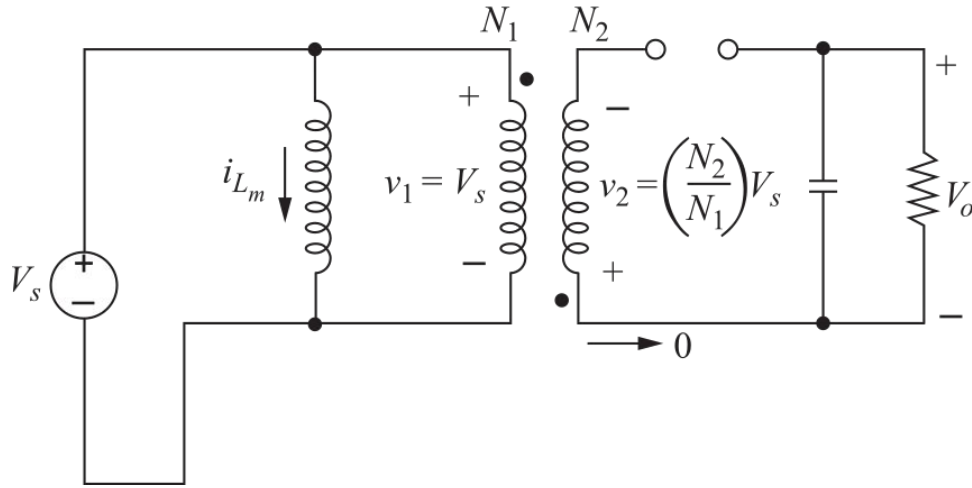
Convertidor Flyback: formas de onda



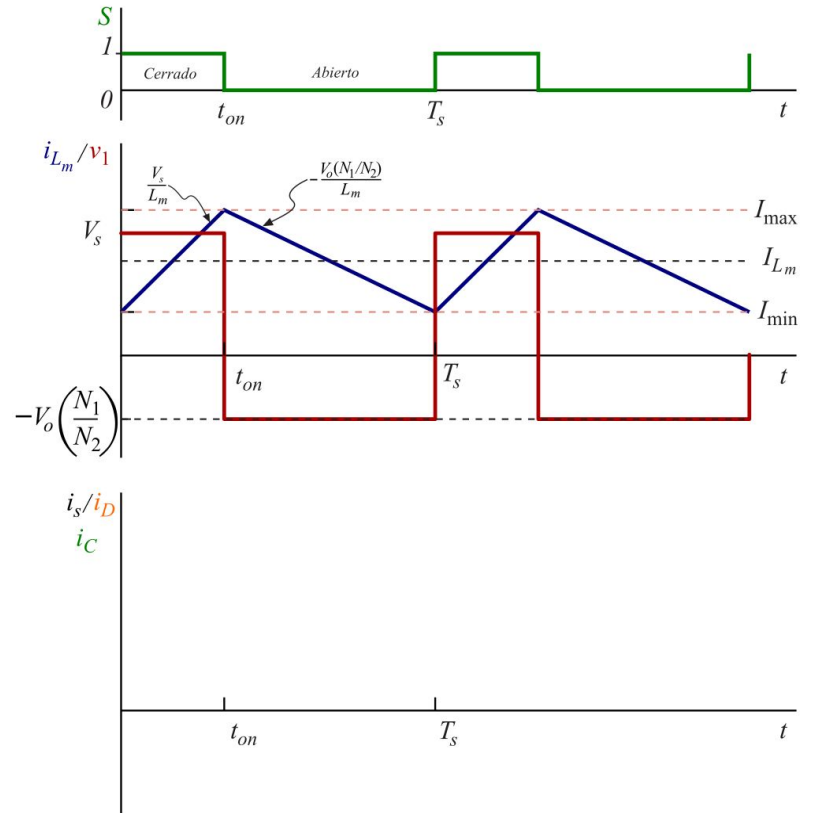
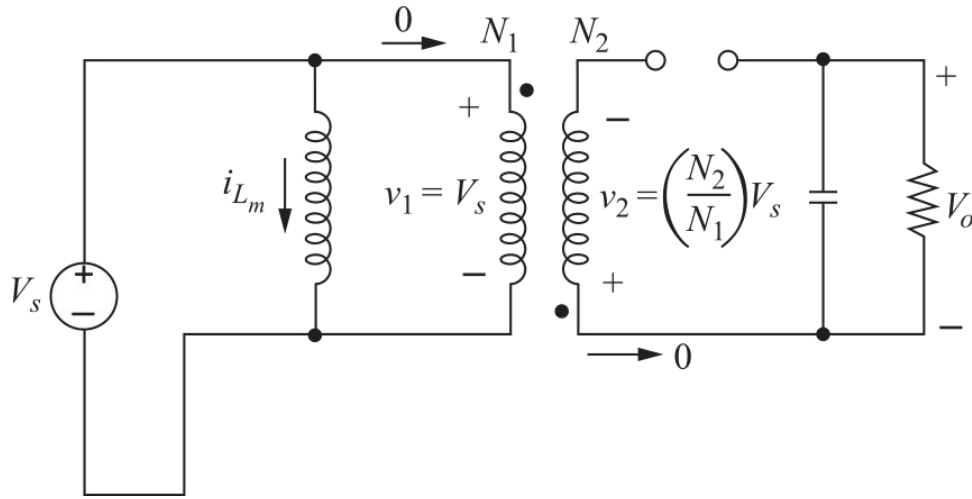
Convertidor Flyback: formas de onda



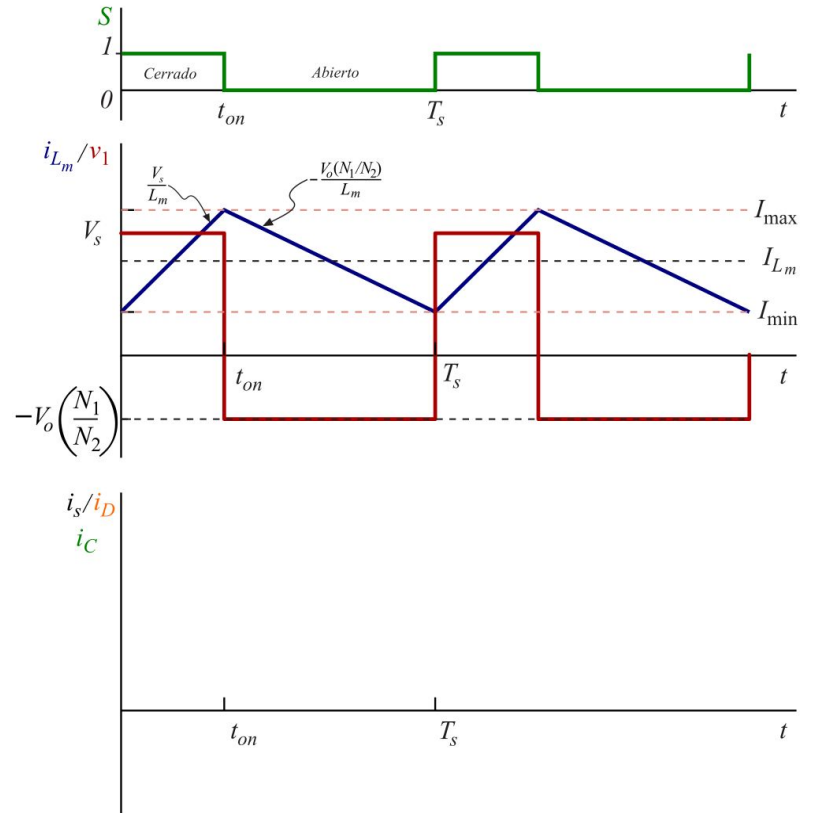
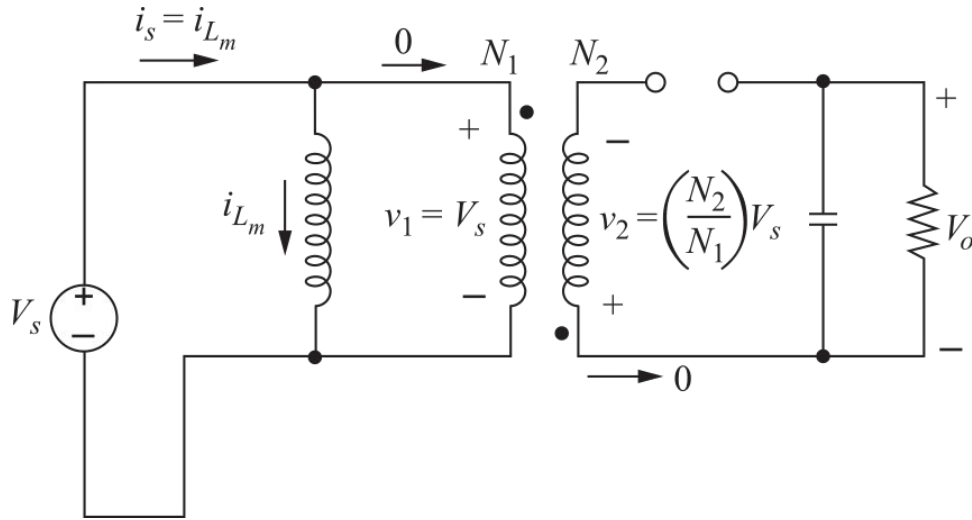
Convertidor Flyback: formas de onda



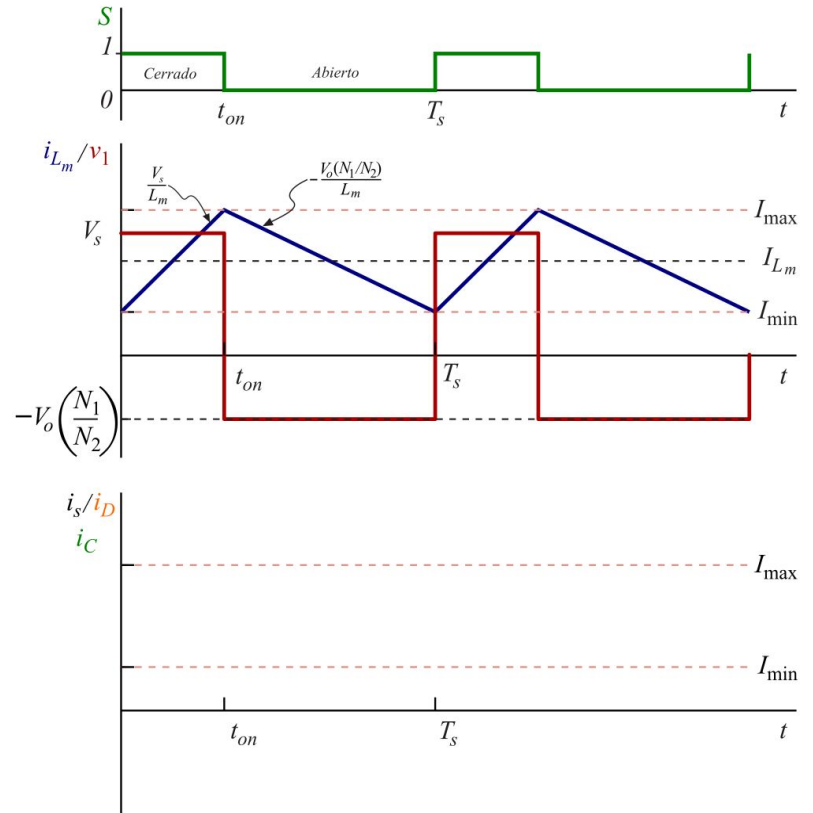
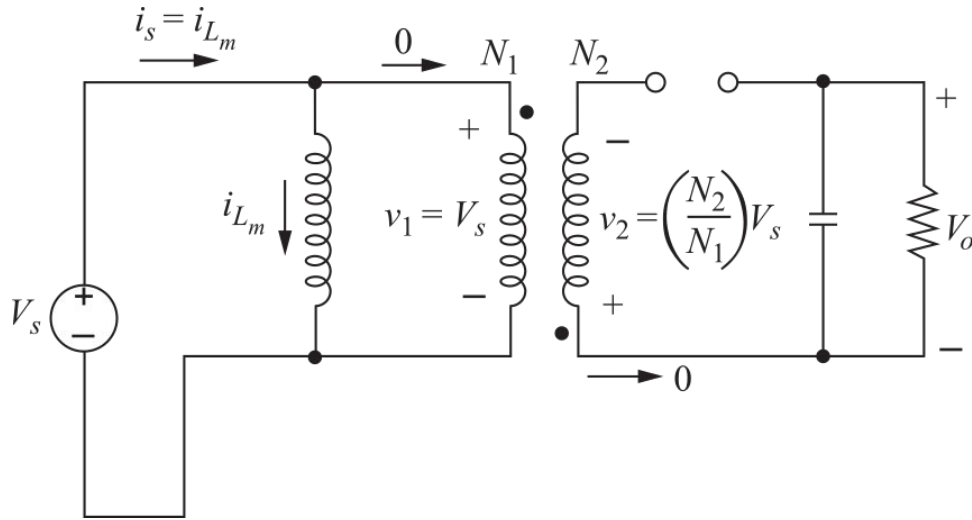
Convertidor Flyback: formas de onda



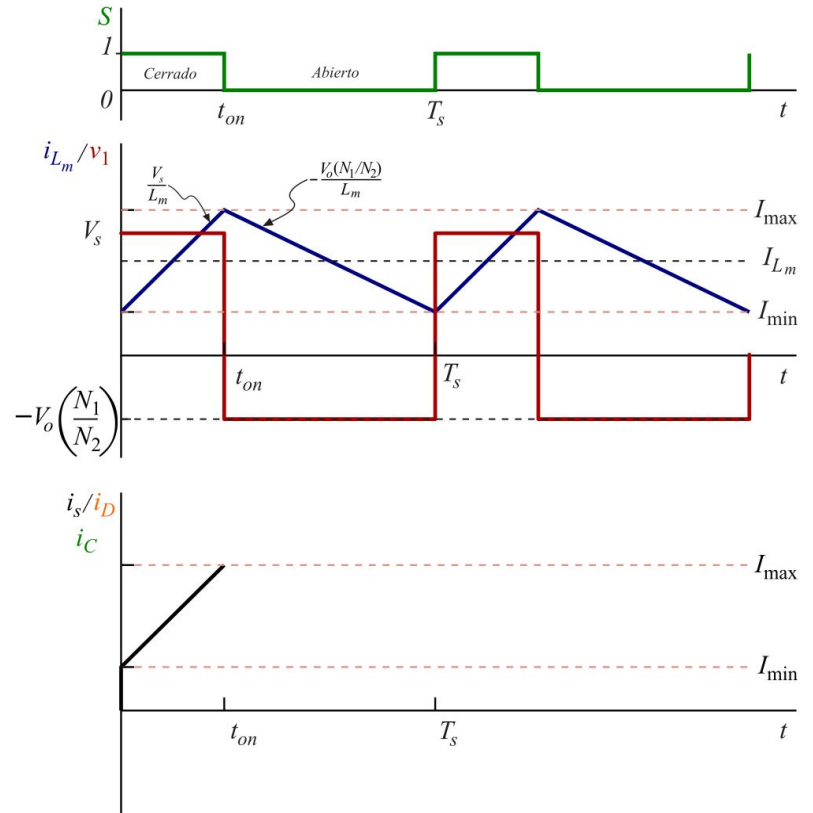
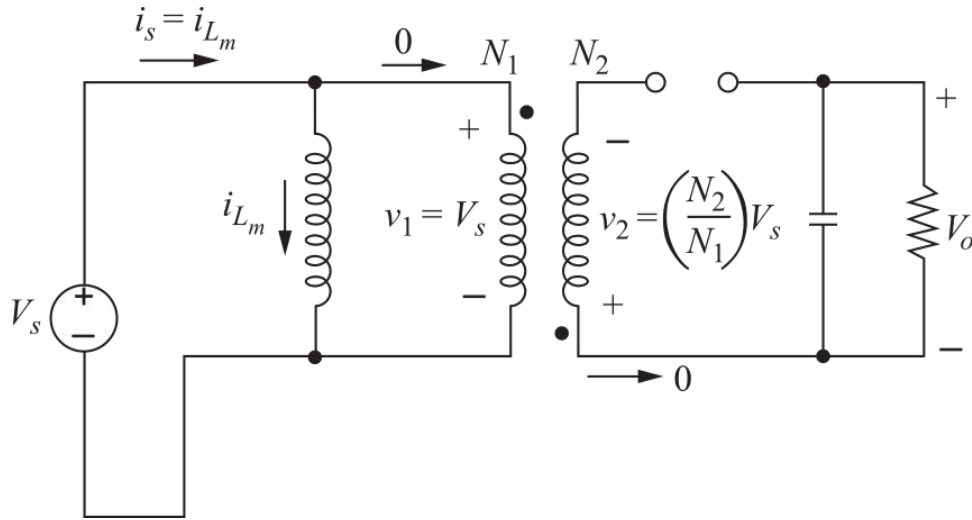
Convertidor Flyback: formas de onda



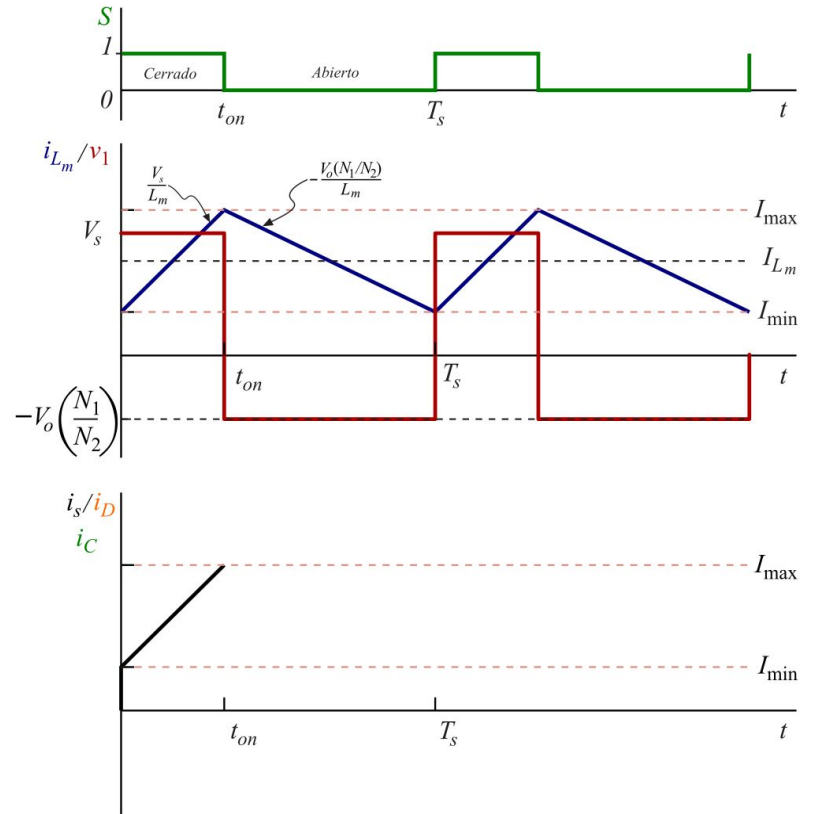
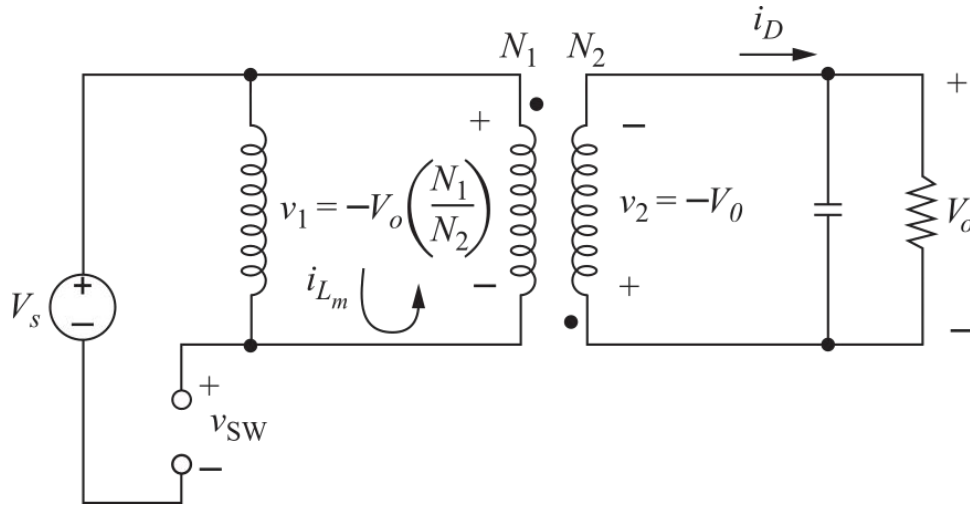
Convertidor Flyback: formas de onda



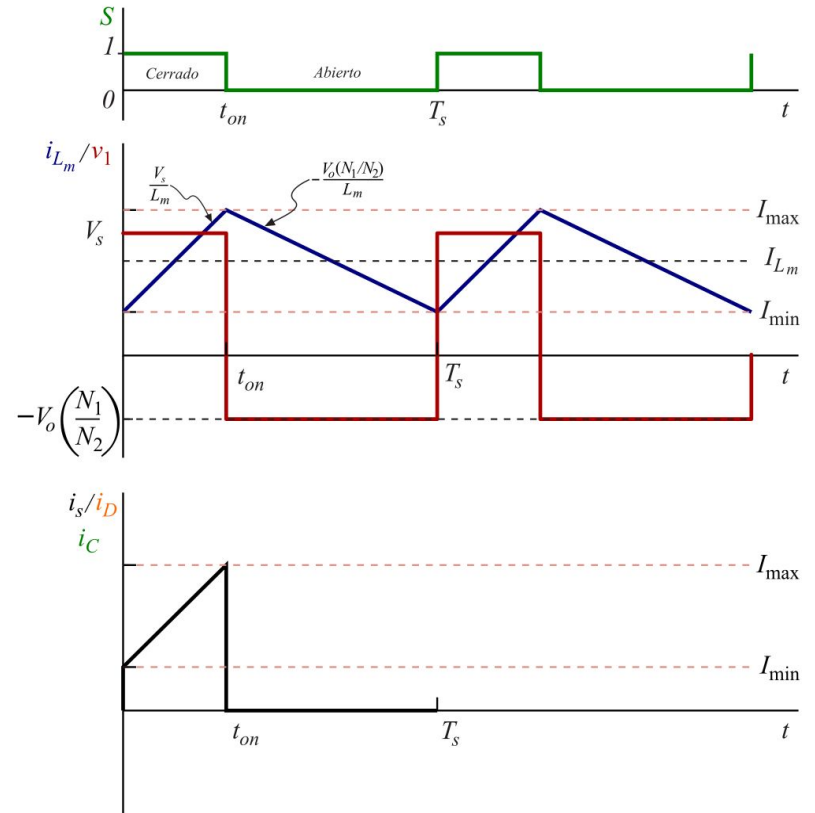
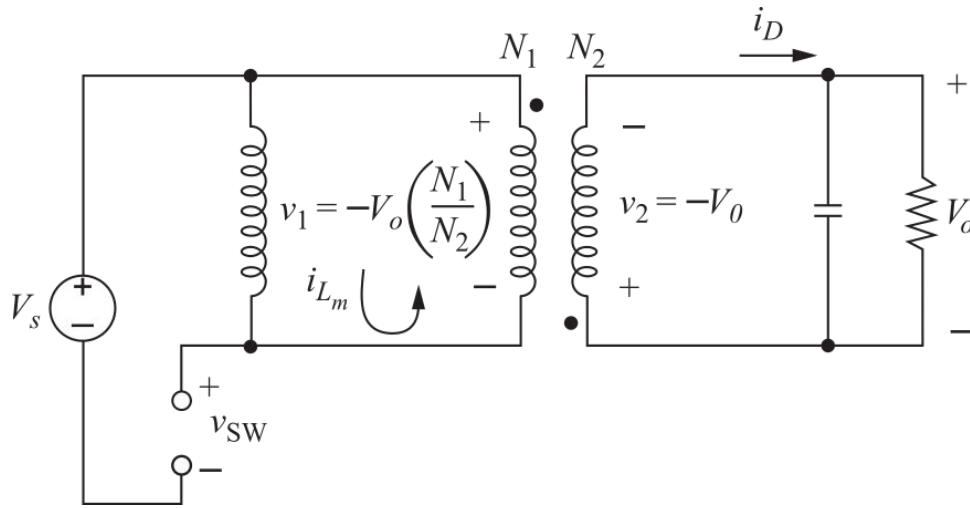
Convertidor Flyback: formas de onda



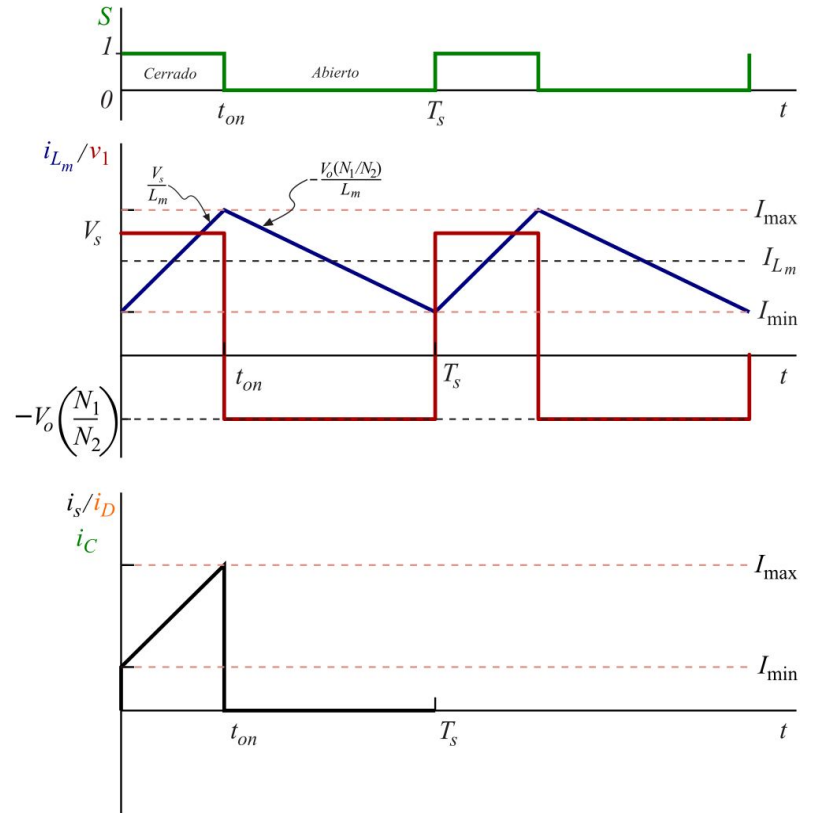
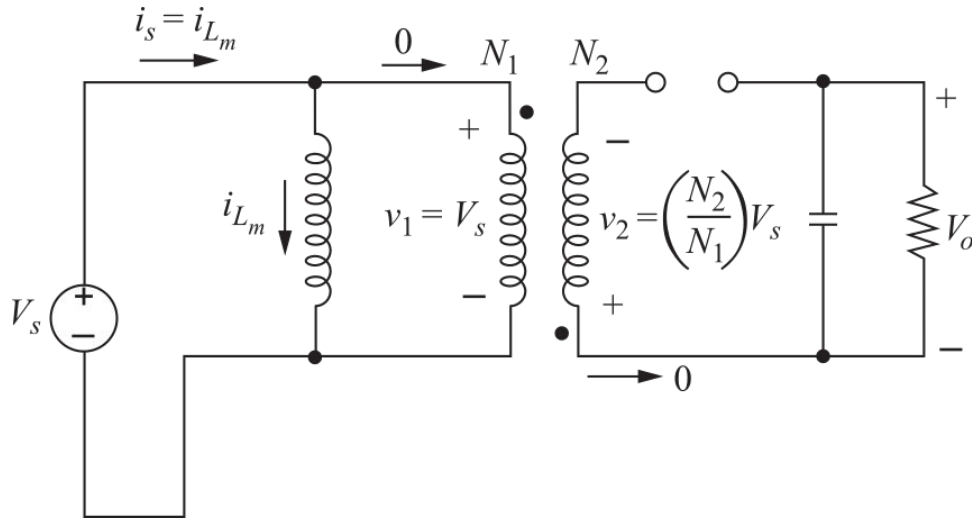
Convertidor Flyback: formas de onda



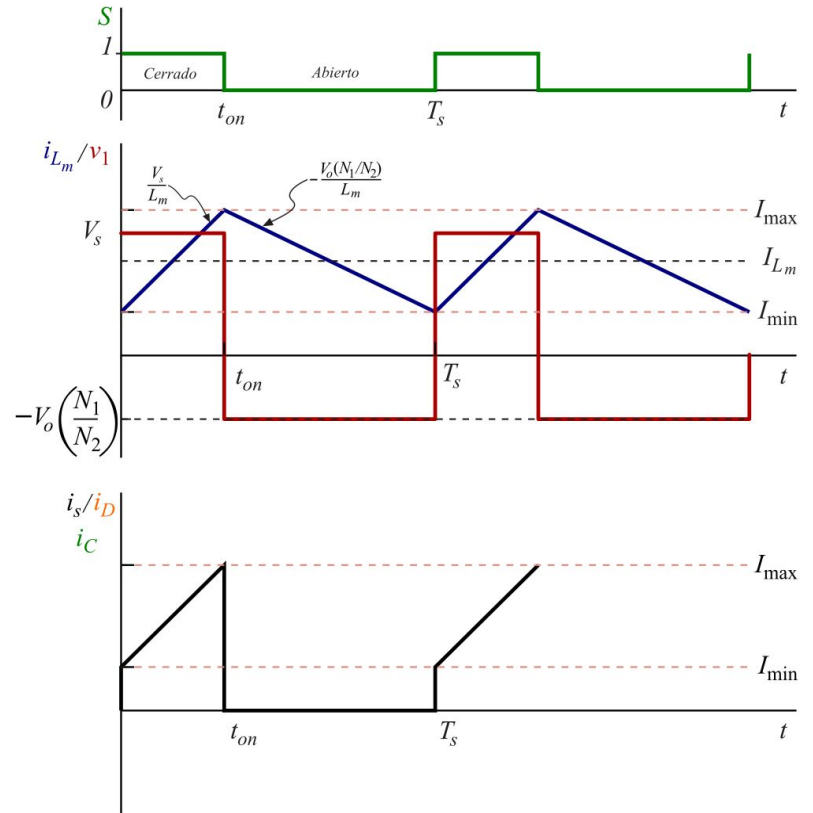
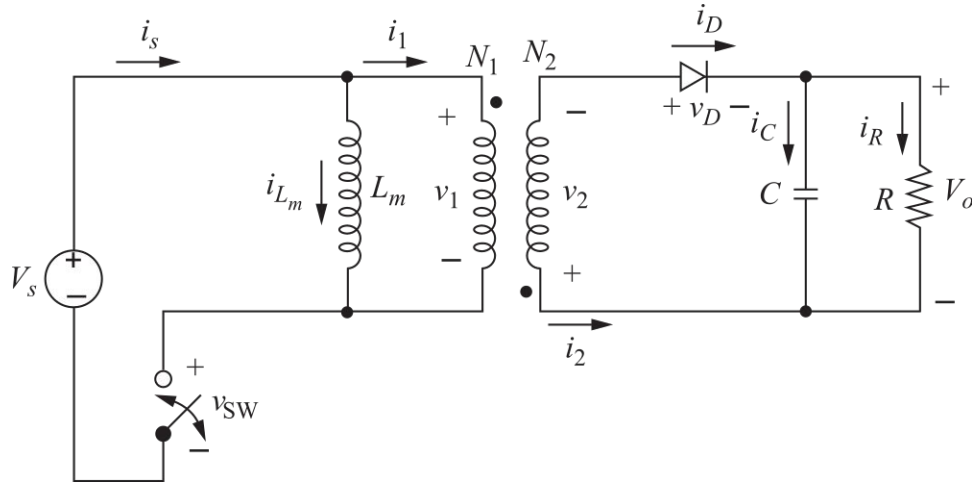
Convertidor Flyback: formas de onda



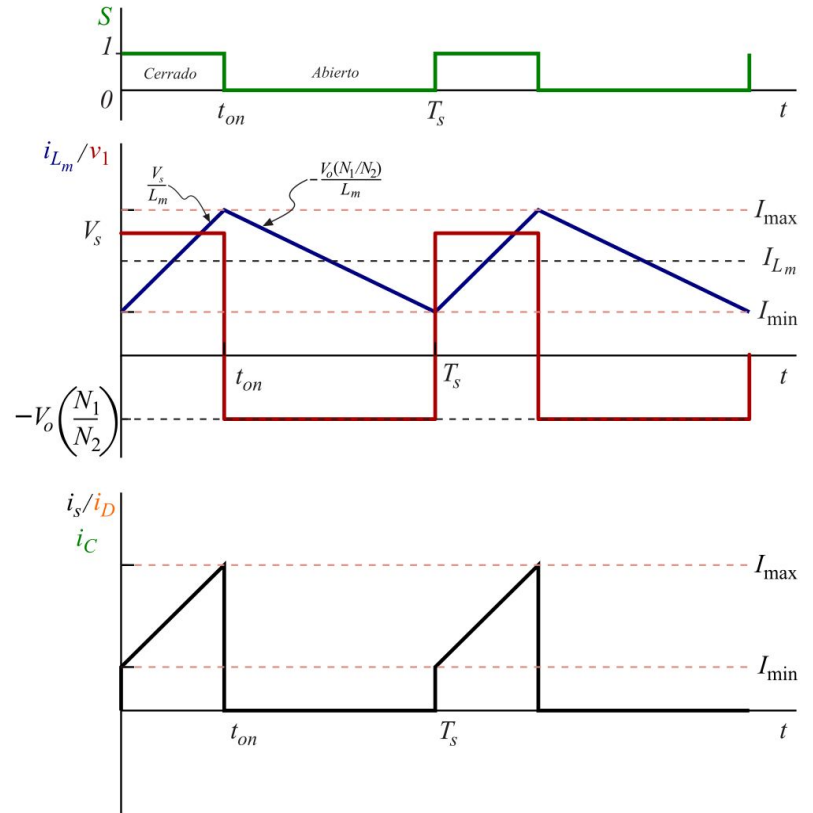
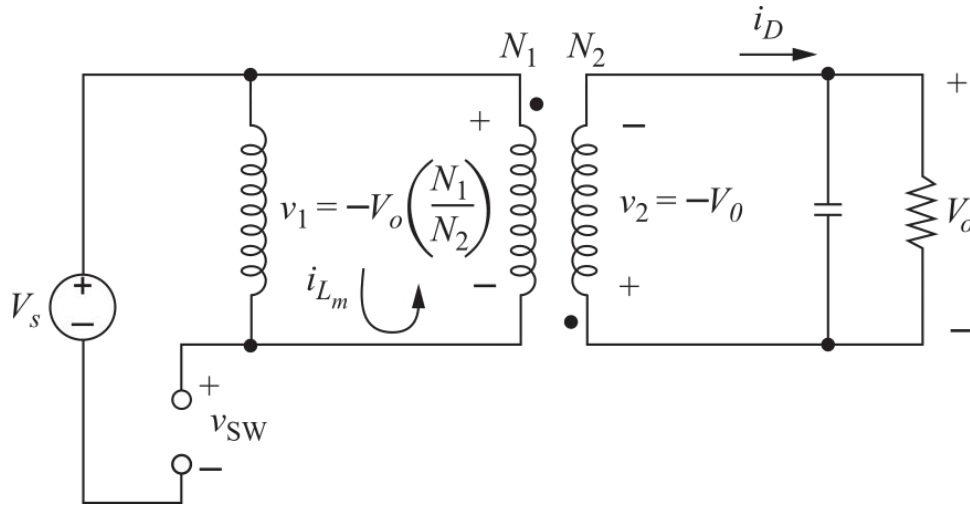
Convertidor Flyback: formas de onda



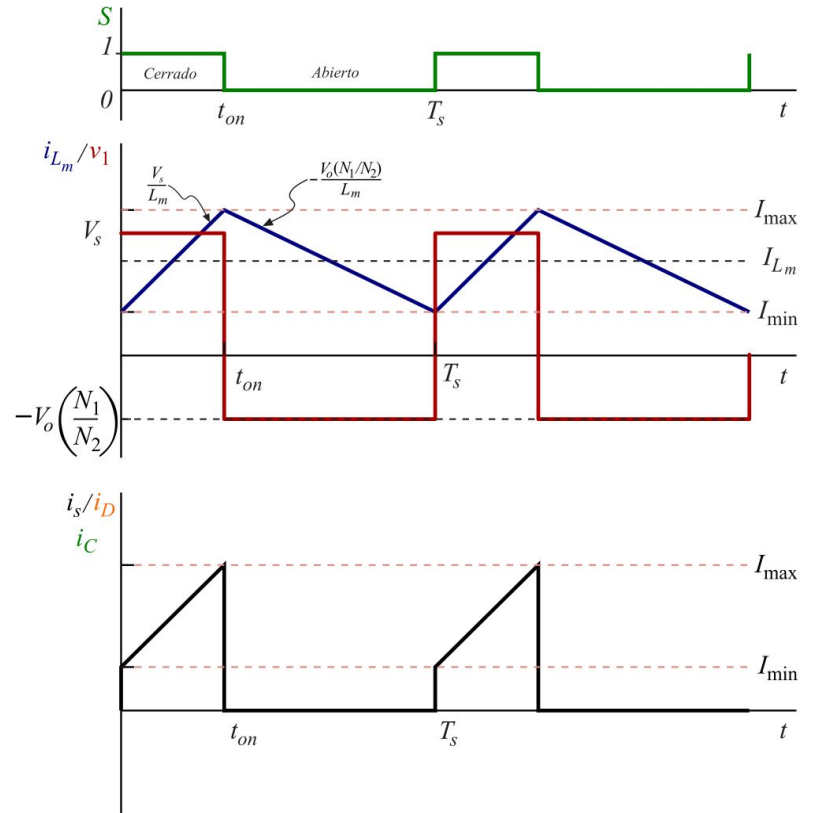
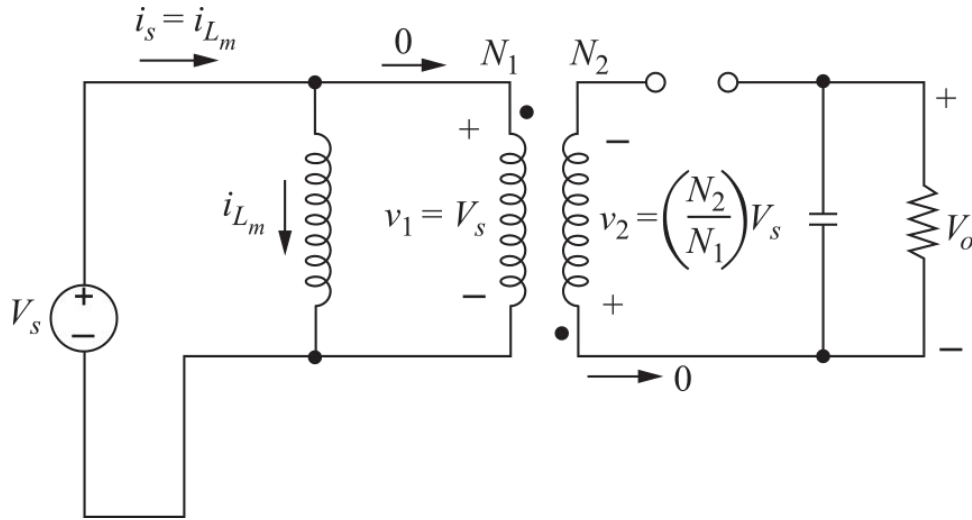
Convertidor Flyback: formas de onda



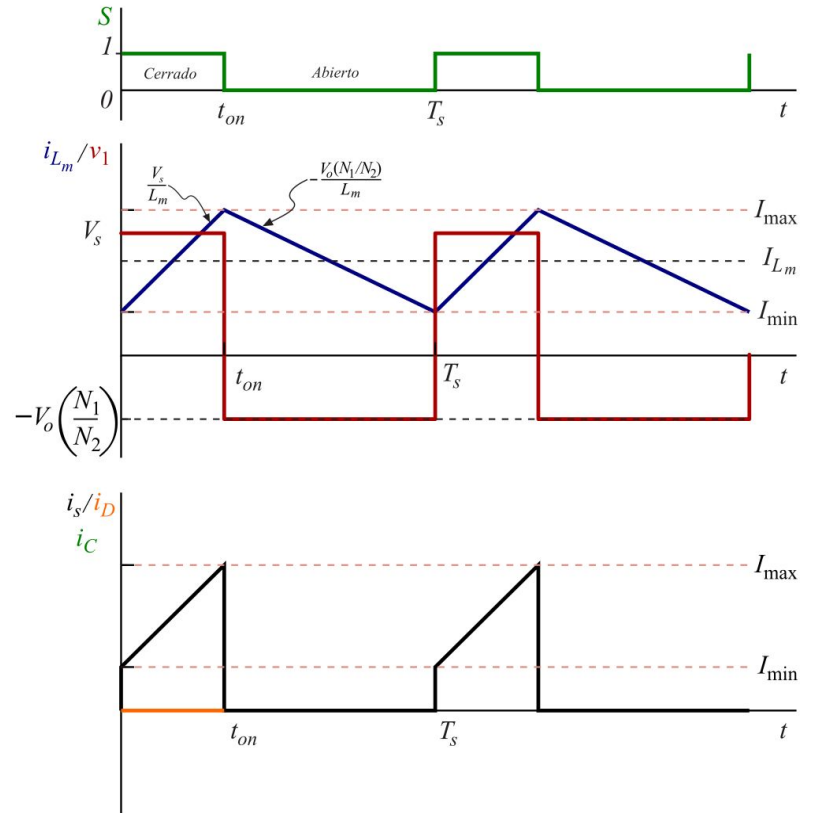
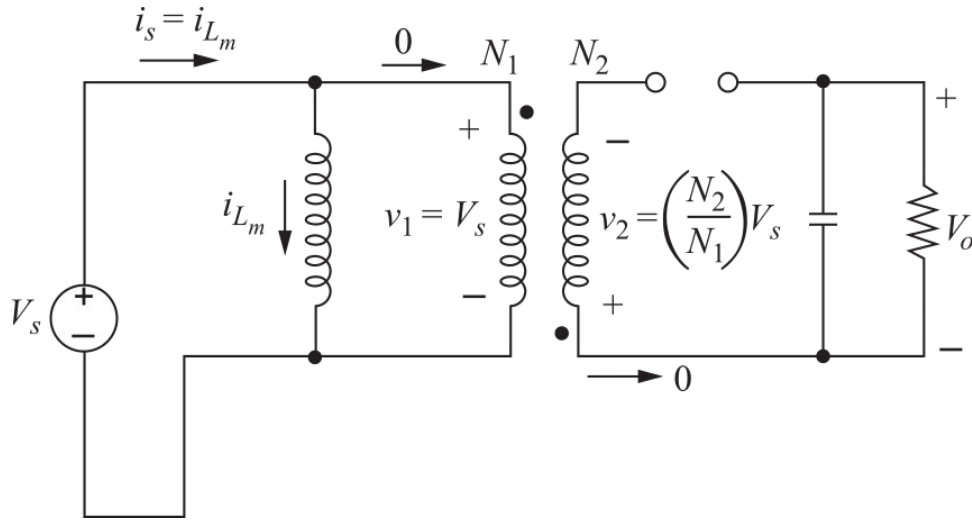
Convertidor Flyback: formas de onda



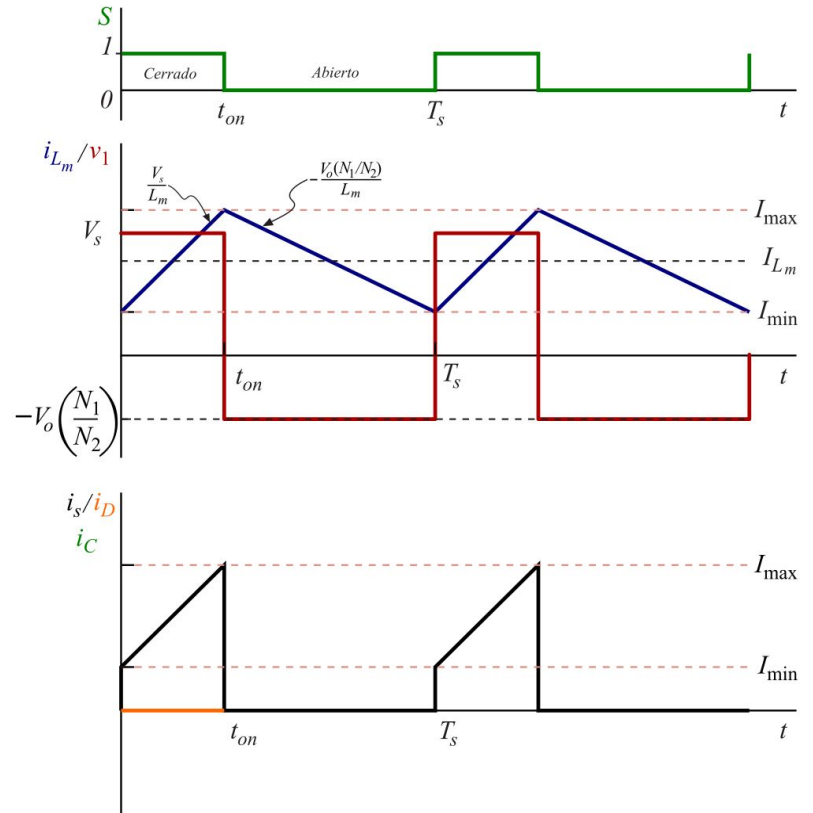
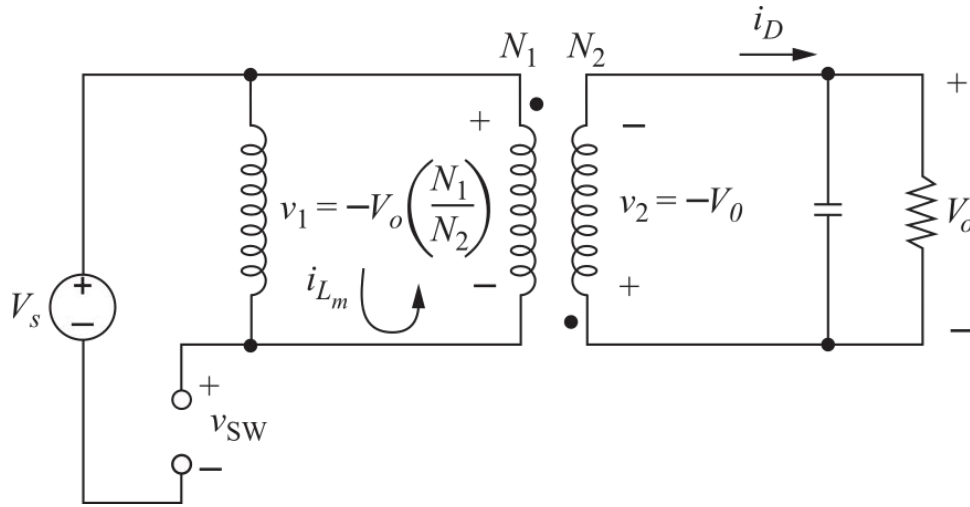
Convertidor Flyback: formas de onda



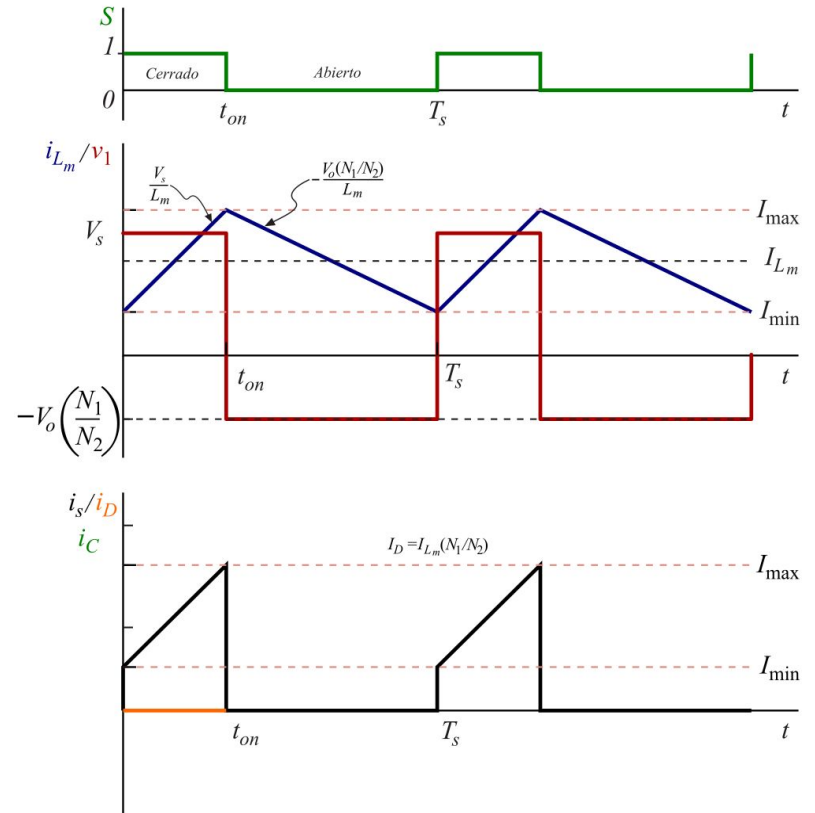
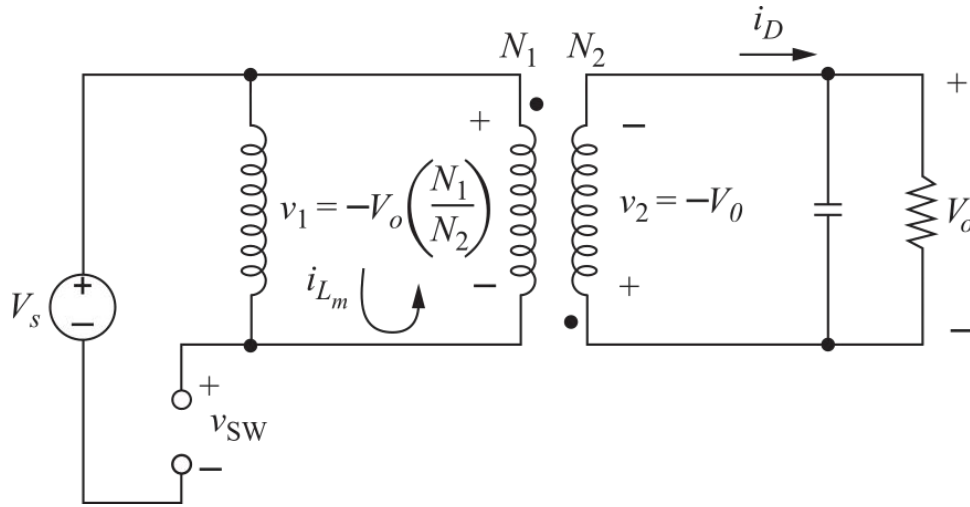
Convertidor Flyback: formas de onda



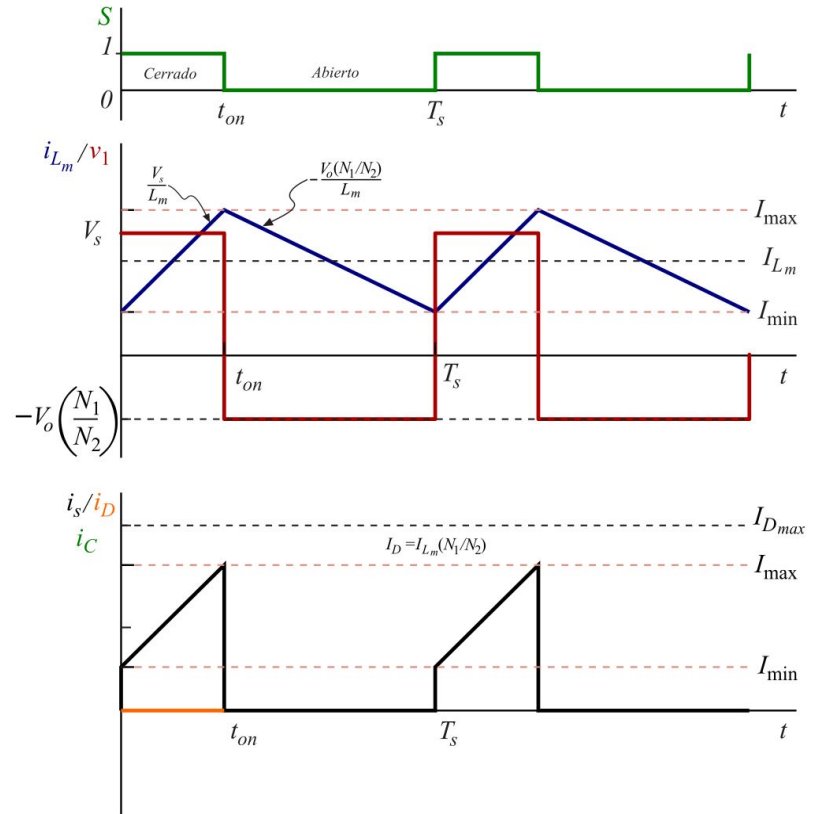
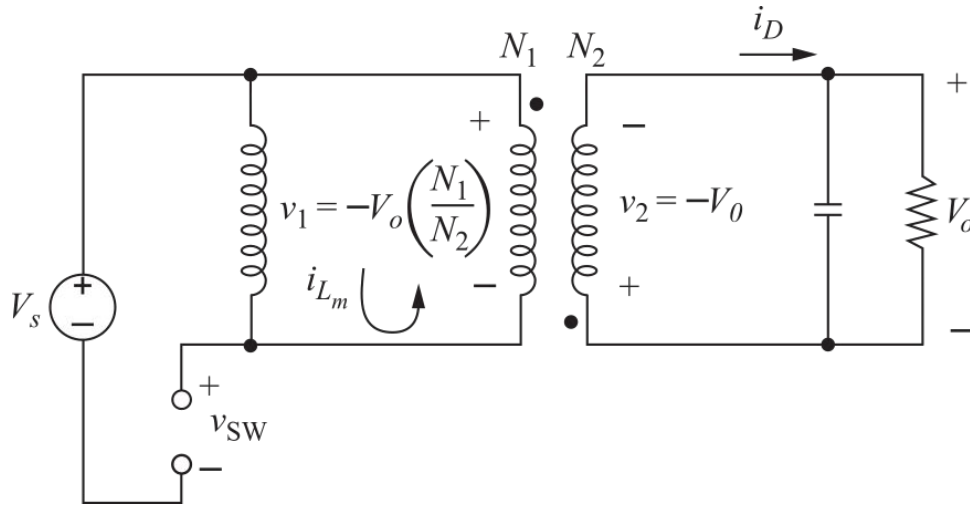
Convertidor Flyback: formas de onda



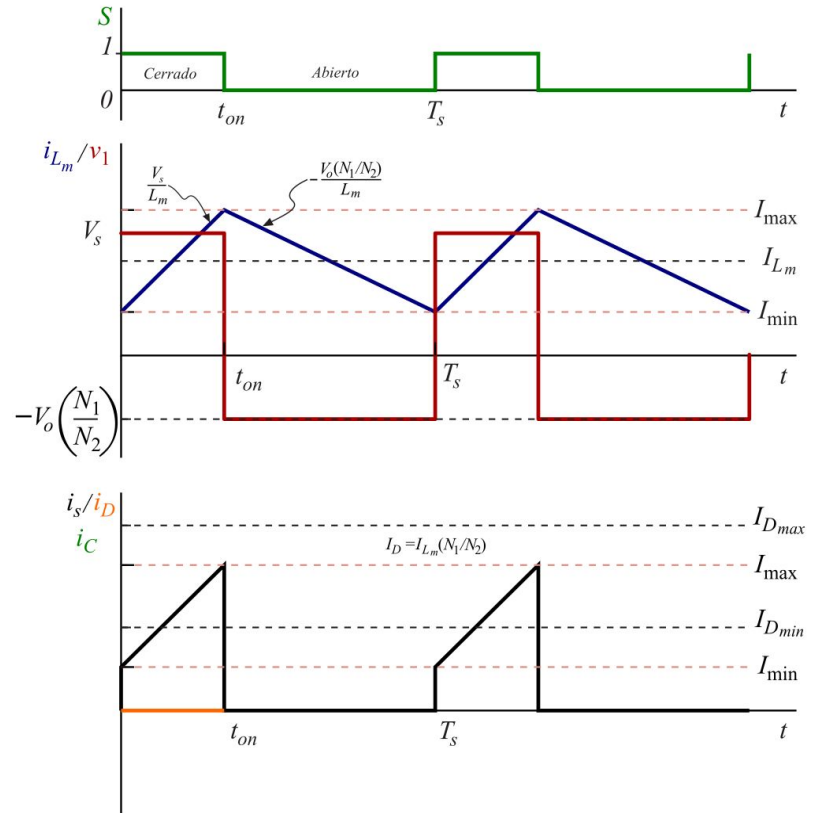
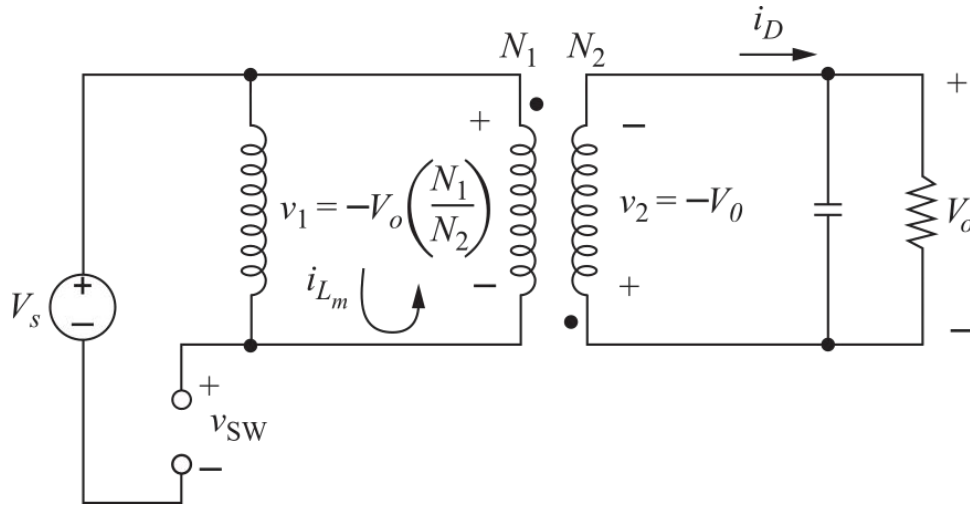
Convertidor Flyback: formas de onda



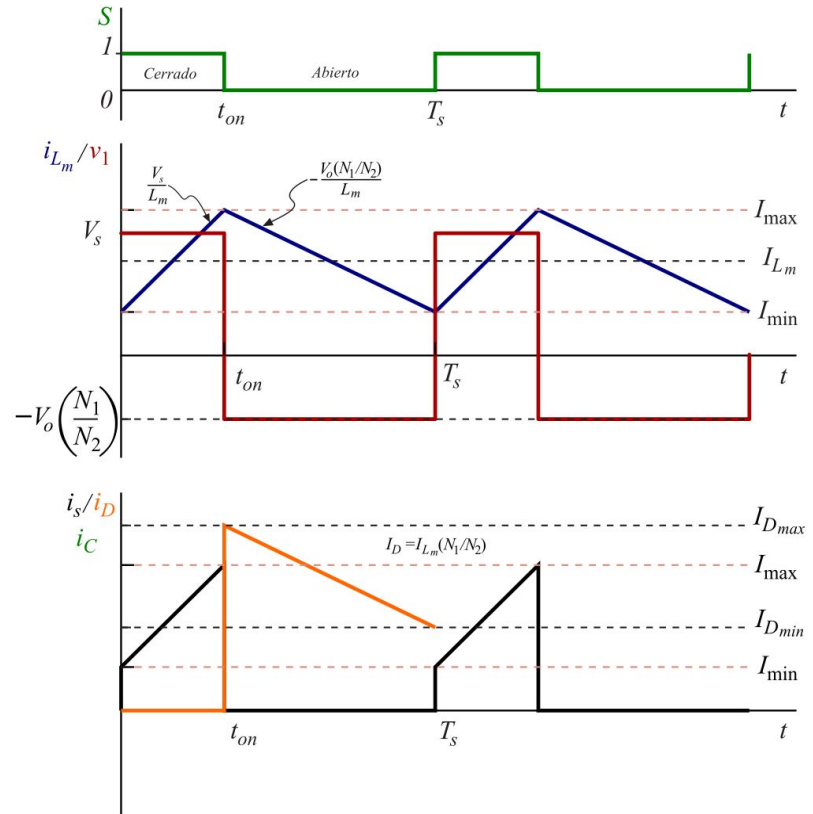
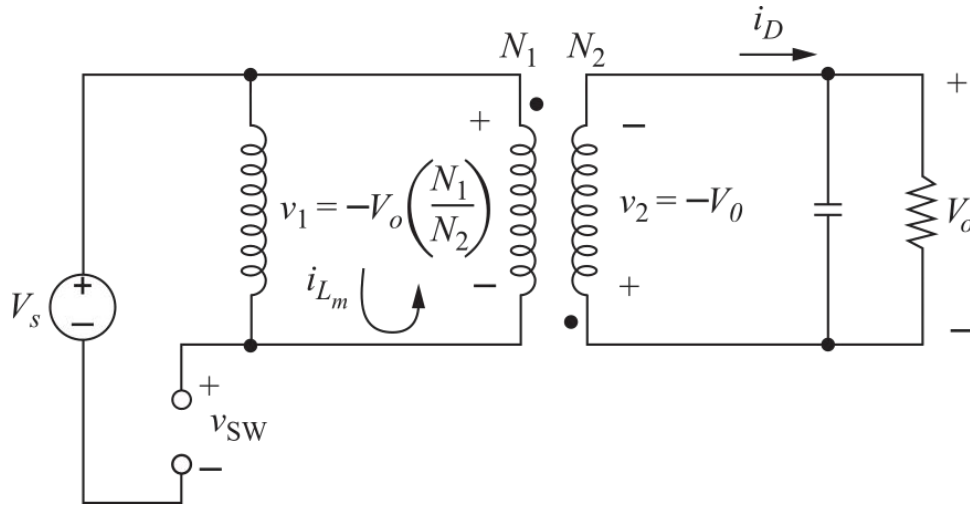
Convertidor Flyback: formas de onda



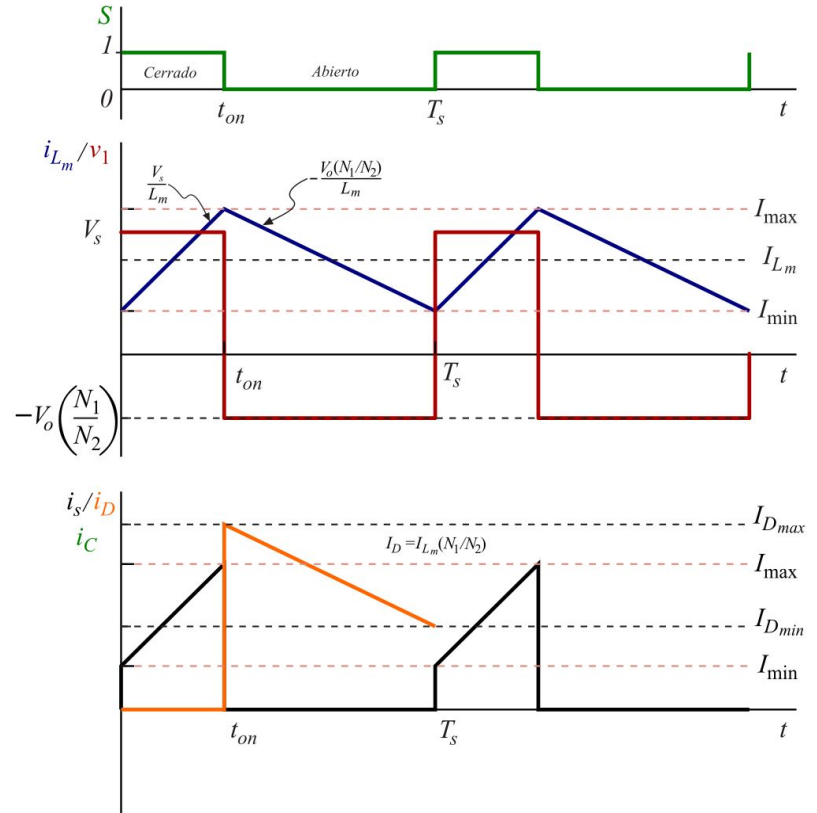
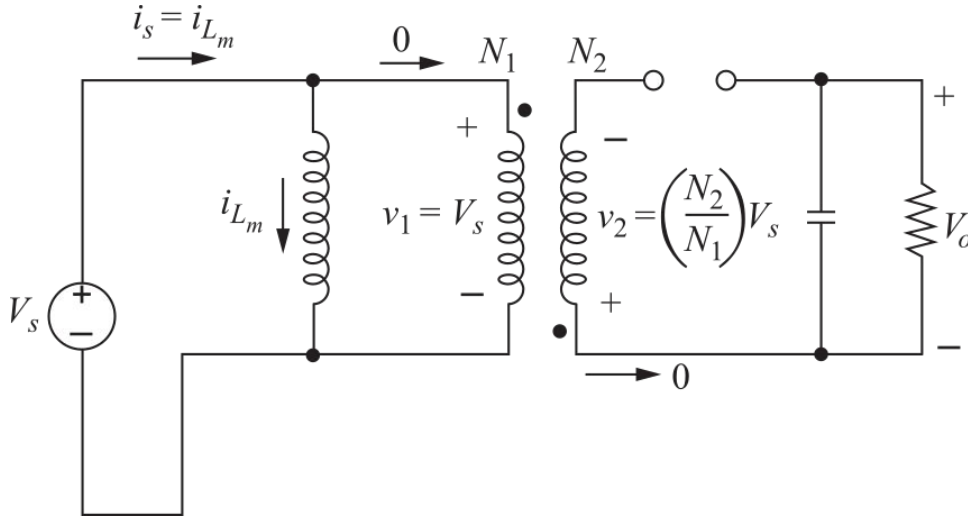
Convertidor Flyback: formas de onda



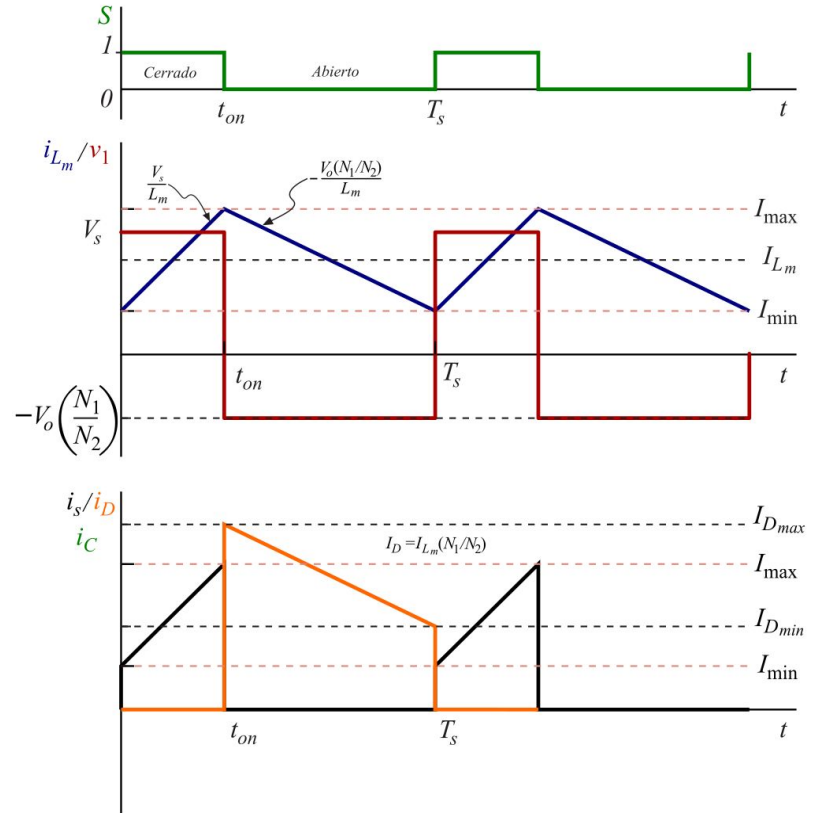
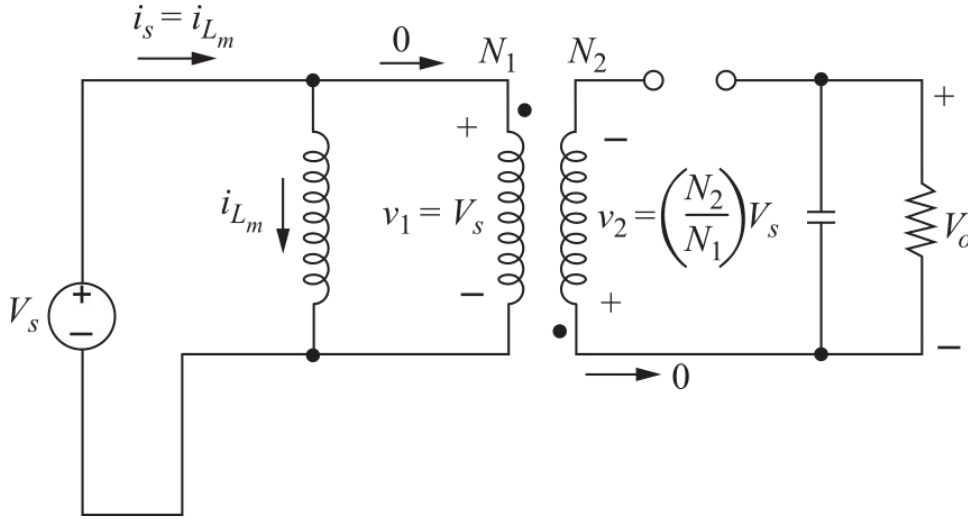
Convertidor Flyback: formas de onda



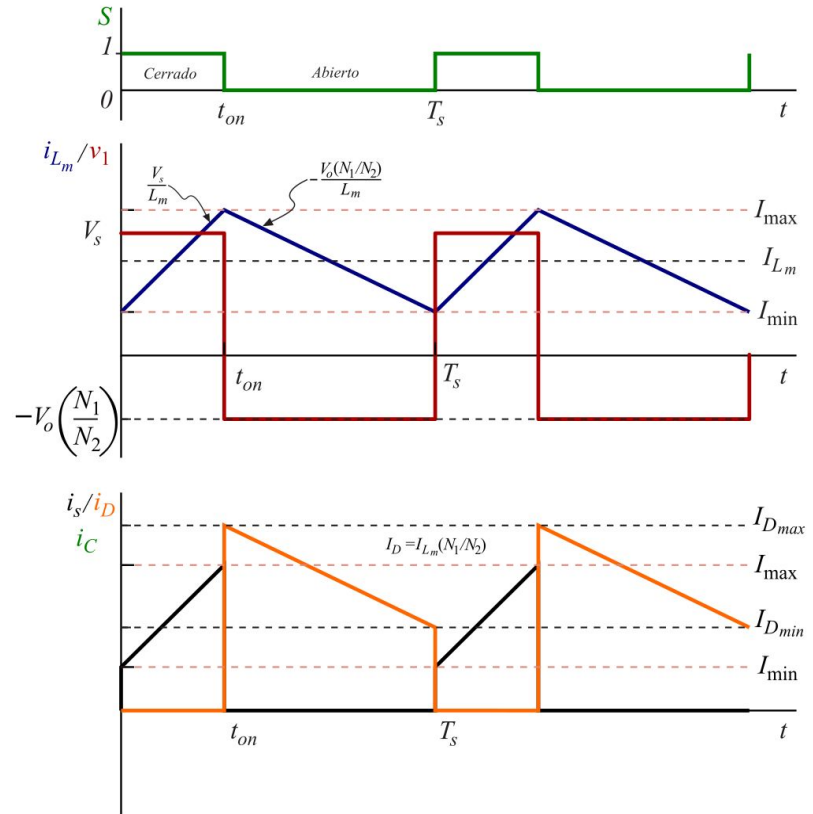
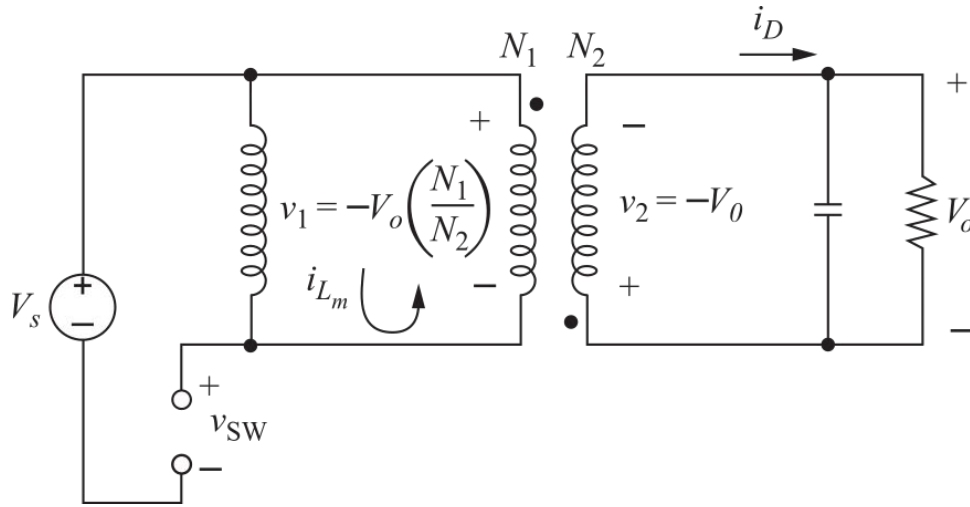
Convertidor Flyback: formas de onda



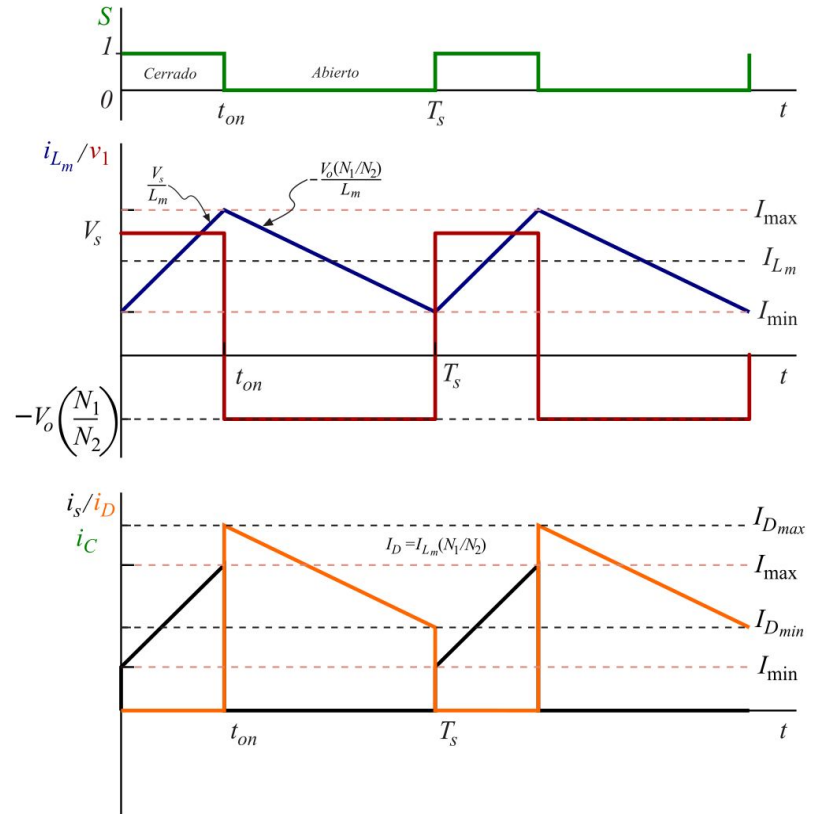
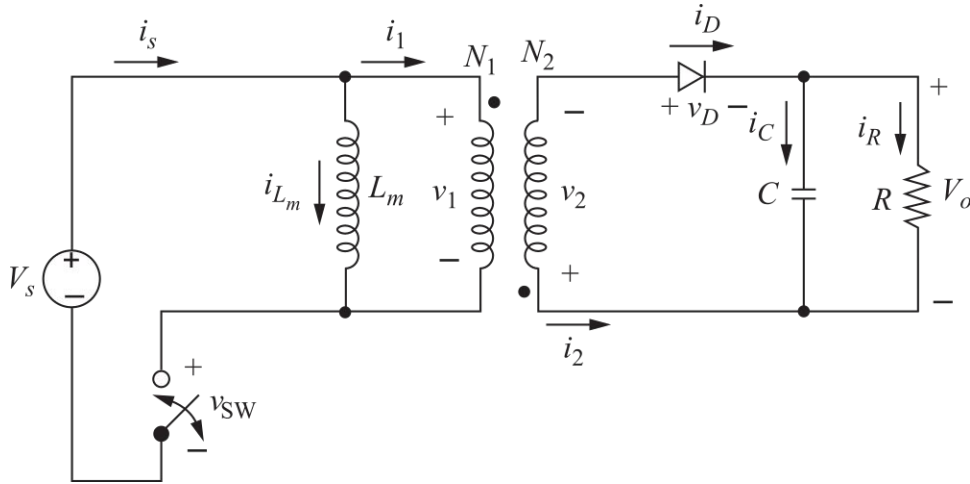
Convertidor Flyback: formas de onda



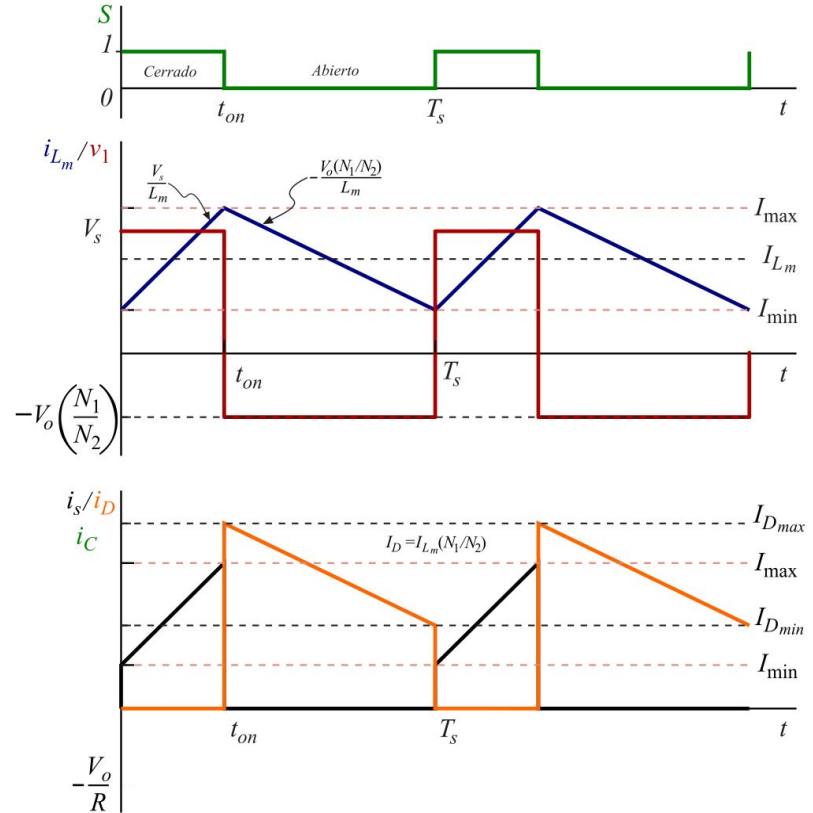
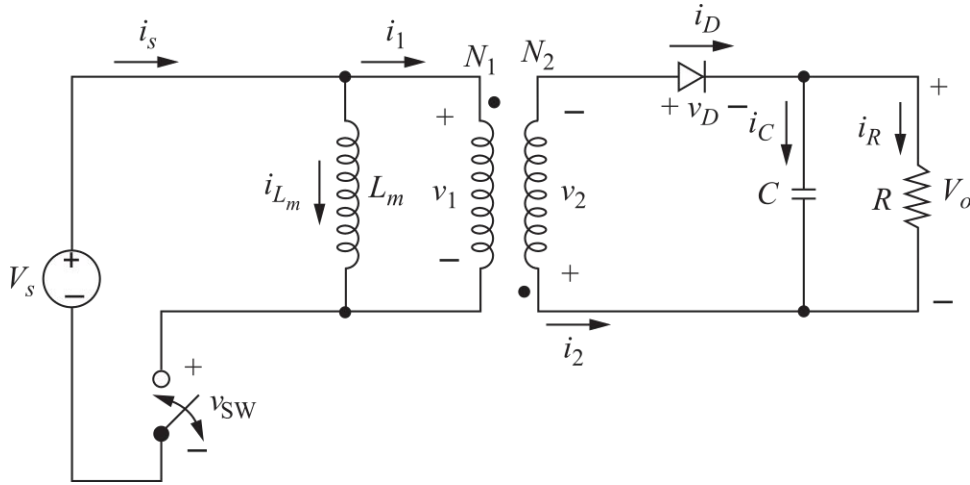
Convertidor Flyback: formas de onda



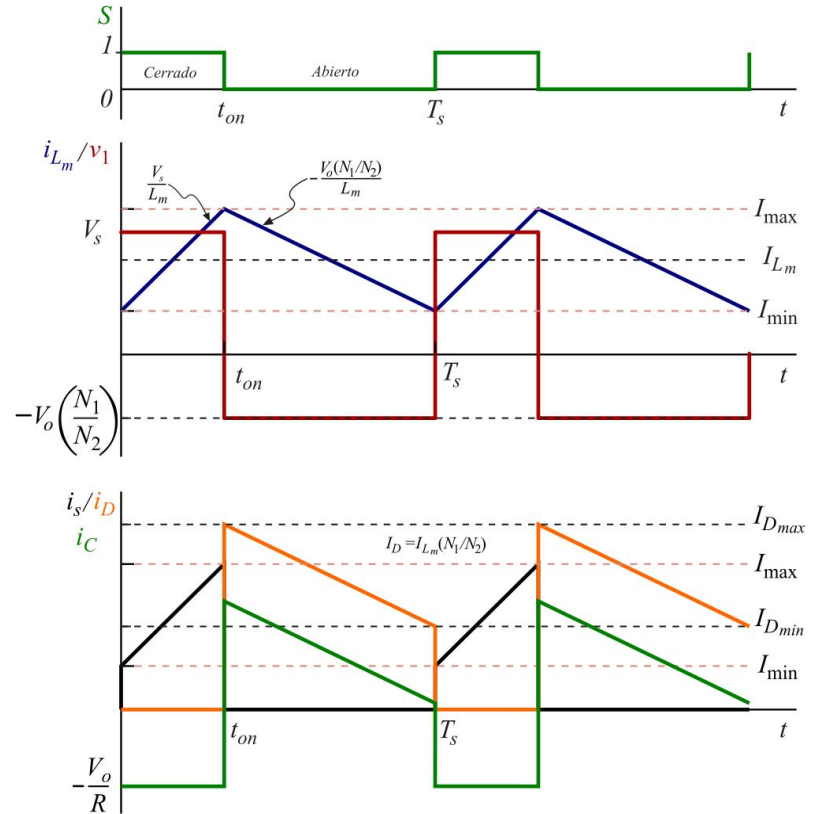
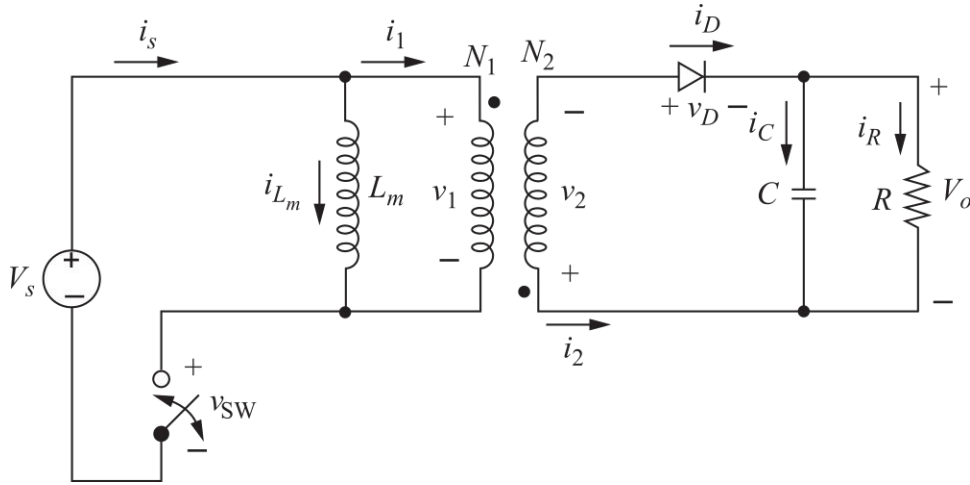
Convertidor Flyback: formas de onda



Convertidor Flyback: formas de onda

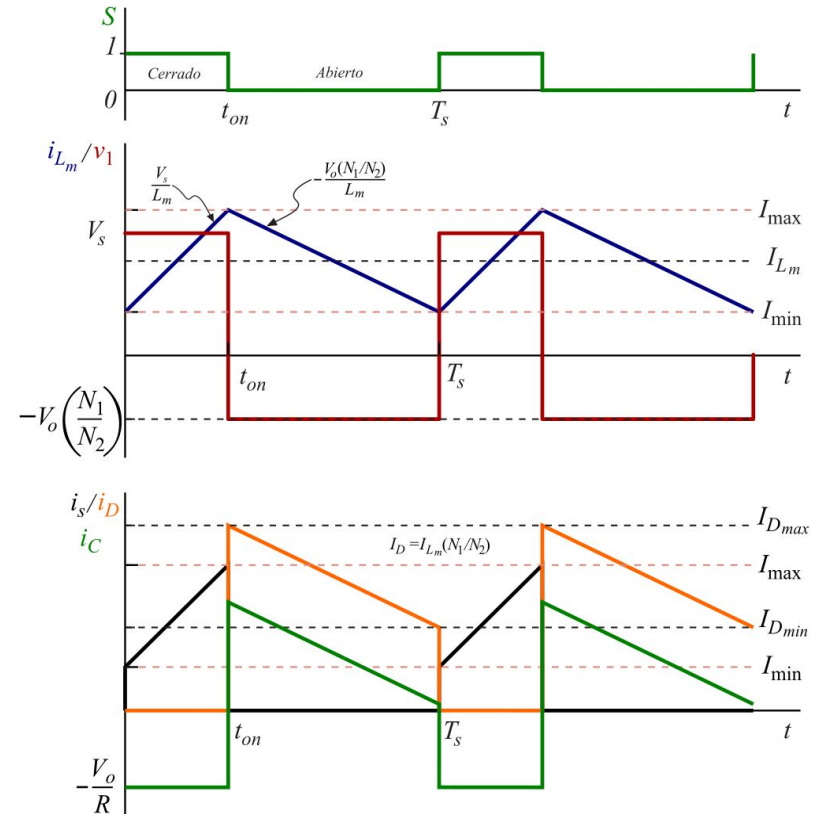


Convertidor Flyback: formas de onda



Convertidor Flyback: relación de conversión

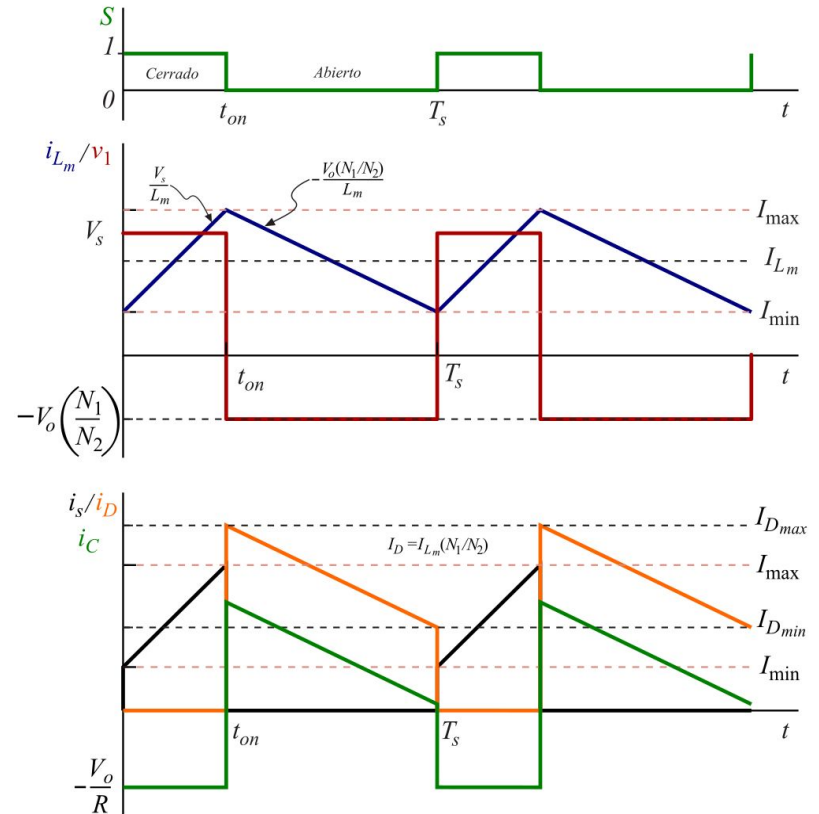
La tensión media en el inductor debe ser cero:



Convertidor Flyback: relación de conversión

La tensión media en el inductor debe ser cero:

$$V_s t_{on} = \frac{N_1}{N_2} V_o (T_s - t_{on})$$

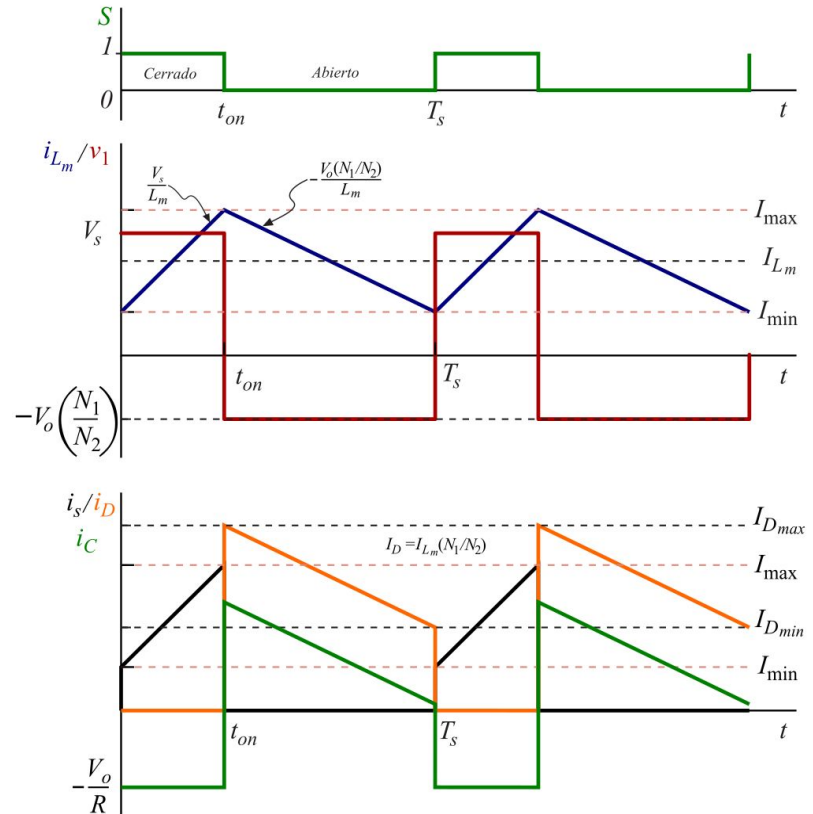


Convertidor Flyback: relación de conversión

La tensión media en el inductor debe ser cero:

$$V_s t_{on} = \frac{N_1}{N_2} V_o (T_s - t_{on})$$

$$V_s \frac{t_{on}}{T_s} = \frac{N_1}{N_2} V_o \left(1 - \frac{t_{on}}{T_s}\right)$$



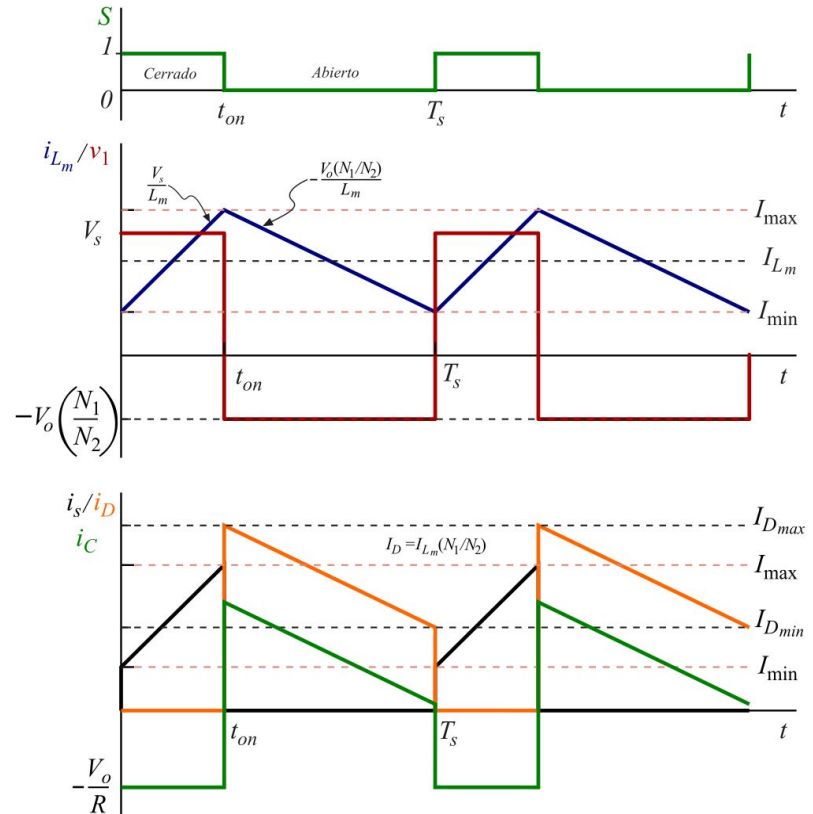
Convertidor Flyback: relación de conversión

La tensión media en el inductor debe ser cero:

$$V_s t_{on} = \frac{N_1}{N_2} V_0 (T_s - t_{on})$$

$$V_s \frac{t_{on}}{T_s} = \frac{N_1}{N_2} V_0 \left(1 - \frac{t_{on}}{T_s}\right)$$

$$V_s D = \frac{N_1}{N_2} V_0 (1 - D)$$



Convertidor Flyback: relación de conversión

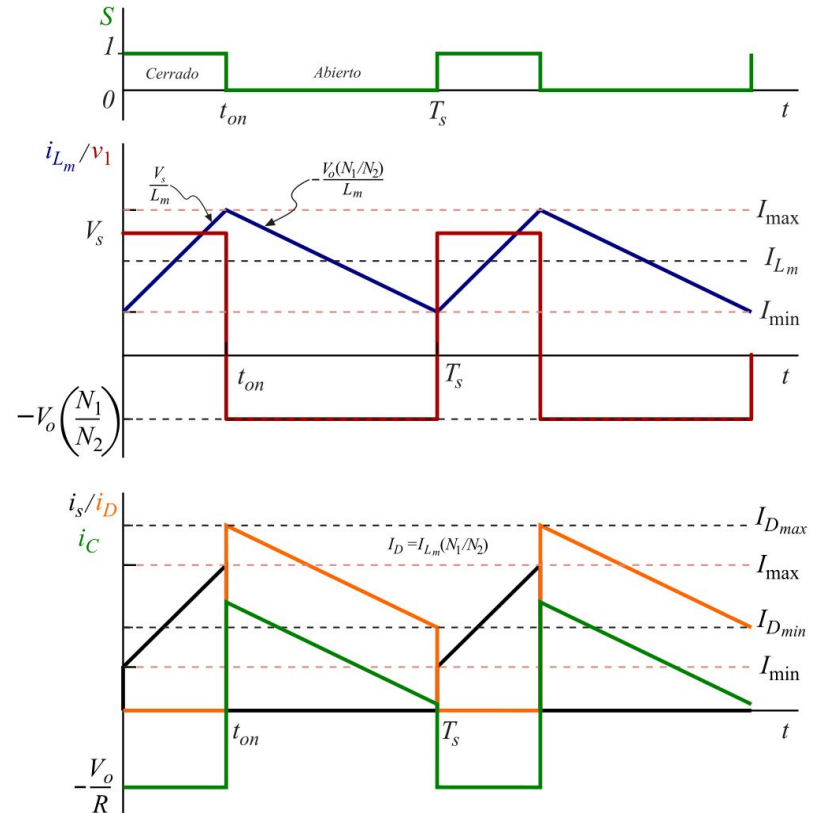
La tensión media en el inductor debe ser cero:

$$V_s t_{on} = \frac{N_1}{N_2} V_0 (T_s - t_{on})$$

$$V_s \frac{t_{on}}{T_s} = \frac{N_1}{N_2} V_0 \left(1 - \frac{t_{on}}{T_s}\right)$$

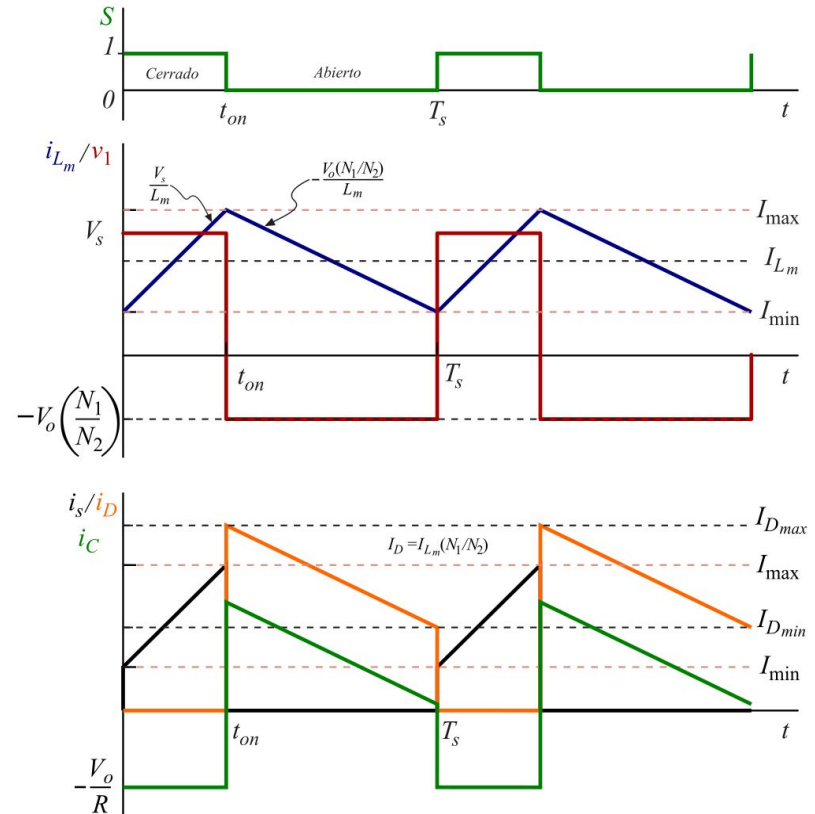
$$V_s D = \frac{N_1}{N_2} V_0 (1 - D)$$

$$V_0 = \frac{N_2}{N_1} \frac{D}{(1 - D)} V_s$$



Convertidor Flyback: relación de conversión

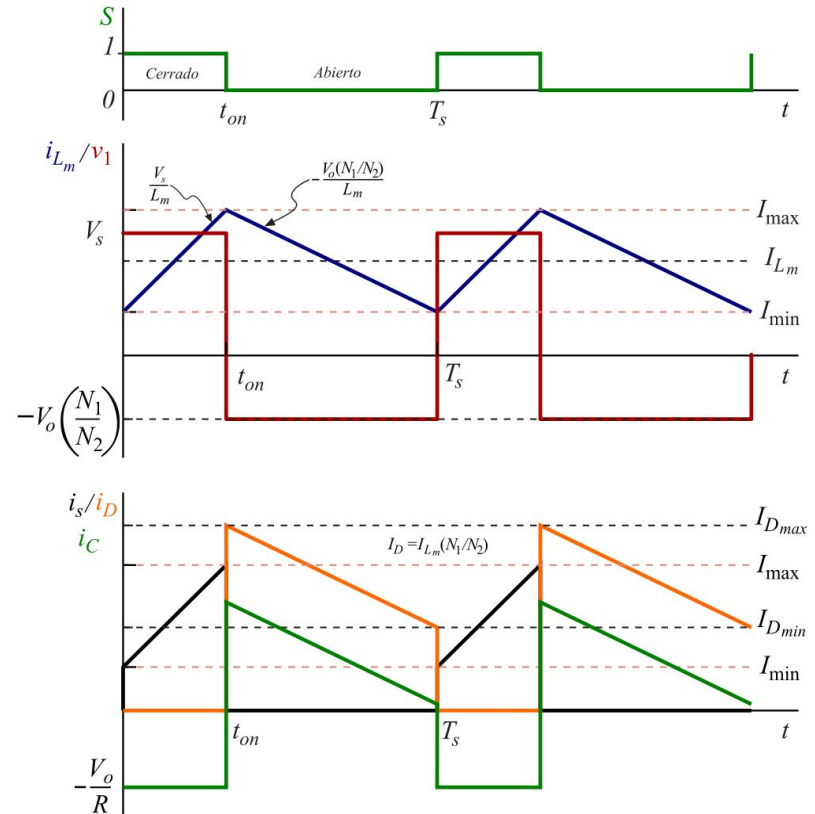
¿Qué pasa con la relación de corrientes?



Convertidor Flyback: relación de conversión

¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

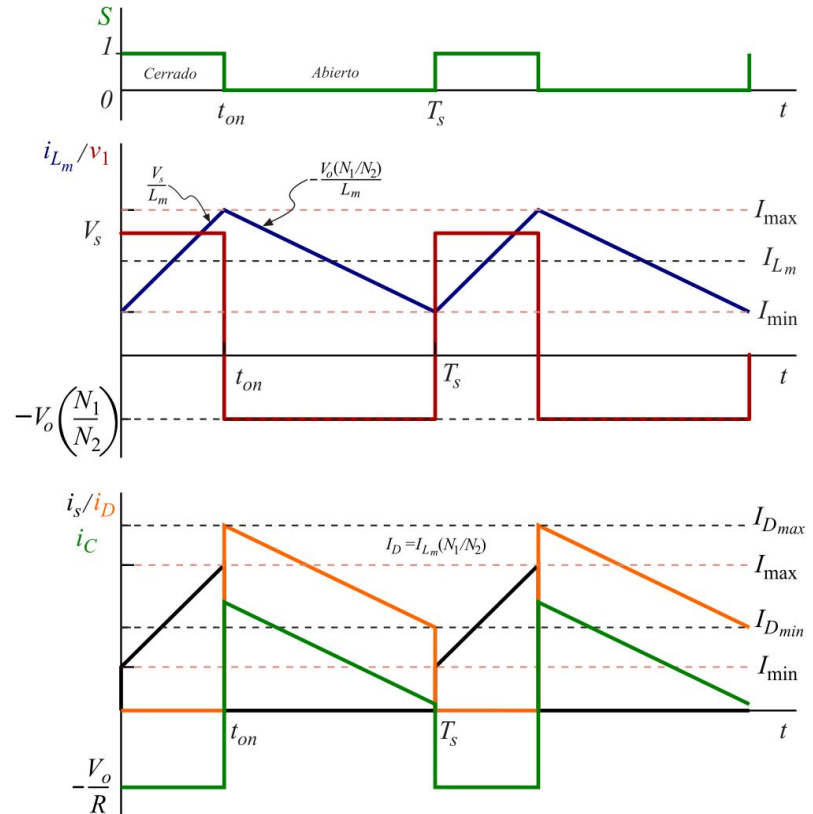


Convertidor Flyback: relación de conversión

¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

$$V_o = \frac{N_2}{N_1} \frac{D}{(1-D)} V_s$$



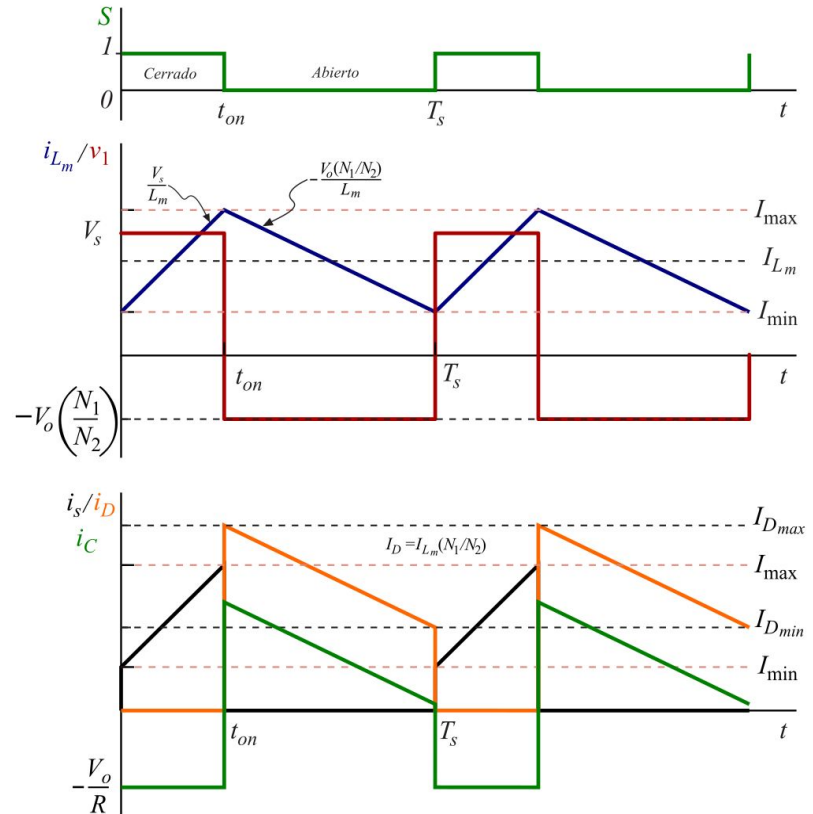
Convertidor Flyback: relación de conversión

¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

$$V_o = \frac{N_2}{N_1} \frac{D}{(1-D)} V_s$$

$$I_s = \frac{N_2}{N_1} \frac{D}{(1-D)} I_o$$



Convertidor Flyback: relación de conversión

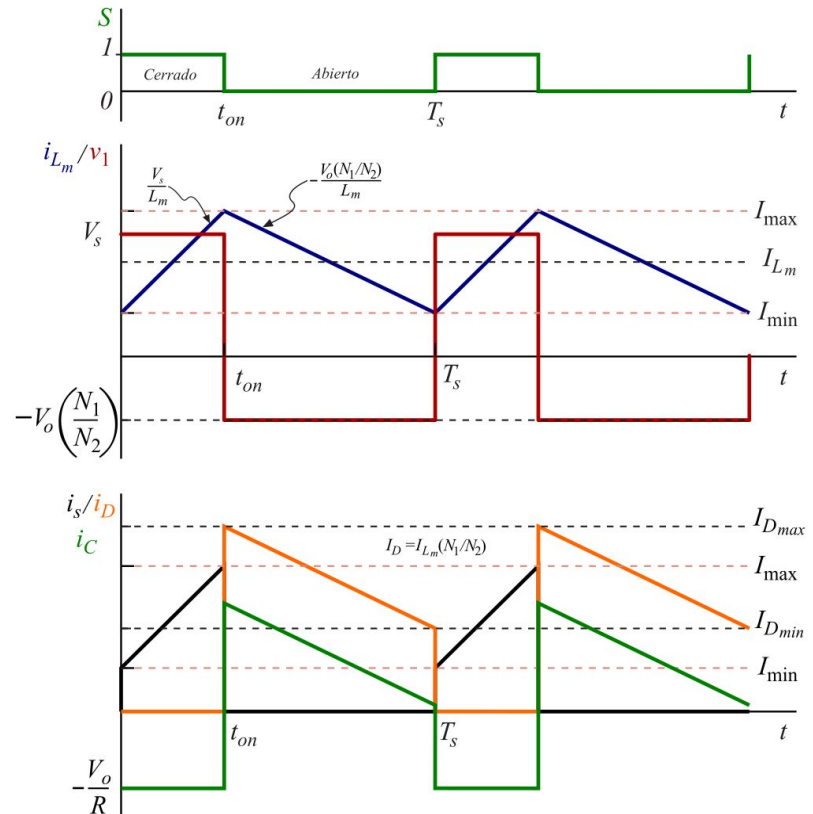
¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

$$V_o = \frac{N_2}{N_1} \frac{D}{(1-D)} V_s$$

$$I_s = \frac{N_2}{N_1} \frac{D}{(1-D)} I_o$$

$$I_s = D I_{L_m}$$



Convertidor Flyback: relación de conversión

¿Qué pasa con la relación de corrientes?

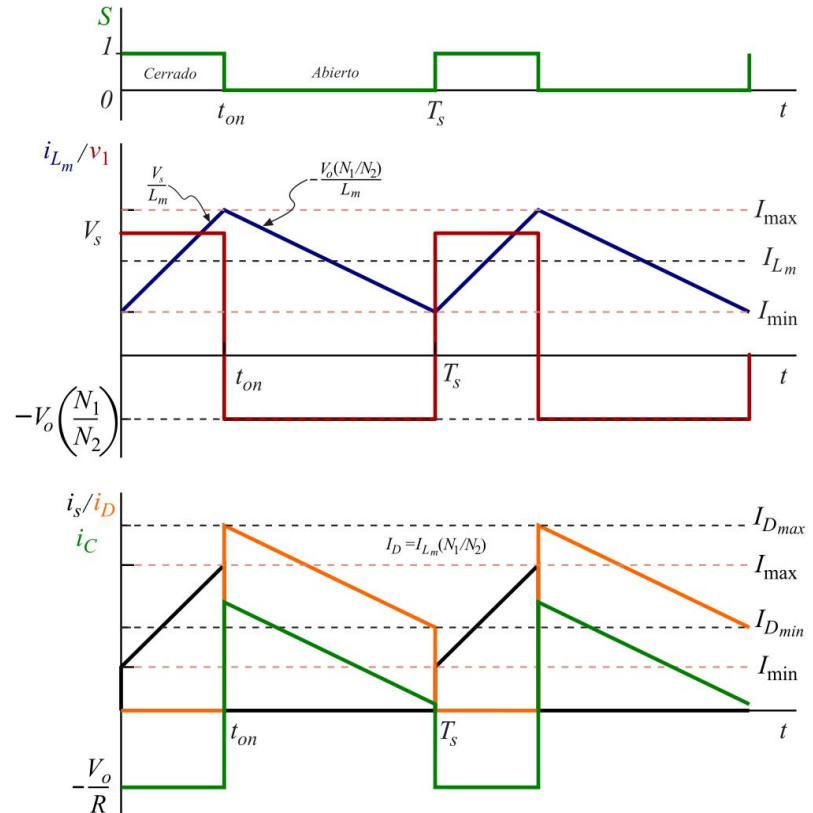
$$P_s = P_o$$

$$V_o = \frac{N_2}{N_1} \frac{D}{(1-D)} V_s$$

$$I_s = \frac{N_2}{N_1} \frac{D}{(1-D)} I_o$$

$$I_s = D I_{L_m}$$

$$I_D = (1-D) \frac{N_1}{N_2} I_{L_m}$$



Convertidor Flyback: relación de conversión

¿Qué pasa con la relación de corrientes?

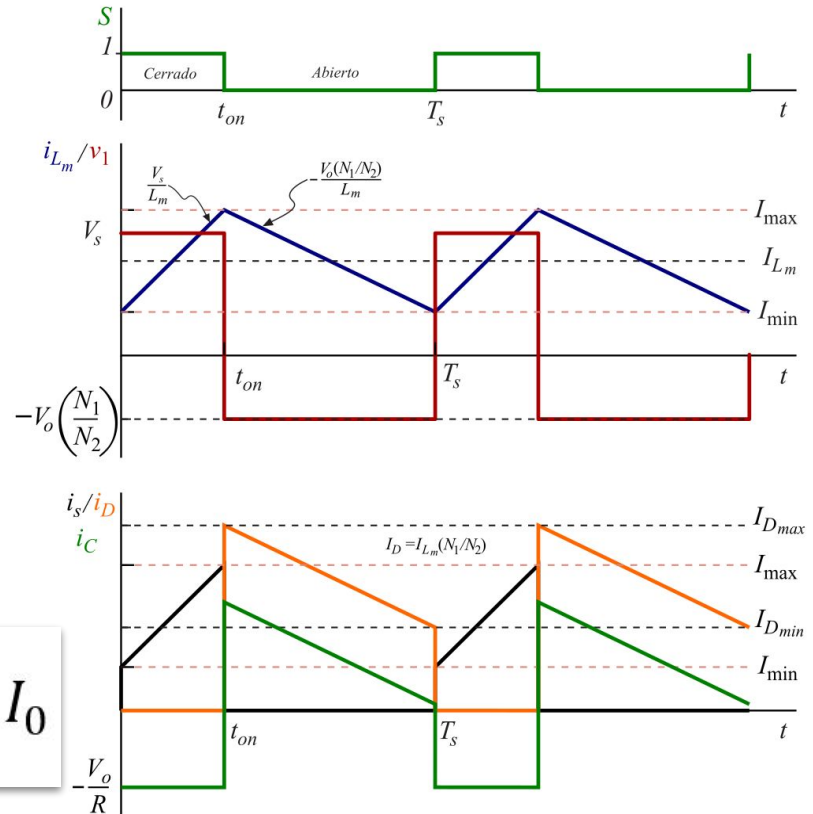
$$P_s = P_o$$

$$V_o = \frac{N_2}{N_1} \frac{D}{(1-D)} V_s$$

$$I_s = \frac{N_2}{N_1} \frac{D}{(1-D)} I_o$$

$$I_s = D I_{L_m}$$

$$I_D = (1-D) \frac{N_1}{N_2} I_{L_m} = I_o$$



Bibliografía

- **Electrónica de Potencia** - Daniel W. Hart (Prentice Hall, 2001)
- **Electrónica de Potencia: Convertidores, Aplicaciones y Diseño** - N. Mohan - T.M.Undeland - W.P. Robbins (McGraw Hill, 2009)
- **Electrónica de Potencia. Circuitos, Dispositivos y Aplicaciones** - M.H. Rashid (2da Ed. Prentice Hall, 1993)
- **Power Electronics. Devices, Drivers, Applications, and Passive Components** - B. W. Williams (University of Strathclyde, Glasgow. 2006).

Descarga gratuita:

→ <http://personal.strath.ac.uk/barry.williams/book.htm>

